Aspects of Classical Chinese grammar

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Chapter 1

Introduction

1.1 The language

This note is about Classical Chinese, often known natively (in Mandarin Chinese) as 文言 ('lit. Literary Chinese'). Regarding the meaning of the relevant terms see § 1.1.1.

The language is sometimes known as *Wen-li* by Western missionaries, especially in Bible translation. This seems to be a misunderstanding of the word 文理, which is a nominal compound and means rhetorics (i.e. 文) and meanings (i.e. 理) of literature works. An educated person therefore would be described as "通文理" ('fluent in rhetorics and meanings'). Such a person of course would have decent understanding of Classical Chinese, and hence 文理 was probably mistranslated as "Classical Chinese", although the word was not natively used to refer to the latter (Zetzsche 2017, p. 161).

Classical Chinese is also sometimes known as 古文 ('lit. ancient articles') or 古汉语 ('lit. ancient Chinese'), and equated with late Old Chinese. This terminology may be misleading, as there was considerable diachronic and regional variances in Old Chinese (§ 1.2).

Further, "old" written materials were definitely *not* always written in 文言. Novels like *Journey to the West* or *Dream of the Red Chamber* were written in earlier forms of Mandarin. Buddhist scriptures translated to Chinese are closer to the Classical paradigm but are also definitely influenced by the grammar of Middle Chinese. The non-Classical written materials, despite being important both culturally and linguistically, are beyond the scope of this work.

1.1.1 Classical Chinese vs. Literary Chinese

The word 文言 is more accurately translated to *Literary Chinese* in English. Strictly speaking, we need to distinguish between *Literary* Chinese, i.e. the high variety of the more than one millennia of diglossia in China, and *Classical* Chinese, i.e. the language of prose texts considered paradigmatic and having the prescriptive authority to dictate the "right" usages of Literary Chinese.

A natural question is then whether it is in fact meaningful to talk about a single, largely homogeneous *Classical Chinese*, when the Classical texts admittedly have diverse features. The goal of this grammar is to analyze the shared grammatical features of these texts. This resembles the stated goal of Huddleston and Pullum (2002), which is to analyze *International Standard English*, i.e. the shared core of all important contemporary varieties of English. The existence of such a shared Classical grammatical

core should not be taken for granted and can only be proven by demonstrating that most texts considered Classical are analyzable under the framework of this work (see also § 1.2.3).[1]

1.2 History, texts and scope

1.2.1 Scope of this grammar

We cover both pre-Han and Han texts in this grammar. This includes the received Classical texts (TODO: table) Excavated texts are invaluable but existing excavated texts often have dialectological issues (§ 1.2.2.2).

An overview of the post-Zhou history of Chinese language(s) is given in the next several sections, in which we also justify the scope of this grammar.

Literary Chinese, i.e. 文言 in its broad sense, also includes post-Classical texts whose grammars are roughly based on the Old Chinese canon but may have a variety of innovations (§ 1.2.4). We do note cite examples from post-Classical texts, except when illustrating diachronic points. The situation here precisely resembles the case of Latin, in which we have Classical Latin and Medieval Latin which imitates the former.

1.2.2 Late Spring and Autumn and Warring States periods

1.2.2.1 Periodization

The early history of Chinese is commonly divided into an Old Chinese period and the Middle Chinese period. The criteria of the division is given in § 1.2.4. The transition between the two periods is typically considered somewhere near the end of the Eastern Han dynasty. Thus, the Old Chinese period contains Shang and Western Zhou, the Spring and Autumn period which filled with chaotic (but not intense) wars between numerous dukedoms, the Warring States period which observed intense wars between seven major states, and the rise and fall of the Qin Empire and finally the Han dynasty.

The Old Chinese period can be divided into the pre-Classical and Classical periods. We consider most texts before Confucius pre-Classical, even considering that some of them have classic status in Confucianism. Thus the Classical period, in the broad sense, spans from the late Spring and Autumn period to late Han dynasty. In a narrower sense it properly begins with Confucius and ends at the founding of the Qin empire (Pulleyblank 1995, p. 3; § 1.2.2). The Han dynasty witnessed the unification and likely koineization towards a common literary standard (Pulleyblank 1995, pp. 3-4; § 1.2.3).

We justify the given pre-Classical/Classical boundary below.

First, pre-Spring and Autumn texts are certainly pre-Classical.

This treatment is consistent with traditional philological scholarship in China: Han Yu, renowned for reviving Classical prose, famously described the *Book of Docu-*

[1] A folk theory about Literary Chinese, which sometimes is generalized to Classical Chinese too, is that the language originated as telegraphese due to the alleged severe lack of writing materials and thus cannot be analyzed as an ordinary natural language. Given the highly redundant style and the rich speech act marking in the *Analect*, this theory is highly unlikely to be correct. Further, telegraphese and abbreviations in later Sinitic languages are markedly different from Classical Chinese (cf. 走美杯 or even 妈见打 in informal Modern Mandarin).

ments as 诘屈聱牙 ('twisted, hard to pronounce') in his 进学解 (*Analysis of academic advancement*). The *Book of Change* similarly is not comprehensible to a scholar trained only in Classical Chinese and requires commentaries to be understood. The *Book of Odes* includes poems spanning a long period of time from early Zhou to Spring and Autumn with considerable regional differences.[2] These texts are not imitated by post-Classical prose writers, except in e,g, imperial proclamations.

Both lexical and grammatical differences can be spotted between Zhou and later texts (received or excavated). The copula 惟 is out of use in Classical texts and the copula construction had been largely replaced by the nominal predication construction (§ 2.2.1). The conjunction 而, uniforming appearing in all later texts, received or excavated, is never seen in pre-Spring and Autumn texts. The pronoun 厥 and the verb 克 (meaning 'be able to') are largely lacking in later texts.

Second, early Spring and Autumn texts are also not a part of the Classical canon.

Limited texts before Confucius are either traditionally passed down in the *Odes*, or are attested in bronze inscriptions archaeologically discovered, and are not imitated by post-Classical prose writers. There lacks a large corpus of early Spring and Autumn continuous prose comparable to, say, the Confucian classics. The bronze inscriptions strongly resemble the Zhou paradigm and are thus pre-Classical.[3]

1.2.2.2 Dialectology

A western/eastern dialectal differences can be observed within the *Odes* (Ma and Sagart 2025) and thus it is not unreasonable to expect regional variances during the Classical period. Indeed, among Warring States texts, the *Mencius* explicitly mentions strong dialectal difference between Qi and Chu dialects (一齐众楚).

A rough tripartite dialectal division between the western Qin and Jin dialects, the eastern Qi and Lu dialects, and the southern Chu dialects in this period has been proposed, based on evidence from phonology (Bian and Huang 2017; Wang 2000), sentence-final particles (Kazuya 2024; Luo 2016), conjunction particles, prepositions, and clause embedding (Onishi 1998). Admixture is however not uncommon (Bian and Huang 2017), and non-phonological dialectal features can sometimes be attributed to orthography (for an example, see Wu 2011)) or register. Thus the precise dialectology in this period is still a challenging topic, which, according to some, cannot be reliably established by currently available evidence (Behr 2023). The discussion below is therefore inevitably partly speculative.

The most specific grammatical features of the Qin dialect are not present in Classical texts from the period and also have limited influences on later prose. As an example, the sentence final particle 殹 never appears in Confucian texts or later formal prose. Classical prose shares quite a lot features with the Chu dialect (the preposition 於, the sentence final particle 也, and the like), but these features are not Chu-specific. The most Chu-specific content words and grammatical markers (like the sentence final particle 些) in Chu poems are largely absent in pre-Han Classical prose texts. Certain orthographical features of excavated Chu texts, which appear to reflect regular sound changes (Bian and Huang 2017), are absent in their received counterparts.

It however is premature to conclude that Classical texts were based on the eastern

[2] This may partly be due to the nature of poetry: regularization in transmission is much easier for prose than for it is for poetry.

[3] As in 秦公钟: 余小子……克明厥心……

dialect group. The main problem is that there is currently no substantial excavated Warring States Qi or Lu bamboo manuscript corpus: the Qin and Chu languages are attested directly in excavated texts,[4] while Qi/Lu dialects are not and we know "extremely little" about their written forms (Feng and Yu 2024). No excavated pre-Han texts are included to represent eastern dialects in Bian and Huang (2017): their sample instead include *Zuozhuan*, *Xunzi*, and Han bamboo manuscripts from Yinqueshan 银雀山, and *Odes* poems thought to represent eastern dialects.

The currently available *Analect* text were widely considered to be from Lu (i.e. 魯論語) and presumably contains eastern dialectal features, and the *Mencius* largely resembles the *Analect* grammatically (Karlgren 1936, pp. 63-77,90). The shared grammatical features between the works may be attributed to the shared language of early Confucianism, but the possibility of later regularization cannot be outright ruled out.

On the other hand, despite being traditionally attributed to Lu historians and included into the eastern group in Bian and Huang (2017) in their comparative study, the language of *Zuozhuan* is well known to be different from that of the *Analect* in terms of the syntax of prepositions, conditional markers, and demonstratives (Karlgren 1936, pp. 63-77).

The peculiarities of *Zuozhuan* may indicate existence of yet another dialect, possibly a pre-Han koine used by historians (Pulleyblank 1995, p. 3). Alternatively, some of the diverging behaviors observed by Karlgren, like choices of prepositions, may just be due to historical development (Venture 2020). We should also keep in mind that received texts are layered and often have a rather complex textual history.

Slightly later texts, include *Zhuangzi*, *Xunzi* and *Han Feizi*, have largely comparable grammatical features with resemblance to both the Lu dialect and the language of *Zuozhuan* (Karlgren 1936, pp. 92-93). The exact nature of the similarities and differences between these texts, the *Analect* and *Zuozhuan* again has multiple possibilities, ranging from convergence towards a common literary koine (Pulleyblank 1995, p. 3) to retaining of weakening archaism in the same dialectal tradition; later artificial normalization also cannot be confidently ruled out.

In conclusion, we can safety state that the Qin and Chu dialects attested in excavated texts have marked differences from the Classical paradigm, even though they were written in the Classical period. Determining the exact origin of the Classical paradigm known today however is much harder. On one hand, there is evidence suggesting a disproportionate large eastern contribution to later Classical prose: the idiosyncratic features of Qin and Chu do not appear in later prose; the traditional narratives attribute the *Analect* and *Zuozhuan* and related texts to Lu; there is phonological evidence for them being in a eastern dialect group. On the other hand, the internal diversity of received texts from the Classical period cannot ignored, which has multiple explanations, and the lack of excavated eastern texts make it a shaky practice to call Classical texts eastern without qualification.[5]

[4]Note that administrative texts should better reflect the regional dialects; texts considered classics may just retain archaic features (Zhang 2011, pp. 45-46).

[5]It would be a major breakthrough to rediscover a corpus of Lu texts. Our eastern-dominated model of received Classical texts would face a major challenge if documentary or scholarly manuscripts written in eastern dialects were discovered and demonstrated to be markedly different from received texts. In this case, the formation of the Classical standard would need to be postponed to the Han dynasty koineization and canonization.

Interestingly and unfortunately, manuscripts that were highly likely from Lu were once found from the walls of Confucius's old residence in Qufu during the last half of the 2nd century BC, and later

A final issue is that students of Confucius noticed that when he recited texts he considered classics and presided rituals, he used 雅言 or 'elegant speech' (*Analects* 7:18). One may wonder what this elegant speech exactly was (presumably it should have Zhou components), but due to lack of materials, it is hard to pin down its source.

1.2.3 The Han dynasty

1.2.3.1 Emergence of a literary koine

The last batch of works that may be reasonably considered Classical were composed during Han dynasty, among which is *Shiji* 史记 ('lit. historical records'), also known as 太史公记 *Records of the Grand Historian*, which laid the paradigm of official historiography of all Chinese dynasties after Han.

The dialectal distinctions in lexicon or grammar discussed in the last section are largely absent in the most prestigious Han prose texts (Karlgren 1936, p. 94; see also § 1.2.3.2).

The overall grammar of prestigious Han prose is consistent with pre-Qin Classical texts. For instance, He (2005) is a comprehensive comparison between *Shiji* and *Zuozhuan*, which finds notable but largely quantitative differences, mainly an increase of complex forms in the texts. Two noticeable possibly qualitative developments are more pre-verbal prepositional constituents, and more forms that can be analyzed as verbal complements (补语; § 8.3), which are likely natural developments towards Middle Chinese. The change was therefore not as radical as the change from the *Documents* to Spring and Autumn texts but also noticeable enough to justify separating Han as a unique linguistic period from the Warring State period (He 2005, pp. 260-264).

However, Han prose should not be regarded as a simple continuation of the proposed eastern Classical standard. The short-lived Qin dynasty substantially shaped the standard of *writing*, and likely also phonology of at least some dialects. Bian and Huang (2017) for instance classify loangraphs in *Shiji* as evidence of western dialects, which can be interpreted as a reflection of the phonology of Sima Qian's own language or the writing standard in his time. On the other hand, as is said above, Qin influence on Classical prose is much more modest in grammar. One clear influence of the language of Qin on *Shiji* is the comitative-only usage of 与 (Onishi 1998).

The lexicon of Western Han prose shows much stronger admixture (TODO: 周秦两汉楚方言词研究).

What we observe above appears to be significant koineization during the Western Han dynasty. The overall grammar and lexicon of this literary koine emerging in the Western Han are largely based on the emerging common elite prose standard attested in linguistically converging late Warring States texts, possibly originating from eastern dialects (see caveats in § 1.2.2.2). It has also received lexical influences from

lost. 汉书 • 艺文志 records Liu Xiang's textual criticism of the excavated texts, observing ~ 700 textual variants. The received *Analect* text, edited by 郑玄 in the Eastern Han dynasty and was known to be partly based on these archaic texts (but with no received critical apparatus). 郑玄 also made a critical apparatus of 仪礼, which contains largely differences in content words and orthography.

Evidence listed above is consistent with the hypothesis that the texts from the walls (presumably in eastern dialects) had no substantial linguistic differences from received texts, and hence our model that the Classical paradigm was predominantly shaped by eastern dialects. But the evidence is also consistent with the later Han koineization model: it may be the case that Zheng Xuan intentionally ignored all grammatical variances he observed.

other dialects. Its phonology and orthography, on the other hand, are likely strongly influenced by the *western* dialects.

The final formation of such a koine is presumably partly a consequence of mass destruction of non-Legalism texts during the Qin dynasty (though the nature and scale of the destruction has been long debated): surviving texts, regardless of their dialectal origins, were revered as the golden standard of prose, and a great admixture of grammatical features thus happened (Karlgren 1936, p. 94).

Another factor is institutional. The Han Empire had a centralized system of governance and actively sought ancient texts and appointed erudites to profess Confucian classics; an obvious but not overly strong preference for scholars from the Qi-Lu area can be seen (Cai 2019). This created an ideal environment for koineization. Presumably, the existing Qi-Lu scholarship tradition and its elite language was strengthened by its incorporation into Han officialdom, and western (recall that the capital city of Han, Changan, was in the former Qin territory) and southern scholars were absorbed into the network, bringing some but not all of their regional features into the koine.

The Han dynasty also had a national education system for future officials. Excavated legal documents describe orthographic training and examinations in handwriting and recitation of texts (Li 2007).

1.2.3.2 Dialectology and genres

The literary koine for prose, of course, is far from the full picture of the language situation of the Han dynasty. Lexically, this has already been clearly indicated by Yang Xiong's 方言, and in this section we discuss the phonological and grammatical aspects of Han dynasty dialectology. Relevant topics are beyond the scope of this grammar and here we can merely skim through them to illustrate the nature of the Classical paradigm.

First, literature in the Han dynasty is not limited to prose. Especially worth noticing is the *fu* genre. Traditionally, *fu* is believed to originate from Chu poems, for the obvious reason that Chu words appear frequently in *fu*. The *fu* genre also likely involved koineization, given the tendency for *fu* authors to use rare words and their diverse regional origins.

A natural question is whether the koine of *fu* differs in a meaningful way from the koine of Han prose. The grammar of *fu* is beyond the scope of this grammar.

Second, the rhyme pattern of poems may indicate dialectal phonological differences of the oral language(s), though it is well known that rhyme patterns are often also strongly influenced by the literary tradition. Dialectal differences, largely consistent with Yang Xiong's 方言, can indeed be observed (Zhang 2023). Obviously, the literary koine was not always identical to the phonology of oral speeches.

Third, we may also wonder the overall difference between the prose koine and the oral language(s) of the period. This also involves grammatical continuation of the second caveat: certain types of excavated Han manuscripts presumably better reflect the oral language(s), and thus inconsistencies between the prose koine and the oral language(s) and regional variances may appear in them.

The excavated administrative texts and private letters do possess certain divergences from the Classical paradigm. Ren (2022) notices a possible possessive pronoun 央 attested in contemporary southern Sinitic languages but not in Classical Chinese and also a dialectal term recorded by Yang Xiong. Another interesting feature is that

the character 遘 appears regularly in the manuscript as a demonstrative before temporal phrases, a use rare in received texts but somehow has precedent in pre-Classical texts (Zhang 2004). More often than not, though, the overall grammatical framework largely resembles the Classical paradigm, once orthographical inconsistencies (e.g. 得毋 vs. 得无) are resolved. Certain usages seemingly unusual to a reader trained only in Classical Chinese, like person names followed by 辞 before long narratives (Ren 2022), can be reasonable explained within the Classical framework (in this case, the noun 辞 takes as an argument a preceding noun phrase denoting the speaker).

It remains to be seen if a systematic classification of grammatical peculiarities in these documents is possible, if there is any. This can be perceived as evidence for the Classical prose paradigm being the literary language of not only elite scholars but also low-level officials and scribes (see the end of § 1.2.3.1).

1.2.4 On the status of post-Han texts

The end of Old Chinese – and hence the end of the Classical period – is marked by, among others, the emergence of 是 as a copula (and not just a demonstrative), the appearance of the disposal construction and the long passive construction (Wang 2013, p. 35).

Thus the mental grammar of an author in post-Han Literary Chinese prose almost certainly differed from any possible grammar of Classical writers. Expectedly, despite purification attempts, vernacular elements made their ways into not only administrative documents but also pure literature and scholar works.

A radical example is texts from the Republican period of modern Chinese history, when official documents were written largely in a pseudo-Classical style: the abstract syntax behind these documents is almost certainly within the framework of modern Mandarin, with the possessive marker 的 replaced by 之 and the relative marker 的 in headless relative clauses replaced by 者 (不服管教的 'who does not conform to instructions' is replaced by 不服管教者). What is Classical in these documents is primarily the superficial forms.

Still, because of the highly formulaic nature of these documents (for instance, usually no sentence final particle appears, and valency alternation besides the passive is discouraged), readers of these texts still benefit from studying the grammar of Classical Chinese. More details can be found in § 19.2.2.

1.2.4.1 Decay and revival of Classical prose

The fall of the Han Empire was accompanied by the fall of Classical prose except in historiography. The post-Han literature history witnessed proliferation of *Fu* and later the emergence of *pianwen*, and the emergence of Chinese translations of Buddhist texts (which retains a lot of Classical features but already contains colloquial features) and later the 变文 genre (containing strongly oral features).

Literary Chinese, as we know it today (§ 1.1.1), can be attributed to the mid-Tang Classical Prose Movement. Proponents of Classical prose, the most famous ones being Han Yu, Liu Zongyuan, and the rest of the Eight Masters of the Tang and Song, rejecting *pianwen* which they considered overly decorative and empty in substance, strongly advocated for a return to both Classical prose and the Old Way, i.e. Confucian ideals.

The movement was incredibly successfully and successfully redefined what prose — 文 — meant for subsequent generations. Thus, we can say that despite that the Classical prose paradigm had stabilizedin the Han dynasty, the concept of a Literary Chinese, independent of oral Sinitic languages, with its prescriptive authority being the Classical texts, competent of writing about everything from history to epitaph, was not in place until the Classical Prose Movement.

Literary Chinese inevitably has vernacular influences from all ages. Its diachronic development is not covered in this grammar. Its existence, however, greatly strengths the cultural importance of Classical Chinese.

1.3 Previous studies

1.3.1 Native grammatical traditions

Classical Chinese authors had conversations about grammaticality and uses of grammatical particles reminiscent of how English native speakers with some exposure to the study of English grammar but no formal training: "delete the *the* here and your sentence looks more concise". No attempts were made to establish intermediate concepts between words and utterances, like structural templates of phrases or grammatical relations, and to organize the grammar as a machine producing acceptable utterances: discussions on grammatical topics were either for education or for rhetorics.

The grammatical awareness of Classical Chinese authors was somehow comparable to what an ancient Roman grammarian or *grammaticus* did, who set his main role as a secondary educator, refrained from analyzing some sort of "underlying" or "internalized" system behind the surface forms and was satisfied by mostly surface-oriented patterns, and would not set up any intermediate concepts between the word and the utterance (Matthews 2019, pp. 7,35,47-48). On the other hand, this approach is contrary to the practice of the Paninian Sanskrit grammatical tradition, which, in today's terminology, starts with dependency relations and abstract features and uses a set of morphophonological rewriting rules to produce the corresponding surface forms (Kiparsky 2009).

The Classical Chinese grammatical tradition appears even looser compared with the Roman tradition in that the former did not even attempt to recognize parts of speech; this however was deeply rooted in the structure of Classical Chinese in that

> **Box 1.1: Ancient Chinese grammatical tradition and Roman tradition**
>
> Is the situation somehow close to what a Roman grammarian (*grammaticus*) would do? It seems that Roman grammarians also didn't care about abstract structures. See:
>
> - Use and Function of Grammatical Examples in Roman Grammarians
> - Quintilian's 'Grammar' (Inst.1.4-8) and its Importance for the History of Roman Grammar
> - What Graeco-Roman Grammar was about

On the other hand, phonology was an active topic in ancient China. This was probably due to influence of Buddhism: the need of translating Buddhist terminology into Chinese motivated systematic studies on Chinese phonology, and clear Sanskrit influences can be found in traditional works on phonology.

1.3.2 Perspectives of European missionaries

Large-scale, systematic and reliable grammatical description of Classical Chinese had unfortunately been largely lacking for quite a while, and until recently, when Classical Chinese was studied linguistically, it was the historical phonology that was studied (Pulleyblank 1995, p. xiii). Several missionary grammars however were still produced

Box 1.2: List of notable works

马若瑟汉语札记

1652 年，卫匡国完成了《中国文法》一书。他将汉语划分出十大词类，并在动词部分对汉语主动语态和被动语态有所介绍。1703 年万济国的《华语官话语法》，依旧沿袭拉丁语法，进行词类划分，并对其形态和范畴进行描写。除此之外，对汉语的一些造句规则（主要是主动句型和被动句型）也进行了归纳说明。1728 年，马若瑟的《汉语札记》首次区分了汉语口语和书面语的语法规则，指出了汉语中“实字”与“虚字”的区别。不过马若瑟并没有依照拉丁语法构造汉语语法规则，而是从语言事实出发对汉语进行描写，这样的做法更符合汉语言的实际情况。

进入 19 世纪，随着欧洲汉学的兴起，越来越多的传教士和专业学者投身于汉语的语法研究中。法国汉学家雷慕莎的《汉文启蒙》在词类划分的基础上，探讨了一些句法规则，并指出汉语的语法不能用西方传统的拉丁语法去生搬硬套。在传教士方面，马礼逊的《通用汉言之法》其系统性比之前的语法专著更强，但主要内容还是集中在词法上，句法部分只归纳了九条规则，仅占五页篇幅。马士曼的《中国言法》在汉语语法本体研究上，也是由八大词类入手着重讨论词法，对句法的讨论较为简单，甚至有时句法与词法的界限并不清晰，将许多构词法放入了句法部分。艾约瑟的《中国官话语法》在前人基础上有所进步。除词法外，他对汉语句法规则的考察也颇为详细，这是此前西方人研究中所少见的。

1.3.3 Modern descriptions

Modern linguistic description of Classical Chinese has a clear starting point: the publication of 《马氏文通》‘Ma’s ’ by Ma Jianzhong (馬建忠).

TODO: 中等国文典, American structuralism in China for the study of Mandarin, Any other grammars in English besides Pulleyblank?

1.4 Remarkable features

1.4.1 Morphological typology and wordhood

Although we are confident that Classical Chinese has words (§ 2.1), the distinction between words and phrases is often hard to put down, because of Classical Chinese being incredibly analytic. Indeed, Classical Chinese is much more analytic than any modern Sinitic language: it does not uncontroversially have resultative complex predication; it definitely lacks any aspectual suffixation in its verbal system (cf. Standard Mandarin aspectual *-le*).

This is interesting because

1.4.2 Flexible parts of speech

1.4.3 Topic-comment constructions

1.4.4 Small inventory of grammatical markers

A folklore about Classical Chinese states that it has a rather simple grammar. This cannot be literally true: we still need multiple pages to only give a sketch of its grammar (Ch. 2). However, it *is* true that Classical Chinese has a rather small inventory of overt grammatical exponents.

For instance, most valency alternations in Classical Chinese are completely unmarked (§ 2.2.2). The prepositions 于 and 於 cover multiple functions that are otherwise marked by multiple prepositions in other languages.

1.4.5 The grammar of characters?

A misconception commonly found among people not well informed about sinology is that Chinese characters directly represent ideas. This is wrong, especially considering the strongly phonetic uses of Chinese characters in excavated texts and existence of phonetic series. However, a question worth asking is whether the complex and compositional structure of Chinese characters resembles the architecture of spoken natural languages.

This issue has been explored in Myers (2019), with a largely affirmative answer: in the book he identifies a word-like grammar (consisting of not only morphology but also phonology) of Chinese characters,[6] and that these grammar-like rules are likely still active even today. Sub-lexical processing of Chinese characters has indeed been identified in independent neurolinguistic experiments (e.g. Chinese readers find it easier to process characters with valid semantic or phonetic components; Duan and Cai 2024, § 2.2).

This naturally raises the question on the mental capacity origin of these rules, which is ultimately related to the big questions in theoretical linguistics and cognitive science and is far beyond the scope of this grammar.[7]

[6] Worth noticing is that similar patterns can be found in all orthographic systems (Myers 2019, pp. 23-25). English orthography, for instance, has phonology-like rules like *-y* + suffix → *-i*-suffix (Myers 2019, p. 26).

[7] To illustrate where these questions may lead to: the brain region primarily responsible for recognizing Chinese characters is the Visual Word Form Area, which is not a part of the language network in the brain. One may thus argue that the "grammar" of Chinese characters and the grammars of spoken languages have different neurological origins (Myers 2019, pp. 209-210), but then the resemblance between the two requires explanation. One may explain the resemblance by domain-general factors, which, if valid, means that what we thought were unique to languages probably are not domain-specific to languages at all. But alternatively this can be interpreted as (surface) morphology and phonology being fundamentally different from syntax. The relevant debates are not essential for understanding the structure of Classical Chinese.

Chapter 2

Grammatical overview

In this chapter we summarize the syntax of the clause (§ 2.2) and the syntax of the noun phrase (NP) (§ 2.3). Form classes in Classical Chinese are analyzed in § 2.4 based on our analysis of structural properties of noun phrases and clauses.

2.1 Morphological typology and wordhood

2.1.1 Wordhood and morphology

Grammatical wordhood (as opposed to phonological wordhood) has been a controversial topic in linguistics. We overview relevant issues in this section.

Grammatical wordhood is perhaps most uncontroversially defined in terms of domain phenomena without clear phonological or combinatorial syntactic motivation. A prototypical example is fixed and obligatory inflectional affixes, whose boundaries demarcate word boundaries. Because Classical Chinese has no inflectional morphology, it is not possible to demarcate word boundaries based on inflectional templates.

A candidate of such phenomena in Classical Chinese is reduplication, which sometimes alters the consonant of the second syllable and thus are not analyzable as syntactic reduplication (Travis 2001). These rules would be known as morphological in other languages, and it is reasonable to call the domain in which they operate the grammatical word.[1] This however is not a criterion applicable in most situations.

Indirect evidence from morphosyntax-sensitive phonological and prosodic phenomena, like the criterion of stress domain (Zhu 2009, pp. 41-42) or the criterion of pause insertion, is obviously not available for a dead language with limited suprasegmental information.

2.1.2 Wordhood and unpredictable grammatical behaviors

Cross-linguistically we observe lexical specification of grammatical behaviors of lexical entries,[2] which imposes strong constraints on their surface appearance, distribution, complementation (e.g. verb valency) and modification. This is in contrast with

[1] This still does not rule out the possibility of morphology without wordhood but the grammar of Classical Chinese does not appear to require such an analysis.

[2] Terms like *lexicalization* are used to refer both in the literature: Harley and Noyer (1999) for example uses the term *lexicalization* to refer to idiomization (i.e. a form having a non-compositional semantic reading, not necessarily grammatical properties). They may even refer to the diachronic collapse of the internal structure of a form to a simpler one.

(semantic) idiomization, as it is unusual if not impossible for a phrasal idiom (like *kick the bucket*) to have such combinatorial constraints.[3] Such lexical entries further undergo multiple possible morphological processes (like those in § 2.1.1) to become a full-fledged concrete grammatical word, and are thus related to grammatical wordhood.

In using this criterion to probe wordhood, attention should be paid to the maximality issue: a form with lexically specified, non-inferrable grammatical properties may be only a *part* of a word; even a maximal form with non-inferrable grammatical properties is not necessarily always a single word, because it can be augmented by regular morphological processes. The phenomenon of lexical specification of grammatical behaviors therefore gives strong hint about where to look for word boundaries but does not on its own settle the problem.

Following the tradition of Sinitic linguistics, we treat all maximal forms with non-inferrable grammatical behaviors as grammatical words, unless there are reasons to think otherwise. The Mandarin verb 關心 has a regular verb-object structure, but it as a whole being transitive cannot be regularly inferred. Thus it is a lexical entry with lexically specified valency, and from this we tentatively assume it is a grammatical word. Similarly, if a constituent at the core of a clause or a noun phrase has lexically specified grammatical properties which decide the structure of the larger constituent, we assume that it is a word. We apply this analysis in § 2.2.2.1, § 2.3.1.1.

2.1.3 Some other criteria

Composite forms are sometimes considered words because they contain bound roots, or have parts that cannot be modified, or have unpredictable meanings (Zhu 2009, pp. 41-43), or are built by processes with features not attested in prototypical phrasal syntax (e.g. lack of case inflection). These criteria have theoretical issues and sometimes risk circular reasoning.[4] The last criterion is also not applicable in Classical Chinese, as words in Sinitic languages are well known to have phrase-like structures (Zhu 2009, p. 41).

In summary, Classical Chinese likely has grammatical words just like other languages, but wordhood tests are not always completely reliably applicable to composite forms in Classical Chinese. This is consistent with the previous observations: Pulleyblank (1995) does not make any distinction between word formation and phrasal syntax, while Unger (2019) has such a distinction but admits that "the boundary between a compound word and an occasional phrase is not clearly definable" (p. 1). Based on existing materials, wordhood of composite forms has to be determined on the basis of inference of best explanation on top of the less reliable diagnostics above. For instance, given that there are far more words than idioms, and that cross-linguistically,

[3] Idioms easily have valency too (as in *take advantage of*) but the valency can almost always be located to a single word (in the example above, *advantage*).

[4] As is admitted by Zhu, unpredictable meanings can be explained by idiomization. The criterion of bound forms is circular as it is repeating the definition of bound form, which then has to be defined, and it is not an easy task: a transitive verb obligatorily takes an argument, which does not make it a bound form. The criterion of non-phrase-like grammatical behaviors is also circular. Phenomena related to the criterion of inability of modification can be simply explained by assuming that a *different kind of modification* is needed: consider *[ethno-[methodology]]-ical*, which structurally resembles phrases with modification. Without inflectional or phonological criteria, it is hard to explain how this form is fundamentally different from prototypical noun phrases.

structure-building processes with strong complexity constraints do tend to produce grammatical words, a form with a non-combinatorial meaning whose parts also do not accept modification is highly likely a word, as alternative explanations are unlikely (see e.g. § 3.2).

We further remark that a language can be reasonably well described by focusing purely on roots, formatives, constructions combining them and lexically specified grammatical properties of simplex and composite forms (as in § 2.1.2). The exact position of word boundaries or the existence of multiple types of domains is perhaps less important in the overall architecture of the grammar of the language.

2.2 Clause

Like all other languages, a Classical Chinese clause can be a simple clause or a complex one constructed from subordination (§ 2.2.8) and coordination (§ 2.2.9).

A simple Classical Chinese clause is a nucleus clause plus discourse-related devices, including its speech act marked by sentence final particles (§ 2.2.7), and topicalization or focalization (§ 2.2.6). Topicalization can also happen for a complex clause (§ 2.2.9.1). These discourse-related devices typically only appear in sentential clauses and are absent in embedded clauses (except for direct quotations; § 2.2.2.8).

A nucleus clause may have nominal (§ 2.2.1) or verbal predication (§ 2.2.2). We call it a *nominal clause*[5] or a *verb clause*, respectively. Both constructions seem to have a well-defined subject position (which is not the same as topic), which however is often left blank (§ 2.2.5). Classical Chinese lacks obligatory tense, aspect, mood (TAM) marking but does have a TAM system marked by adverbs and possibly some sentence final particles (§ 2.2.3).

2.2.1 Nominal predication

The top-level structure of a Classical Chinese clause may contain an optional subject and a noun phrase (NP) acting as the predicate (1, 2). A nominal predicate typically expresses an "is-a" relation between the subject (§ 2.2.5.3) and the predicate (1). Some nominal clauses however express a possessive relation between the two (2).

(1) [此 人]$_{\text{subject}}$ [力士]$_{\text{nominal predicate:NP}}$
cǐ rén lì-shì
this person strength-man
'This person is a man of great strength.' (史記・魏公子列傳)

(2) [蟹]$_{\text{subject}}$ [六 跪 而 二 螯]$_{\text{nominal predicate:NP}}$
xiè liù guì ér èr áo
crab six leg CONJ two claw
'A crab has six legs and two claws.' (荀子・勸學)

We note that the nominal predication is nominal in the structural sense: a verbal clause can have a predicate headed by a form only recorded as a noun in the lexicon (cf. § 2.2.2.1).

[5] We do not use the term to refer to a nominalized clause or a complement clause (which often does not have many nominal features anyway).

Negation in Classical Chinese nominal clauses is usually expressed by 非, placed before the nominal predicate (3).

(3) [[非]$_{\text{negation}}$ 忠臣]$_{\text{nominal predicate:NP}}$ 也
fēi zhōng-chén yě
NEG loyal-minister SFP
'[He] is not a loyal minister.' (說苑・立節)

The nominal predicate is compatible with TAM adverbs (4).

(4) 且 相如 [素]$_{\text{TAM}}$ 賤人
qiě Xiàngrú sù jiàn-rén
and NAME always humble-person
'...and Lin Xiangru has always been a person with a very low social status.' (史記・廉頗藺相如列傳)

2.2.1.1 Copula constructions

All the constructions mentioned above are without a copula. In the pre-Classical age there was a copula 惟, which however had largely died out of use in Classical texts. Meanwhile, grammaticalization had added several copulas to Classical Chinese (Pulleyblank 1995, pp. 20-22).

2.2.2 Verbal predication

A verbal clause consists of a subject and a verb phrase (VP), which is the predicate of the clause. Verb phrases can be coordinated (§ 2.2.9), demonstrating their constituent status.

A verbal predicate consists of a main verb, arguments and adjuncts (§ 2.2.2.2), and TAM adverbs (§ 2.2.3), as illustrated in (5). Hierarchical phenomena can be observed in this example: TAM adverbs always proceeds arguments or circumstantial adjuncts, and among the latter, the instrument typically appears before the P argument.

(5) [三王]$_{\text{subject,A}}$ [[既]$_{\text{TAM}}$ [以 一]$_{\text{instrument}}$ [定]$_{\text{main verb}}$
sān-wáng jì yǐ dìng
three-king already using determine
[法度]$_{\text{P}}$]$_{\text{verbal predicate:VP}}$
fǎ-dù
law-system
'The three great kings had already established social institutions by [it].' (荀子・大略)

In describing non-TAM clausal arguments, it is wise to separate (deep) syntactic argument or adjunct slots and (surface) grammatical relations (subject, etc.). We use the standard S, A, P, G, T etc. labels to refer to the former. This does not imply the unity of morphosyntactic properties of these slots represented by a single label (S roles famously split into A-like and P-like ones, for instance). Slots with no standard letter labels are represented as *location* or *instrument* etc., as in (5).

The mapping from the deep argument slots to grammatical functions and surface morphological case marking is known as alignment. A further subtlety is that the

mapping from the former two to surface morphological case marking may also be heterogeneous (Maling 2001).

The diagnostics below supports treating Classical Chinese as having nominative-accusative alignment and hence has a well-defined subject (§ 2.2.5). This, together with the fact that the verb phrase is a constituent active in other parts of the grammar, means that the clause structure of Classical Chinese resembles that of English, in that a verbal clause consists of a verb phrase and a subject, which, being outside the verb phrase, may be known as the external argument (Huddleston and Pullum 2002, p. 53). Classical Chinese has no case system and does not have quirky subjects marked by adpositions.

On the other hand, it is cross-linguistically harder to establish the *object* as a recurring grammatical function across different clausal constructions (cf. the case in Latin; Pinkster 2015, pp. 28-29). In studying Classical Chinese, the term *object* is traditionally often used to refer to any argument without an explicit preposition for convenience (Li 2004, p. 102).

2.2.2.1 The main verb

Classical Chinese completely lacks prototypical verb inflectional morphology, rendering wordhood judgement difficult. Whether a monomorphemic form can be the main verb of a clause, what valency it has, and its possible valency alternations (§ 2.2.2.3) appear to be lexically specified. This supports wordhood of monomorphemic verbs (§ 2.1.2).

Whether certain multimorphemic forms should be considered words and hence instances of verb derivation is less clear. As an instance, in the sequence 問曰, it is hard to determine if the second verb had already been incorporated into the first verb. We do not find any instance in Classical texts that require a verbal morphological analysis (問曰 for instance at least originally came from argument structure-level coordination; cf. 18).

A peculiarity of Classical Chinese is that a root typically appearing in non-verbal environments can undergo ad hoc categorization and serve as the main verb.[6] (6) is a perfectly well-formed verbal clause: it expresses a dynamic event and not just a state and has a modality auxiliary (illustrating that the predicate is not a noun phrase). (8) is a stative verbal clause involving ad hoc categorization.

(6) [假 舟 楫 者]$_{\text{subject}}$，[非 能 水]$_{\text{verbal predicate:VP}}$ 也……
jiǎ zhōu jí zhě fēi néng shuǐ yě
draw.help boat paddle REL NEG can swim SFP
'Those who draw help from boats and paddles cannot swim, …' (荀子・勸學)

This ad hoc categorization process differs from prototypical derivation in that it is irregular (and thus it is not possible to generalize from the pattern in 6 to a generic process; Dixon 2010, § 11.3). Unlike the case in formal English,[7] in Classical Chinese

[6] The main verb is sometimes known as the *predicator* i.e. the core of the predicate, to highlight that we are talking about a slot in the clausal structure and not a form class in the lexicon. For convenience this term is not used in this grammar.

[7] Ad hoc categorization however is possible in informal English, as in *I might [guinea pig] it for you* and even the dephrasal *"Sappho and her friend"'d.*

ad hoc categorization of forms, which produces usages that are not lexically well established, is considered stylistically acceptable: the appearance of 水 in (6) does not necessarily mean creation of a new verb *lexical entry*; indeed its usage as a verb meaning 'to swim' is in fact rather rare.[8]

In other clauses, the meaning of the root filling the main verb position of the clause can be regularly inferred, and should thus be considered verbalizing with zero marking (also known as conversion, in the terminology of Huddleston and Pullum (2002, pp. 1640-1641)). An important process of this kind is the tropative (31).

2.2.2.2 Arguments vs. adjuncts

Argument-like clausal dependents traditionally are classified into arguments and adjuncts. This distinction traditionally is made according to a heterogeneous array of criteria.

First, we have the parameter of whether the clausal dependent is lexically licensed by the main verb. This is often tested by criteria like obligatoriness or specification of e.g. the preposition of the clausal dependent, though these phenomena can often (but not always) be explained by semantic and pragmatic reasons and/or idiomization (McInnerney 2022, pp. 112-117, 125, 128, 212). We consider this to be the defining criterion of argumenthood.

Verb valency and its alternation possibilities in Classical Chinese verbs appear to be at least partially lexically specified (see e.g. § 2.2.2.3).

Second, there is the core/peripheral parameter: adjuncts are often (but not always) structurally more external or peripheral, attached to an already complete structure. This parameter is tested by diagnostics based on pro-form substitution of the verbal predicate, extraction constraints, and the like.

The instrument > P linear order in (5) can be understood as the P argument being structurally more internal. This however should not be taken to mean that the instrument is not lexically specified.

Third, there is the parameter whether an argument is *oblique*: an oblique agreement is often either a prepositional phrase or bears an inherent case, and often does not participate in verbal agreement. This parameter is in principle independent to the two parameters above in that it is possible to have a structurally peripheral clausal dependent with no preposition or inherent case marking; this is typically known as an applicative construction, and the clausal dependent in question is conventionally considered an argument, even though it can sometimes be attached freely.[9]

Classical Chinese has no agreement or inherent cases. Prepositional constituents (which are thus oblique) that are adjunct-like (5) or argument-like (11) are attested. Applicative constructions are abundant in Classical Chinese (§ 2.2.2.6).

[8] In traditional terms, 詞類活用 'word class flexibility', i.e. ad hoc categorization of a root, does not mean 一詞多義.

[9] In the typological literature, applicative is often analyzed as a peripheral argument being taken into the core (Dixon 2009, p. 169). The core/peripheral distinction here are not necessarily about the second parameter above, and should not be understood as meaning that the applicative argument is always structurally internal after the valency change.

2.2.2.3 Intransitive and transitive constructions

The constituent order of transitive clauses is almost always SVO (7; but see § 2.2.2.10). Intransitive clauses have a SV constituent order (8).

(7) [子]$_{\text{subject,A}}$ [[奚]$_{\text{reason}}$ 不 [為]$_{\text{verb,DO}}$ [政]$_{\text{P}}$]$_{\text{verbal predicate:VP}}$?
zǐ xī bù wéi zhèng
2 why NEG conduct government
'Why do you not take part in government?' (論語 • 為政)

(8) [君子]$_{\text{subject,S}}$ [不 [器]$_{\text{verb,BE}}$]$_{\text{verbal predicate:VP}}$
jūn-zǐ bù qì
lord-son NEG instrument
'A gentleman does not resemble an instrument/container (i.e. his skills are broad).' (論語 • 為政)

Consistent with cross-linguistically attested patterns, in Classical Chinese, a mono-transitive or intransitive clause can be about an intentionally initiated event (DO; 7, § 9.1.1), a state (BE; 8, § 9.1.2) or a change of the state (BECOME; 32, § 9.1.2). The DO type can further be divided into the transitive and intransitive classes. BE and BECOME clauses are intransitive by definition. Finally, BECOME and BE clauses are often inputs to the causative construction, resulting in CAUSE-BECOME/BE clauses.

The distinction between these constructions has long been noticed (Li 2004, pp. 92-96)[10] and has consequences for animacy and volition of the subject as well as the viability of argument structure alternation processes: S=A labile valency alternation naturally exists for DO clauses (§ 9.2.1.1), and S=P labile valency alternation naturally exists for CAUSE clauses (§ 9.2.2.1).

Whether a verb is able to head a DO, BE, BECOME or CAUSE clause is lexically specified and is an important parameter in subclassifying verbs: for instance, some causative verbs have no intransitive counterparts (§ 9.2.2.2) and eventually are reanalyzed as a DO verb (Mei 2015, pp. 269-271), and not all DO verbs have S=A labile valency (Li 2004, p. 93).

Some verbs also allow quantitative objects (29), and are "transitive" superficially with a structure different from that of (7).

2.2.2.4 Passive and causative constructions

Classical Chinese has a regular causative valency increasing construction producing CAUSE clauses (§ 10.1). It typically is not compatible with a verb frame that already has an A-like argument (Mei 2015, p. 364).

There are also productive passive-like valency decreasing constructions, which suppress the subject and promote an internal argument to the subject position (§ 10.2). These constructions are different from prototypical passive *voice* constructions in that they are much more selective about the original verb frame.

(9) is an example of causativization on top of passive-like valency decreasing. The verb 殺 is otherwise a DO verb, without the meaning of 'cause sb. to be killed'.

[10] More accurately, Li (2004) classifies subtypes of objects (受動賓語 vs. 致動賓語). His classification maps exactly to the DO vs. CAUSE-BECOME distinction. He also emphasizes what is essentially the difference between the S=A DO labile valency and the S=P BECOME labile valency.

(9) …[殺]$_{\text{main verb,CAUSE-“PASSIVE”}}$ 御叔…
shā Yùshū
kill NAME
‘…caused Yushu to be killed.’ (左傳・成公二年)

Classical Chinese seems to already have a prototype of what is later known as the disposal construction in later Sinitic languages (e.g. the 把 construction in Mandarin) (10). Other verb phrase structure alternations with explicit grammatical markers include the 為…所 construction and more.

(10) 盡 以 [其 寶器]$_{\text{T}}$ 賂獻 [于 周 厘王]$_{\text{G}}$
jìn yǐ qí bǎo-qì lù-xiàn yú Zhōu Xīwáng
all DISPOSAL 3.GEN precious-vessel bribe-offer to Zhou NAME
‘[They] offered all their precious vessels as gifts to King Xi of Zhou.’ (史記・晉世家)

2.2.2.5 Prepositional arguments

(11, 12) are examples of prepositional constituents, carrying a wide range of semantic roles. When an object is present, a rigid prepositional argument-after-object linear order can be observed after the main verb (11, 12). Prepositional phrases (PPs) can also be fronted (§ 2.2.2.10).

(11) 故 賜 [之]$_{\text{G}}$ [以 重 祭]$_{\text{T}}$
gù cì zhī yǐ zhòng jì
therefore give 3 PREP solemn offering
‘…and therefore gave him solemn venerations.’ (禮記・祭統)

(12) 季康子 問 政 [于 孔子]$_{\textbf{addressee}}$
Jì-Kāngzǐ wèn zhèng yú Kǒngzǐ
NAME ask politics PREP NAME
‘J. asks Confucius about politics.’ (論語・顏淵)

One issue surrounding licensing of prepositional arguments is whether some of them are licensed in so-called small clauses (Punske 2013). No convincing evidence for a small clause analysis is found in Classical Chinese; clauses with structures comparable to (12) cannot be analyzed in the small clause framework anyway (Mei 2015, pp. 417-419).

Regardless of the validity of the small clause analysis, prepositional arguments stay in situ in every type of valency alternation, except for the applicative. Many preposition arguments can be safely viewed as the least pivotal and the most internal argument (Mei 2015, p. 404), though some, like those in passive-like constructions, are likely structurally more external (Mei 2015, pp. 301-302).

Some prepositional arguments can be replaced by non-prepositional objects (秦 in 13).

(13) [入 [秦]$_{\text{target}}$]$_{\text{temporal}}$ [何 見]$_{\text{predicate}}$
rù Qín hé jiàn
enter Qin what see
‘On entering Qin, what did you see?’ (荀子・強國)

2.2.2.6 Applicative constructions

Besides post-verbal preposition omission, Classical Chinese has applicative constructions that turn a clause containing a prepositional argument into a double object clause (Mei 2015, Ch. 10).

(14) illustrates an applicative clause where the target of the question becomes an object (cf. 12). The prepositional argument appears to be promoted to a more pivotal position (§ 2.2.5.2), as demonstrated by the linear order.

(14) 上 問 [上林尉]$_{\text{applicative addressee}}$ 諸 禽獸 簿
shàng wèn Shàng-lín-wèi zhū qín-shòu bù
emperor ask upper-forest-administrator various bird-beast register
'The emperor asked the superintendent of Shanglin about the registers of the various animals.' (史記・張釋之馮唐列傳)

Other structurally comparable constructions include the tropative and the benefactive (Mei 2015, pp. 389-415). Mei (2015, p. 420-421) also notes that ditransitive clauses of giving and receiving in Classical Chinese may have a similar structure.

Not all verb frames have applicative counterparts. T-before-G applicative counterparts of (11) for instance are not attested. Double objects may still appear by removing the preposition (cf. 13; § 9.3.1).

2.2.2.7 Possession in argument structure

External possession is a heterogeneous phenomenon attested cross-linguistically (Deal 2017). Classical Chinese has a subject external possession construction, apparently structurally parallel to Japanese external possession (15).

(15) [子路]$_{\text{subject,possessor}}$ [[性]$_{\text{possessed}}$ [鄙]$_{\text{predicate}}$]$_{\text{predicate}}$
Zǐlù xìng bǐ
NAME nature uncultivated
'Zilu was uncultivated by nature.' (史記・仲尼弟子列傳)

(15) should not be analyzed as topicalization, because typically a pause cannot be inserted after 子路 (§ 2.2.5.1). It should also not be analyzed as ordinary possession, as the sequence *X* 性 meaning 'the nature of *X*' is rare if not impossible.

2.2.2.8 Complement clause constructions

Besides raising and control, Classical Chinese has at least two complement clause constructions, distinguishable by a structural complexity constraint: it is not possible for the prototypical complement clause in (16) to have sentence final particles; on the other hand, the direct quotation (see § 9.6) in (17) is sentential and has a sentence final particle.

(16) 臣 竊 以為 [不 便 於 君]$_{\textbf{finite complement clause}}$
chén qiè yǐwéi bù biàn yú jūn
minister humbly consider NEG advantageous to lord
'I humbly consider [this] disadvantageous to you, my lord.' (韓詩外傳・卷四)

(17) 子 曰：[“不 患 人 之 不 己 知，患 不 知 人
zǐ yuē bù huàn rén zhī bù jǐ zhī huàn bù zhī rén
master say NEG worry person GEN NEG self know worry NEG know person
也。”]**direct quotation**
yě
SFP
‘The Master said: “Do not worry that others do not know you; worry that you do not know others.”’ (論語・學而)

Whether the complement clause is a direct quotation is lexically determined. In Classical Chinese, direct quotation is taken by a handful of verbs, the most frequent one being 曰.

Other verbs of speech, which themselves do not license complement clauses, can introduce direct quotation by argument structure-level coordination (§ 2.2.9; 18).

(18) [子禽]$_{\text{subject: NP}}$ [[問]$_{\text{verb}}$ [于 子貢]$_{\text{target}}$]$_{\text{VP branch 1}}$
Zǐqín wèn yú Zǐgòng
NAME ask at NAME
[[曰：“…”]$_{\text{direct quotation}}$]$_{\text{VP branch 2}}$
yuē
say
‘Ziqin asked Zigong: …’ (論語・學而)

2.2.2.9 Raising and control

It is possible for an argument of the complement clause (usually the A argument; § 2.2.5.2) to appear as a clausal dependent in the main clause. Depending on when that argument is a part of the deep argument structure of the main clause or not, such an argument-sharing construction can be classified as control or raising. The underlying mechanisms of control constructions may again be heterogeneous, a topic beyond the scope of this work.

Raising-to-object, object control, and similar argument sharing constructions are collectively known as 兼語式 or the *pivot construction* (Pulleyblank 1995, p. 40). Uncontroversial pivot constructions are limited in number in early texts, because many of them can also be analyzed as argument structure-level coordination (Mei 2015, p. 376; § 2.2.9). Raising is rare in Classical Chinese (Mei 2015, p. 375-376). (19) is an example of the pivot construction in Classical Chinese, and is also an example of lexical causative (§ 10.1.2), analyzed as object raising in Mei (2015, p. 375) but may be alternatively analyzed as object control.

(19) [令]$_{\text{main verb}}$ [軍]$_{\text{P,A}}$ [勿 敢 犯]$_{\text{complement clause}}$
lìng jūn wù gǎn fàn
order army NEG dare violate
‘[He] ordered/made the army not to dare violate [the prohibition].’ (韓非子・十過)

Subject control can also be observed in Classical Chinese (20).

(20) [吾]$_{\text{subject,A,A}}$ 不 [欲]$_{\text{main verb}}$ [觀 之]$_{\text{complement clause}}$ 矣
wú bù yù guān zhī yǐ
1 NEG want watch 3 SFP
'I no longer wish to watch it.' (論語・八佾)

2.2.2.10 Verb phrase structure

All arguments (He 2005, pp. 3-5), include prepositional ones (60), can be omitted when they are identifiable from the context.

Certain verb phrase constituents can appear before the main verb. TAM adverbs almost appear before the main verb. Below we discuss fronting of arguments and adjuncts.

First, the object pronoun is often regularly moved before the verb in a negative clause, sometimes hopping from a control clause into the main clause (21), leading to a SOV order. An interrogative object pronoun can also be fronted (13). This may be analyzed as the pronoun clitic being attached to the negator or the main verb and has diachronic variation (Aldridge 2010).

(21) 余 [不 女$_i$ 忍 [殺 −$_i$]]$_{\text{predicate}}$
yú bù rǔ rěn shā
1 NEG 2 endure kill
'I can't bear to kill you.' (左傳・昭公元年)

Second, Classical Chinese appears to have a verb phrase internal focus position (Mei 2015, pp. 323-324; cf. Danckaert 2011).

In (22), the pre-verbal constituent 以其妹 seems to be a focalized argument. It cannot be the sentential focus, because the subject 季康子 seems to stay in-situ with no pause after it and therefore cannot be the sentential topic.

(22) 季康子 [[以 其 妹]$_{\text{VP-focus}_i}$ 妻 之 −$_i$]$_{\text{predicate:VP}}$
Jì-Kāngzǐ yǐ qí mèi qì zhī
NAME with 3.GEN younger.sister marry.to 3
'Ji Kangzi gave him his younger sister in marriage.' (左傳・哀公八年)

The fronted prepositional phrase may further undergo preposition-object inversion (23). This is much rarer than regular fronting of object pronouns, consistent with the analysis that fronting in (23) has information structure values and is due to focalization.

(23) [[[室]$_{\text{PP focus:NP}_j}$ 於 −$_j$]$_{\text{VP focus:PP}_i}$ 怒 −$_i$]$_{\text{VP}}$, 市 於
shì yú nù shì yú
home in be.angry market in
色
sè
show.expression
'Being angry at home and showing it in the marketplace' (左傳・昭公十九年)

Object fronting is also possible (之三子告 TODO)

2.2.3 Tense, aspect, and modality

Syntactic TAM categories (not just devices with semantic TAM meaning) are within a single clause and form a system with a hierarchy of auxiliaries and adverbs. Thus English *should have been doing* has grammaticalized TAM marking by auxiliaries, as the relative order of the auxiliaries cannot be shifted even with semantic motivations. Adverbs like *now* or *frankly* follow a similar order constraint and are a part of the TAM system (Cinque 1999).[11]

Because Classical Chinese have no native speakers, it is not possible to know if certain forms are impossible, and therefore there is always room to argue that an alleged auxiliary verb is in fact a content verb or certain unattested word orders are in fact possible. Further, some temporal clausal modifiers may be circumstantial adjuncts and not TAM adverbs (Cinque 1999, pp. 29-30; Mei 2015, pp. 430-431). The analyses below therefore are all tentative to a certain degree.

2.2.3.1 Speaker-oriented categories

The most wide-scope TAM-like categories are usually categories related to the speaker's evaluation of the situation being talked about; some of them can no longer be considered prototypically TAM categories. Categories falling under this type include evaluative adverbs (like *frankly* at the start of a sentence), evidentiality, and epistemic modality.

Classical Chinese has a category somewhere between evidentiality and epistemic modality, which may be known as speaker's commitment and has a subjective/objective distinction. Strong speaker's commitment to objectivity of the state of affair being reported can be marked by the adverb 固 (24). That the adverb 固 is always before the anteriority adverb 已 demonstrates its wide-scope, speaker-oriented nature.

(24) [固]$_{\text{objectivity}}$ [已]$_{\text{anteriority}}$ 明 矣
gù yǐ míng yǐ
certainly already clear SFP
'[This] certainly is already clear.' (春秋繁露・楚莊王)

On the other hand, Li (2004, p. 359) notes that clauses with certain sentence final particles, which he calls 說明句, are more subjective. The sentence initial particle 蓋 also marks an inferential attitude towards the situation being reported.

Some speaker-oriented adverbs, like 殆 in (26), may be interpreted as epistemic modality.

Classical Chinese has one category known as the *sentential aspect*, marked by a sentence final particle (25). Details about the category and how its cross-linguistic comparison can be found in § 14.1.3.

(25) 此 迫 矣$_{\text{sentential aspect}}$
cǐ pò yǐ
this urgent SFP
'This has become urgent.' (史記・項羽本紀)

[11] On the other hand, *be able to* is not a grammaticalized TAM marker (yet), as *many of us would have been able to have made this kind of bomb* is attested, where the PERFECT marking appears both before and after *able*, modifying both the fact many of us being able to do something and the thing being done (i.e. making this kind of bomb), strongly suggesting that we are looking at a biclausal construction.

2.2.3.2 Tense and modality

Tense adverbs are attested in Classical Chinese. (26) is an instance of adverbial marking of future.

(26) 夫子 [殆]$_{\text{epistemic}}$ [將]$_{\text{future}}$ 病 也
fū-zǐ dài jiāng bìng yě
master-HON probably FUT become.ill SFP
'The Master will probably become ill.' (禮記・檀弓上)

Mei (2015, p. 428) proposes that the sentence final particle 矣 introduces an explicit deictic time and therefore should be considered a (weak) tense marker. Pulleyblank (1995, p. 116), on the other hand, analyzes the particle as a "new information" sentential aspect.

Non-epistemic wide-scope modality categories about whether something has a chance to happen exist in Classical Chinese. (27) illustrates a possibility adverb. A realis/irrealis distinction can be found in conditional constructions (44), relevant to the licensing of TAM markers (Mei 2015, p. 81).

(27) 我 [固]$_{\text{objectivity}}$ [能]$_{\text{possibility}}$ 治 偏枯
wǒ gù néng zhì piān-kū
1 certainly can cure one.sided-paralysis
'I certainly can cure paralysis on one side of the body.' (呂氏春秋・似順論・別類)

2.2.3.3 Aspect

What is known as aspect includes anteriority i.e. secondary tense (English *have done*; Huddleston and Pullum 2002, p. 140), (Rizzi and Cinque 2016), (im)perfectivity i.e. point of view aspect or progressive aspect, the habitual aspect, and a lot more. These categories often have scope comparable to that of volition (Rizzi and Cinque 2016).

We have already seen an instance of anteriority (24). 方 in (28) is a progressive adverb, demonstrating existence of point of view aspect in Classical Chinese. The adverb also has a prospective meaning.

(28) 秦王 方 環 柱 走
Qín-wáng fāng huán zhù zǒu
Qin-king PROG circle pillar run
'The king of Qin was running around a pillar.' (史記・刺客列傳)

2.2.3.4 Lexical aspect

The lexical aspect is correlated to both argument structure and TAM categories (Meisterernst 2013; Meisterernst 2016). (29) illustrates how presence of a quantity argument turns the clause telic. (30) illustrates how the habitual aspect coerces an atelic reading. It has also been observed that stative verbs are not compatible with perfective adverbs (Meisterernst 2016).

(29) 行 [二 百 餘 里]$_{\text{quantity}}$
xíng èr bǎi yú lǐ
travel two hundred more li

'[They] travelled more than two hundred li.' (史記・衛將軍驃騎列傳)

(30) [常]$_{\text{habitual}}$ 封 三 錢 之 府
cháng fēng sān qián zhī fǔ
HAB seal three currency GEN treasury
'[He] regularly sealed the treasury of the three currencies.' (史記・越王句踐世家)

2.2.4 Negation

The structural positions and semantic interpretation of negators show considerable cross-linguistic variances (Moscati et al. 2012). Multiple negators are attested in Classical Chinese (6, 19, 35). Their distributions are obvious conditioned by morphosyntactic factors: the negator 非 for instance appears limited to negation of stative predicates (3, 6). We further note that Classical Chinese has double negation (TODO: 不能不).

2.2.5 Subjecthood

In § 2.2.1 and in § 2.2.2, we both mention the concept of *subject*. This concept of a single, unitary pivot in a clause should not be taken for granted.

Two types of *pivots* exist in a clause: one based on argument structure properties like obligatory argument omission in control constructions or binding of reflexive pronouns (§ 2.2.5.2), another based on clausal pivot properties like subject sharing in coordination and relativization (§ 2.2.5.3). Classical Chinese has a well-defined subject with both clausal and argument structure pivot properties and hence has no syntactic ergativity (Aldridge 2008).

Another problem is the distinction between subject and topic, both appearing at the beginning of a clause with unclear structural differences. Some scholars treat any seemingly pivotal constituent a subject (e.g. Li 2004, p. 41) and hence the topic is a "big subject" (Li 2004, p. 42). This opinion appears to suggest that Classical Chinese has only information structure and no argument structure in its syntax (Mei 2015, p. 122). We discuss the issue in § 2.2.5.1.

2.2.5.1 Subject vs. topic

We argue that identifying subject and topic (as in Li 2004, pp. 41-42) is not desirable for describing Classical Chinese.

First, in reading traditions of Classical Chinese (as indicated by traditional 句讀 punctuations), some pivotal constituents are followed by a pause, while others are obligatorily not followed by a pause (Mei 2015, pp. 121-122). No existing theory equating subject and topic naturally explains the distinction; on the other hand, availability of the pause is a cross-linguistic indication of topicalization.

Second, the constituent closest to the main verb often has an obligatorily semantic role.

In tropative clauses, the pivotal constituent closest to the main verb obligatorily refers to the person holding the opinion. (31), where the main verb is obviously a tropative use of the noun 客 'guest', can only be interpreted as 'Lord Mengchang considers me as a guest' (and never 'I consider Lord Mengchang as a guest'). This implies

that the argument structure of the clause imposes a hard constraint on what argument can be promoted to a pivotal constituent, which is unusual for topicalization.

(31) 孟嘗君 [客]$_{\text{main verb,tropative}}$ 我
Mèngcháng-jūn kè wǒ
NAME-lord guest 1
'Lord Mengchang treats me as a guest.' (戰國策・齊人有馮諼者)

When the pre-verbal pivotal constituent closest to the main verb is semantically not the agent, it is often *obligatorily* so according to the lexical properties of the verb (32), which again is unusual for topicalization (but not unusual if interpreted as valency alternation).

(32) [國 [定]$_{\text{main verb,BECOME}}$]$_{\text{clause branch 1}}$ 而 [天下 定]$_{\text{clause branch 2}}$
guó dìng ér tiān-xià dìng
state become.settled CONJ heaven-under become.settled
'The state becomes settled, and all under heaven becomes settled.' (荀子・王霸)
'*The state settles someone,...'

Therefore, pivotal constituents that are not topics and are structurally closer to the main verb than prototypical topics exist in Classical Chinese.

2.2.5.2 Pivot in argument structure

Cross-linguistically, the pivot at the level of argument structure, distinct from the clausal pivot, is often the A-like argument in a clause even in syntactically ergative languages. In Classical Chinese this can be demonstrated by examining the shared argument in control constructions (§ 2.2.2.9). The reflexive pronoun binding test is not applicable in Classical Chinese, as reflexive-like forms may have logophoric or emphatic distributions.

Further, the existence of an argument structure pivot can be confirmed by observing valency alternation constructions, as most of them involve either promotion of an argument to an A-like position or removal of the A-like argument.

It is sometimes also possible to define the second most pivotal argument. In applicative clauses, the promoted argument appears in the pre-verbal position, which can be interpreted as it being promoted to a structurally more pivotal position (§ 2.2.2.6, 14). In double object 'giving' clauses, it is usually the G argument being promoted to the subject position in valency decreasing (33), which again can be interpreted as the G argument being structurally more pivotal.

(33) [諸侯]$_{\text{G}}$, 賜 [弓矢]$_{\text{T}}$ 然後 征
zhū-hóu cì gōng-shǐ rán-hòu zhēng
various-lord grant bow-arrow then campaign
'The feudal lords are granted bows and arrows and only then undertake military campaigns.' (禮記・王制)

Hierarchies of pivotality or "externality" of arguments are not uncommon in world languages.[12]

[12]In Japhug, a language with direct-inverse alignment, it is possible to have a clause containing a

2.2.5.3 Clausal pivot

We then argue that the A-like, most pivotal argument in a verbal clause and the initial constituent in a nominal clause are clause pivots, around which everything else "revolves".

The clearest criterion that defines clausal pivotal subjecthood is probably coordination: when clauses are coordinated, one constituent seems to be shared by all of them, then this constituent is probably the clausal pivot, which is usually the argument structure pivot in nominative-accusative languages.

In Classical Chinese, clauses with 而 coordination overwhelmingly have S=A coordination. This is illustrated in (46), in which the two coordinated clauses share the argument structure pivot, meaning that the S and A arguments are in comparable clausal grammatical functions.

S=P coordination, on the other hand, can be attributed to either topicalization or passivization (e.g. § 2.2.9, 47). Therefore, Classical Chinese clauses appear to uniformly promote the most pivotal argument to the clausal pivot position, consistent with the nominative-accusative alignment.

2.2.6 Information packaging

2.2.6.1 Topicalization

Topicalization is ubiquitous in Classical Chinese. Indeed, the vanilla nominal predication construction (1) is much less frequent than the so-called 判斷句 'judgmental clause', in which the subject is topicalized, often with the sentence final particle 也 (34), or the topic particle 者 (35), or both.

(34) [城 北 徐-公]$_{\text{topic:NP}}$， [齊-國 之 美-麗 者]$_{\text{comment}}$
chéng běi Xú-gōng Qí-guó zhī měi-lì zhě
city north NAME-GONG Qi-country GEN beautiful-beautiful REL
也
yě
SFP
'Mr. Xu from the north of the city is a handsome man in the country Qi.' (戰國策・鄒忌脩八尺有餘)

(35) [兵]$_{\text{topic: NP}}$ 者， [不 祥 之 器]$_{\text{comment, predicate}}$
bīng zhě bù xiáng zhī qì
weapon TOPIC NEG fortunate GEN instrument
'Weapons are not auspicious.' (老子)

Topicalization, and the topic marker 者, is not restricted to the nominal predication construction (36).

causer, an instrument, an agent and a patient. The personal indexation marker appears to be decided on the basis of the hierarchy *causer*→*instrument*→*agent*→*patient*, by first taking the two most internal arguments and deciding which is more salient on the empathy hierarchy, then comparing the result with the third most internal argument and deciding which is more salient, and finally comparing the result of the last step with the most external argument; hence 1→3→2→3 is morphologically equivalent to 1→2 (Jacques 2021, p. 310, p. 584, (116), p. 848, (67)). Similar mechanisms are attested cross-linguistically Wiltschko (2014, § 7.4.1).

(36) 此 二 人 者， 實 弒 寡君
cǐ èr rén zhě shí shì guǎ-jūn
this two person TOPIC indeed murder.ruler our-lord
'These two men indeed murdered our lord.' (左傳・隱公四年)

What is topicalized is of course not restricted to the subject. This fact is a piece of evidence supporting the distinction between subject and topic in Classical Chinese. In (37), the comment clearly has a nominal predicate. What is promoted to the topic position however is not the subject of the nominal predicate construction, but the predicate. The subject is likely the *focus* and not the topic (Mei 2015, p. 138). Topicalization of other clausal constituents is also possible (e.g. § 2.2.9, 47).

(37) [仁 之 實]$_{\text{topic:NP}_i}$， [[事 親]$_{\text{subject}}$ [是]$_{\text{predicate}_i}$]$_{\text{comment}}$ 也
rén zhī shí shì qīn shì yě
benevolence GEN substance serve parent this SFP
'As for the substance of benevolence, serving your parents is it.' (孟子・離婁上)

Some topic-like constituents, mainly prepositional phrases, are analyzed as *frames* in Mei (2015, pp. 129,323).

2.2.6.2 Focalization

Topicalization is marked by fronting, but what is fronted is not necessarily a topic. In (38), for instance, the verbal predicate is fronted, which likely is not topical. Note that the sentence final particle is fronted as well.

(38) [[甚]$_{\text{VP}_i}$ 矣]$_{\text{focus}}$， [[汝 之 不 惠]$_{\text{subject:clause (§ 2.2.7.2)}}$ ⸺$_{\text{predicate}_i}$]
shèn yǐ rǔ zhī bù huì
extreme SFP 2 GEN NEG smart
'You are so stupid! (lit. So extreme is your being unintelligent!)' (列子・湯問)

> **Box 2.1: A complete overview of the left periphery**
>
> See https://referenceworks.brill.com/display/entries/ECLO/COM-000248.xml

2.2.7 Speech acts and clause types

This section discusses high-level grammatical categories which, in the case of sentential clauses, are often related to the speech act of the clause. These categories are traditionally known as *clause types* or 句類, as opposed to 句型 i.e. the internal structure of clauses (Li 2004, p. 58).

2.2.7.1 Sentential clauses: interrogative, exclamative, and imperative

The illocutionary force of sentential clauses are marked by sentence final particles. In fact, these particles almost only appear in sentential clauses.

Interrogative clauses may completely lack formal marking except interrogative pro-forms (13). The interrogative illocutionary force is optionally marked by 乎 and

other particles (39). The exclamative speech act is similarly marked by sentence final particles (40).

(39) [其 能 久]$_{\text{nucleus}}$ 乎$_{\text{illocutionary force}}$?
qí néng jiǔ hū
MODAL can last.long Q
'Can [this] last long?' (史記・周本紀)

(40) [吾 死]$_{\text{nucleus}}$ 矣$_{\text{sentential aspect}}$ 夫$_{\text{illocutionary force}}$!
wú sǐ yǐ fú
1 die SFP EXCL
'I am going to die!' (孟子・離婁下)

We note that in (40), the sentential aspect marker 矣 (§ 2.2.3.1) appears before the exclamative 夫, which means that the two systems of particles can coexist. The reverse order * 夫矣 is not attested.

2.2.7.2 Nonfinite constructions

Classical Chinese has one clausal construction that is structurally clearly different from sentential clauses, and may be known as nonfinite.

It is possible to insert 之 between the subject and the predicate of a clause. The result appear as the object (41) or as a conditional or temporal clause (42; § 2.2.8). Sentence final particles are never attested in this type of clauses.

This process is sometimes known as nominalization (Pulleyblank 1995, p. 64). This term cannot be taken literally in all situations: in (42), the "nominalized" clause does not admit adjectival modification and takes an object, both unusual for a noun phrase. The example therefore has a structure comparable to the English gerundive non-finite clause *his playing the national anthem* and is structurally different from the noun phrase.

(41) 王 如 知 此，則 無 望 [民 之 多 于 鄰
wáng rú zhī cǐ zé wú wàng mín zhī duō yú lín
king if know this then NEG hope people GEN more than neighbor
國]$_{\text{object:}zhī\text{-construction}}$ 也
guó yě
country SFP
'If Your Majesty knows this, then don't expect your people to be more plentiful than your neighboring countries' people.' (孟子・梁惠王上)

(42) [父母 之 愛 子]$_{\text{condition:}zhī\text{-construction}}$，則 為 之 計
fù-mǔ zhī ài zǐ zé wèi zhī jì
father-mother GEN love child then for 3 plan
深遠
shēn-yuǎn
deep-far
'When parents love their children, they plan far ahead for them.' (戰國策・趙太后新用事)

2.2.8 Subordination

The term *subordination* sometimes means all kinds of clause embedding. In this section we primarily focus on bipartite clauses with the structure and meaning of 'if ...then ...' or 'when ...'. Complement clauses are discussed in § 2.2.2.8 and § 2.2.2.9. Relative clauses are discussed in § 2.3.4.

An overview of subordination constructions in Classical Chinese can be found in Mei (2015, Ch. 3). In all Classical Chinese conditional constructions, the condition usually appears before the consequence (43, 44). The consequence can be marked by a particle (e.g. 則; 43).

The condition clause can also be marked but zero marking is possible. Classical Chinese distinguishes between realis and irrealis conditional constructions (§ 2.2.3.2): the former are marked by e.g. 既 (43), while the latter are marked by e.g. 若 (44).

(43) [既 來 之]$_{\text{condition}}$，[則 安 之]$_{\text{consequence}}$
jì lái zhī zé ān zhī
already cause.to.come 3 then settle 3
'Once you have brought them here, settle them.' (論語・季氏)

(44) [若 已 食] 則 退
ruò yǐ shí zé tuì
if already eat then withdraw
'If [the parent] has already eaten, then withdraw.' (禮記・內則)

An interesting phenomenon is that the subordinated clause can be a *zhī*-construction clause (42). This is cross-linguistically unsurprising (as in Japanese or Turkish). Note that the marker 若 can be attached to the gerundive condition clause as well (Mei 2015, p. 98). Other interesting features of the conditional clause include that the condition marker can sometimes appear after the subject.

Besides clausal-level subordination, Classical Chinese also likely has verb phrase-level subordination. In (45), the first verb phrase (whose agent is *obligatorily* the subject of the clause) describes the manner of the second, and thus should be understood as subordination.

(45) 吾 嘗 [跂]$_{\text{modifier verb phrase}}$ 而 [望]$_{\text{main verb phrase}}$ 矣
wú cháng qì ér wàng yǐ
1 once stand.on.tiptoe CONJ gaze SFP
'I once stood on tiptoe and gazed into the distance.' (荀子・勸學)

2.2.9 Coordination

Coordination happens at the level of clause, (extended) verb phrase or argument structure. The distinction between the last two is that coordination of two argument structures results in a *single* situation of affairs, usually with shared TAM marking, while coordination of two verb phrases is more comparable to clausal coordination (Mei 2015, pp. 192-201). The two also have differences in relativization (Mei 2015, p. 207).

In Classical Chinese, certain examples are clearly argument structure-level coordination (18). Others are clearly not, including those in which the two verb phrase branches are separated by a pause (Mei 2015, p. 195). However, due to the absence of obligatory TAM marking (and unavailability of acceptability judgement data), marginal

cases in the classification above exist in Classical Chinese. See (TODO) for a detailed analysis.

Coordination is primarily marked by 而 (46), which is not used in noun phrase coordination. Note that the functions of 而 are not restricted to conjunction (Mei 2015, p. 183; 45).

(46)

項王	[按 劍]$_{\text{VP branch 1}}$	而	[跽]$_{\text{VP branch 2}}$
Xiàng-wáng	àn jiàn	ér	jì
NAME-king	hold sword	CONJ	rise.on.knees

'King Xiang held his sword and raised himself on his knees.' (史記・項羽本紀)

2.2.9.1 Topic chains as syntactic coordination

> **Box 2.2**
>
> Where to put this section? And what's the actual relation between topicalization and coordination?

An interesting question is the interaction between topicalization and coordination. Mei (2015, p. 217) contends that "topic chains", i.e. several clauses with a shared topic (Mei 2015, Ch. 4 § 3.3), are discourse structures and not syntactic structures. Therefore topicalization happens first, and coordination happens then: after that no further topicalization is possible. He further argues that clauses in a topic chain cannot be linked together by 而. (47) however seems to be a counterexample. This example clearly contains two coordinated clauses. In the first clause 取之于藍, 之, appearing after the verb, can only be a pronoun, and the only sensible reading of the clause is that 之 (the object) is coreferential with 青 at the initial of the sentence, and 取之于藍 then means '(people) extract it (i.e. indigo dye) from *Indigofera*.' Therefore, 青 at the initial of the sentence is the object of the first clause and the subject of the second clause, meaning it cannot be the shared subject. This, together with the traditional pause after the first 青, means the first 青 likely is a topic, which means here topicalization happens *after* coordination.

(47)

[青]$_{\text{topic: NP}_i}$,	$-_i$ 取	[之]$_{\text{object: Pronoun}_i}$	于	藍	而	$-_i$ 青	于
qīng	qǔ	zhī	yú	lán	ér	qīng	yú
indigo.dye	pick	it	from	*Indigofera*	CONJ	blue	than
藍							
lán							
Indigofera							

'Indigo dye, people extract it from *Indigofera*, but it's bluer than *Indigofera*.' (荀子・勸學)

2.3 The noun phrase

The Classical Chinese NP can be roughly divided into the determiner region and the "core" region, the latter known in Huddleston and Pullum (2002) as the *nominal*.[13] The

[13] In this note, when the term *nominal* is used as a noun, it refers to the determined region in NPs, while when it is used as an adjective, it refers to the status of being the head of a NP.

latter is the head noun plus possible arguments and modifiers, and the former can be left empty or filled by a demonstrative or a "possessor", the role of the latter being not confined to a semantic possessor (Pulleyblank 1995, p. 61; § 2.3.5).

2.3.1 Structure of the noun

2.3.1.1 Wordhood

Like the case of the verb, it is hard to demarcate the boundary of the Classical Chinese noun with absolute certainty based solely on textual evidence.

The golden standard of morphological templates does not apply. As for the criterion of lexical specification of grammatical behaviors, some monomorphemic forms, like 客, typically head noun phrases and when they do head clauses, they have meanings regularly inferrable from their nominal uses (as in the tropative example 31). They thus should be regarded as grammatical words, due to lexical specification of their grammatical behaviors (§ 2.1.2).

Wordhood of composite forms like (49) however are still indeterminate in this lens. Their established and non-inferrable meanings can be explained by idiomization without wordhood. Their obligatorily heading a noun phrase may be simply explained by the *head* being in a noun lexical entry.

To use the lexical specification criterion of wordhood we have to demonstrate e.g. unpredictable lexical specification of the argument structure (cf. the discussion in § 2.2.2). This is not easy (for monomorphemic or composite nominal forms), because there is no obligatory possession in Classical Chinese.[14]

Lexical specification of the noun phrase argument structure, if it exists, produces very subtle effects.[15] The two noun phrases in (48) obligatorily mean 'a minister serving *X*' and never, say, 'a minister who happens to reside in the sovereign of *X*'. This can be taken to indicate that the possessor is selected by the head noun. Another possible piece of evidence for lexical specification of noun phrase argument structure is (51).

(48) 非 [寡人 之 臣]， [社稷 之 臣] 也
fēi guǎ-rén zhī chén shè-jì zhī chén yě
NEG humble-person GEN minister altar.of.soil-altar.of.grain GEN minister SFP
'[He] is not my minister, but a minister serving the state.' (禮記・檀弓下)

Complexity constraints of complex nominal forms is sometimes taken as evidence for wordhood (see also § 3.2). In (49), the modifier usually cannot be modified by adjectives. Structural complexity constraints nonetheless appear in structures that are clearly not words (cf. absence of sentence final particles in complement clauses; § 2.2.2.8).

(49) [舟]$_{\text{modifier}}$ 人
zhōu-rén
boat-person

[14] Pre-Classical Old Chinese might have traits of inalienable possession (Baxter and Sagart 2014, p. 57) (which is speculative; Hill 2019, p. 141) but this did not automatically transmit to the Classical period.

[15] Diagnostic for argument selection is easier for English nouns (Huddleston and Pullum 2002, p. 440), as noun phrase arguments in Classical Chinese typically lack any marking of their grammatical functions.

'boatman' (韓詩外傳・卷四)

2.3.1.2 Nominalization

Multiple constructions have been known as nominalization. The 之-insertion (§ 2.2.7.2) is sometimes considered nominalization. Interestingly, relativization in Classical Chinese may indeed be nominalization (§ 2.3.4).

Classical Chinese also has nominalization constructions targeting structurally simpler forms. In (50), 堅 refers to 'what are hard (i.e. defensive gears)' and 銳 refers to 'offensive weapons', semantically comparable to relative clauses and analyzable as zero derivation (cf. discussion on derivation in § 2.2.2.1).

(50) 被 堅 執 銳
pī jiān zhí ruì
wear hard hold sharp
'wear armour and wield weapons' (史記・陳涉世家)

2.3.2 Argument structure

Classical Chinese noun phrases have rather limited positions for arguments: there is no prepositional argument comparable to English *normalization [of this behavior]*. Typically there is only one dependent that can be reasonably considered an argument. It often appears in the possessor position (48), but not always (51). In fact, the possessive 之 is never inserted into (51). That the noun phrase dependent should not be a possessor appears to be a lexically specified behavior of 獄掾 and indicates existence of core i.e. lexically specified arguments in Classical Chinese noun phrases.

(51) [櫟陽]$_{\text{location}}$ 獄掾
Yuèyáng yù-yuàn
Yueyang prison-clerk
'a prison clerk in Yueyang' (史記・項羽本紀)

We also note the meaning difference between (48) and (52), which suggests that there exists more than one argument-like position in the noun phrase.

(52) 社稷 臣
shè-jì chén
altar.of.soil-altar.of.grain minister
'a minister important for the state (lit. a state-level minister)' (史記・袁盎鼂錯列傳)

2.3.3 Adjectives

Cross-linguistically, what are known as adjectives can be divided into inner direct modifications and outer (reduced) relative clauses (Cinque 2010, p. 23-25). A third class of adjectives, which are structually argument-like (Moreno 2018; Huddleston and Pullum 2002, p. 439), are absent in Classical Chinese.

Existence of direct modification adjectives in Classical Chinese is shown by their relatively fixed order (53; cf. English *a [big] [gray] bear*).

(53) [大]$_{\text{size modifier}}$ [白]$_{\text{color modifier}}$ 狗
dà bái gǒu
big white dog
'a big white dog' (漢書・五行志)

It should be noted that examples involving multiple pre-nominal attributives are scarce. Complex attributives are much more idiomatically constructed using the relative clause construction (55).

2.3.4 Relative clause constructions

Classical Chinese has a fused relative clause construction marked by 者. A fused relative clause can take a determiner (54), and hence may be analyzed as a nominal (Aldridge 2009).

(54) [晉 之 [從政 者]$_{\text{clause nominalization}}$]$_{\text{NP}}$
Jìn zhī cóng-zhèng zhě
Jin GEN engage.in-government REL
'those who take part in government in Jin' (左傳・宣公十二年)

(55) is an instance of a long, complex relative clause in Classical Chinese. This illustrates that we are indeed dealing with a relative clause construction and not a low-level nominalization construction.

(55) 若 [– 至 力 農 畜， – 工
ruò zhì lì nóng xù gōng
as.for utmost exert.oneself farm rear.livestock work.as.artisan
虞 商 賈， – 為 權利 以 成
yú shāng gǔ wéi quán-lì yǐ chéng
manage.natural.resources trade trade pursue power-profit to achieve
富， [– 大 者 傾 郡]， [– 中 者 傾 縣]， [–
fù dà zhě qīng jùn zhōng zhě qīng xiàn
wealth great REL surpass commandery middle REL surpass county
下 者 傾 鄉里]] 者， 不 可 勝 數
xià zhě qīng xiāng-lǐ zhě bù kě shèng shǔ
low REL surpass village-neighbourhood REL NEG can exhaust count
'As for those who devote their efforts to farming and livestock, crafts, natural resources or trade, or pursue power and profit to become rich—the greatest surpassing commanderies, the middling counties, and the least villages and neighbourhoods—they cannot all be counted.' (史記・貨殖列傳)

When the 者 relative clause is used as a non-fused relative clause, it is always postpositive. The structural analysis of the construction is discussed in TODO.

Classical Chinese has pre-nominal relative clauses too (56).

(56) 得 志 之 君子
dé zhì zhī jūn-zǐ
attain aspiration REL lord-son
'a gentleman who has attained his aspirations' (春秋繁露・竹林)

2.3.5 The determiner system

A demonstrative-before-numeral order can be observed in the determiner system (57).

(57) 此 三 姦 者
cǐ sān jiān zhě
this three wicked REL
'these three wicked men' (荀子・非十二子)

The "possessor" (which does not necessarily have possessive meaning), when present, is separated by the particle 之 from the nominal region (58). It appears that the possessor, the demonstrative, and the pre-nominal relative clause (56) are in competition (while they coexist with the numeral).

(58) 君 之 三 臣
jūn zhī sān chén
lord GEN three minister
'your three ministers' (禮記・檀弓下)

2.3.6 Prepositions

What fall under the name of preposition include those that are roughly non-morphological grammatical relation markers, those from directional constructions, and transitive adverbs.

Classical Chinese does not have modern Sinitic directional constructions of the shape of 在……上, and appears to have no transitive adverbs. Prepositions are restricted to markers of oblique clausal dependents.

Prepositions in Classical Chinese often have diachronic verbal etymology, but at least some of them have already fully grammaticalized. The meaning of 於 in (59) cannot be reasonably analyzed as verbal.

(59) 且 貳 於 楚 也
qiě èr yú Chǔ yě
and have.divided.allegiance to Chu SFP
'[Zheng] also owed allegiance to Chu.' (左傳・僖公三十年)

It is possible to omit the complement of a preposition.

(60) 愚 不 足 與 謀 智
yú bù zú yǔ móu zhì
foolish NEG sufficient with discuss wisdom
'The foolish are not worth discussing wisdom with.' (荀子・正論)

2.4 Parts of speech division

The term *word class* (or *parts of speech*) means two things: (a) classification of full-fledged grammatical words, meaning that a noun is a concrete grammatical word heading a noun phrase, etc., and (b) *lexical* form classes, each of which contains lexical entries with comparable lexically specified grammatical properties (see § 2.1.2),

meaning that a noun is a lexical entry with unpredictable grammatical properties and strongly prefer serving as a noun in the first sense.

Both definitions have cross-linguistic variances. Properties of head grammatical words of constructions vary cross-linguistically. Further, the mapping from abstract lexical entries to concrete grammatical words is not always straightforward, and thus the landscape of lexical form classes may differ considerably from the landscape of classes of grammatical words.[16]

Frequent zero derivation and ad hoc categorization of words and the lack of inflectional endings in Classical Chinese have led some to argue that the language lacks lexical parts of speech division. This statement cannot be taken literally: in § 2.2.2.1 and § 2.3.1.1 we have demonstrated that content morphemes have clear preferences for nominal or verbal environments, and thus at least a noun class and a verb class as lexical form classes can be established.

An adjective class can be recognized (often as a subclass of stative verbs) based on the criterion of tropative usage.

An adverb class can also be recognized, which contains forms appearing in clausal modifier positions (§ 2.2.3).

In principle, function words can be introduced when introducing constructions licensing them. In practice, for convenience, and considering grammaticalization, divergences in analyses, and possible real-world divergence of their usages among speakers, they are often assigned part of speech tags and introduced in a surface form-to-function manner. An overview of these words (虛詞 in the traditional terminology) can be found in Li (2004, Ch. 4).

[16] An example is root classes in Maya languages (Siddiqi et al. 2020, p. 372).

Chapter 3

Nouns

We cover forms that appear as the head of noun phrases in this chapter, which may lexically belong to the noun form class, or be regularly derived from other forms, or be a result of ad hoc categorization of other forms.

3.1 Simplex nouns

3.1.1 Nominal roots

Most simplex nouns in Classical Chinese are monosyllabic. Some Classical Chinese nouns are still simplex nouns in contemporary Sinitic languages, like 人. Others have developed into bound roots. The the form 王 in (1) is on its own a noun phrase in Classical Chinese, while its modern day reflex in Mandarin needs to be complemented by a precedent form (e.g. 国王).

(1) 王 如 知 此
wáng rú zhī cǐ
king if know this
'If Your Majesty knows this, ...'

Although 王 is sometimes used as a verb, this should not be regarded as it having a lexically specified verbal function, because this use is better analyzed as denominalization (§ 8.2.2).

Disyllabic nouns without obvious internal structures also exist, traditionally known as 联绵字. Some appear to be reduplications (§ 3.3.1) and are thus not truly simplex. Certain disyllabic place names are likely Kra-Dai loans (§ 3.4.2.1).

Some disyllabic nouns have no settled origins. One instance is 蟋蟀, whose reconstructed Old Chinese pronunciation is, according to Baxter and Sagart (2014, p. 290) *srit-srut*. This form can be explained as onomatopeoia. Similar forms can be found in neighboring languages and the possibility of borrowing (from Old Chinese to neighboring languages, or vice versa, or multiple borrowing) cannot be ruled out.

3.1.2 Roots with multiple functions

3.2 Compounding

Compounding is word formation by combining content forms. The boundary between compounding and derivation, cross-linguistically, is not always clear, as grammatical formatives may also appear in what may be known as compounding (like the linking elements between the constituents of the word).

Compounding should also be compared with prototypical phrasal syntax. Huddleston and Pullum (2002, pp. 448-451) list criteria for a form to be considered a compound, as opposed to a noun phrase. Some criteria they provide have at least marginal counterexamples,[1] and others appear to only demonstrate that the "compounding" processes are structually more restricted than prototypical phrasal syntax, a phenomenon also observed in obvious phrasal forms.

The distinction between compounding and noun phrase structural building is therefore ideally defined only *after* grammatical wordhood is defined based on criteria of (syntactically and phonologically unmotivated) domain boundaries and/or grammatical lexical specification phenomena: a compounding process regularly produces forms that are morphologically words.

Therefore, following the discussion in § 2.1.3, we do not offer completely uncontroversial justification of wordhood of each process deemed as compounding in this section. Instead, based on the principle of inference of best explanation, we rely on multiple indirect criteria like complexity constraints or established meanings, which, cross-linguistically, have strong correlation with grammatical wordhood, but are by no means indicative of wordhood when used individually.

This section is largely based on Unger (2019, § 0.1.2), with substantial reworking.

3.2.1 Subordinative compounds

3.2.1.1 Attributive subordinative compounding

Subordinative compounds in Classical Chinese rigidly have a modifier-head order. The modifier in attributive subordinative compounds is adjective-like in its meaning. An attributive subordinative compound therefore has an identical linear form to that of noun phrases (2-4).

(2) 大夫
dà-fū
great-man
'high-ranking official (lit. great man)'

(3) 小人
xiǎo-rén
small-person
'commoner; petty person (lit. small person)'

(4) 太子
tài-zǐ
great-son

[1] Consider *tear- and rain-drop*, *sun rise and set*, attested online.

‘crown prince (lit. great son)’

Despite the linear expressions, these forms are likely grammatical words.

First, all the examples above have idiomatic, non-transparent meanings, which is unusual if they are phrases.

Second, the wordhood of these forms is sometimes suggested by how they are modified. In (5), the adjective 上 typically describes the abstract property of having an upper status (as in 上赏, 上策, etc.); it clearly does not modify the head 夫, and the form is obligatorily interpreted to mean ‘a high-ranking official with an upper status’, never ‘a great man who is of an upper status in a vague way’. This suggests that 上 and 大 in the form are not in the same modification sequence: instead, 大夫, as a whole, undergoes adjectival modification and is capable of being modified by 上. Thus the form 大夫 is *structually* different from, say, 大狗 ‘big dog’. This does not necessarily means the former is a grammatical word but cross-linguistically often correlates to grammatical wordhood (cf. English *white blackbird*). Thus 大夫 is likely a grammatical word.[2]

(5) 上大夫
shàng-dà-fū
upper-great-man
‘senior official (lit. upper great man)’

3.2.1.2 Relational subordinative compounding

In relational subordinative compounding, the modifier has a substantive meaning, but is not to be interpreted as referential: it instead restricts the meaning of the head with an “aboutness” relation with the head. Therefore, the example (6) can be literally taken to mean ‘a person who has something to do with boats’ (but not ‘a person who has something to do with an aforementioned boat’), and hence a boatman.

(6) 舟人
zhōu-rén
boat-person
‘boatman’

Cross-linguistically, the difference between relational subordinative compounding and attributive subordinative compounding cannot be simply attributed to the lexical form class of the modifier (consider English *sickbed*, which is a relational subordinative compounding and means ‘a bed having something to do with sickness = a hospital bed’). We see the same phenomenon in Classical Chinese. In (7), 凶 does not mean ‘inauspicious’, but ‘related to inauspicious things = related to death’.

(7) 凶服
xiōng-fú
inauspicious-clothing
‘mourning clothes’

[2]We note that a similar analysis appears to suggest that in English, some forms that are phonologically multi-word may need to be analyzed as single grammatical words: consider e.g. *top Great Man* here.

The relational subordinative forms typically have idiomatic and non-combinatorial meanings and thus 舟人 almost never means 'passengers', although this interpretation is combinatorially viable.

3.2.1.3 Argumental subordinative compounding

Compounds like (6) can be distinguished from those resembling (8). The difference is that in (8), the modifier appears to be selected by the head, and in this way is like an argument. This observation is further corroborated by the fact that in quite a few compound-like nominal forms containing 子 as the head, the modifier has an ancestral interpretation (9).

(8) 君子
jūn-zǐ
lord-son
'gentleman (lit. son of a lord)'

(9) a. 皇子
huáng-zǐ
emperor-son
'son of the emperor'

b. 天子
tiān-zǐ
heaven-son
'emperor (lit. Son of Heaven)'

These argumental subordinative forms, like other compounds, frequently have idiomatic, non-combinatorial meanings. Thus a Confucian 君子 (8) does not need to literally be a son of an aristocrat, and 舟人 in (6) rarely means 'a person riding the boat'.

3.2.2 Coordinative compounds

3.2.2.1 "True" coordination

In what may be known as prototypical coordination in Classical Chinese coordinative compounding, two forms are placed together to mean the union of their substantive interpretations. The meanings of (10, 11) are straightforward.

Coordinative compounds formed in this way are always collective, semantically interpreted to refer to the things or people *as a whole*. Thus forms like * 一父母 'one parent' are not attested in Classical texts.

(10) 父母
fù-mǔ
father-mother
'parents'

(11) 禽兽
qín-shòu
bird-beast
'birds and beasts'

This construction is simply known as coordination in Pulleyblank (1995, p. 61), without any discussion on its wordhood. Some coordinated forms, like (10), have rather transparent meanings. Others have idiomatic meanings rather hard to infer (12). The example (12) is especially interesting, because a search of 社, 稷 and 社稷 in the pre-Qin and Han sector of the Chinese Text Project (Table 3.1) shows in more than half of occurences of 社 and 稷, the two forms appear as constituents of 社稷. This is unusual for a prototypical phrasal idiom, and thus we consider it likely that 社稷 is likely a grammatical word, which further suggests that Classical Chinese has a word-forming coordination construction.

(12) 社稷
shè-jì
altar.of.soil-altar.of.grain
'the state (lit. altars of soil and grain)'

Table 3.1: Frequencies of 社, 稷 and 社稷

Form	Occurrence
社	1685
稷	1670
社稷	993

3.2.2.2 Generic-specific compounding

In a generic-specific compound, both constituents are interpreted substantively, and the meaning of the second constituent is a subset of the meaning of the first constituent. The compound is then interpreted identically to how the second constituent is interpreted. In this way, (13) means 'small birds' and not birds in general (and certainly the coverage of the word does not include eagle). This semantic feature distinguishes generic-specific compounding and true coordination.

(13) 鸟雀
niǎo-què
bird-sparrow
'small birds'

The diachronic shift of the meaning of certain roots sometimes means that certain forms can be understood to be a prototypical coordination compound and as a generic-specific compound at the same time.

Some occurences of the form 禽獸 are also better regarded as having a generic-specific structure (14). The root 禽 has an early, broad reading meaning '(captured) animals' (15). Thus in (16), the form 禽鹿 can only be understood as a generic-specific compound meaning 'deers'.

(14) 禽獸 之 皮
qín-shòu zhī pí
animal-beast GEN skin
'the skins of beasts'

(15) 獸 亦 可 曰 禽
shòu yì kě yuē qín
beast also can call animal
'Beasts can also be called *qín*.' (爾雅注疏・卷十・釋鳥第十七)

(16) 或 逐 禽鹿 而 給 食
huò zhú qín-lù ér jǐ shí
some pursue animal-deer CONJ supply food
'Some hunted deer to obtain food.' (吳越春秋・越王無余外傳)

On the other hand, 禽獸 can be understood as a prototypical coordination (11), given the later usage of 禽 as a specific term covering birds (17).

(17) 二 足 而 羽 謂 之 禽
èr zú ér yǔ wèi zhī qín
two foot CONJ feather call 3 bird
'Those with two feet and feathers are called birds.' (尔雅)

3.2.2.3 Tautological coordinative compounding

In another Classical Chinese coordinative compounding process, the two constituents have comparable substantive meanings, and the resulting compound means what the individual roots mean.

(18) is an example of this compounding process. Because the two parts of the words have identical meanings, we expect them to be at least sometimes swappable. The form 友朋 is attested (19),

The uniform semantic transparency of these forms however calls their wordhood status into question. Unger (2019, p. 11) for instance cautions that these forms should probably not be known as compounds (cf. English *rules and regulations*).

(18) 朋友
péng-yǒu
friend-friend
'friends'

(19) 知交 友朋 之 相 助 也 宜
zhī-jiāo yǒu-péng zhī xiāng zhù yě yí
know-associate friend-friend GEN mutually help SFP fitting
'It is fitting for close acquaintances and friends to help one another.' (韓非子・解老)

When the two constituents cannot be swapped, whether the form is truly a tautological coordinative compound is questioanble. 宾客 is attested (20), but * 客宾 is not. This appears consistent with the observation that the constituent denoting things or people that are perceivedly more important appears in the initial position, which appears to indicate that the form should be analyzed as a prototypical coordinative compound (§ 3.2.2.1).

(20) 宾客
bīn-kè
guest-guest
'guests'

3.2.2.4 Property compounds

Two forms can be coordinated to denotea a scale with its two endpoints corresponding to the two forms interpreted as *properties*. Thus 大小 in (21) means 'size', and so on.

(21) 大小、 高卑、 厚薄 之 度
dà-xiǎo gāo-bēi hòu-bó zhī dù
large-small high-low thick-thin GEN measure
'measurements of size, height, and thickness' (禮記・月令)

3.2.2.5 Order of constituents

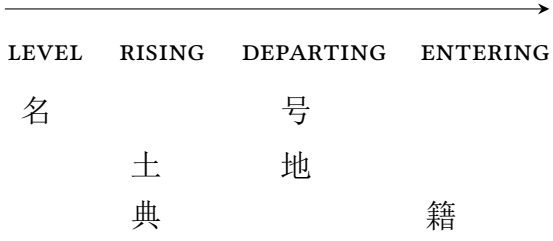
In all coordinative constructions above, the order of the two constituents is not completely aribtrary: for instance, contrary to (10) and (11), * 母父 or * 兽禽 are both not attested in Classical texts.

This is partially explained by the convention to put the constituent traditionally considered more prestigeous and important first, which perhaps is the reason of the absence of * 母父, as under the patriarchal clan system, the father was considered more important.

The case of 禽兽 vs. * 兽禽 on the other hand may be due to historic evolution of the meaning of 禽: 獸禽 has an early, broad reading meaning '(captured) animials' (14), and when the narrow reading of 禽, 'bird', became prevalent and hence a prototypical coordination * 獸禽 would have been possible, the form 禽兽 had already become high-frequency enough to be the default choice when the two roots were to be combined.

However, the order constraints seen in many other forms cannot be explained by the above two reasons. Zhang (1996) quantitatively studies coordinative forms in Classical texts, and demonstrates that the main factor in deciding the order of constituents appears to be the tone of the roots (or, more precisely, the Old Chinese phonological features of these roots that correspond to the tones of these roots' Middle Chinese reflexes): the tone of the first constituent tends to be to the left of the tone of the second constituent in the LEVEL-RISING-DEPARTING-ENTERING hierarchy (Fig. 3.1).

Figure 3.1: The tone hierarchy deciding constituent order in coordinative forms



The origin of this hierarchy is not clear. One possible explanation is avoidance of unnatural consonant clusters in the middle of a phonological word. This explains the LEVEL > RISING, DEPARTING, ENTERING tendencies, and the tendency for ENTERING roots to appear the last (so that no plosives are inserted into the middle of a phonological word, before the initial of the second syllable). The preference for the RISING > DEPARTING order may be explained by stipulating that the **-ʔ* coda of RISING had already started to lose its consonantal status in the Classical period. Thus the whole

hierarchy is tentatively explainable. This explanation, if plausible, still begs the question of the origin of such a constraint, which may be related to the emergence of the shared typological profile of the Mainland Southeast Asia linguistic area. These issues are however beyond the scope of this grammar.

The hierarchy in Fig. 3.1 is not intact in all contemporary Sinitic languages. Modern Standard Mandarin has several notable counterexample: 健康, 构拟, 会道门. Its diachronic evolution is beyond the scope of this grammar.

3.2.3 Dephrasal compounds

3.2.3.1 Agentive nominalization of verb phrases

Official titles in Classical Chinese are often agentive nominalization of verb phrases (22, 23). These forms have idiomatic meanings and cannot be taken literally. A 司馬 for instance was never in charge ordinary horses. There also appears to be a

(22) 司馬
sī-mǎ
manage-horse
'the one who manages [war]horses (i.e. the minister of war)'

(23) 相国
xiàng-guó
assist-state
'the one who assists the state (i.e. the chancellor)'

Classical Chinese lacks uncontroversial patientive nominalization of verb phrases or clauses into words.

An apparent counterexample is 予求 in (24), which is from the early Spring and Autumn period and thus is pre-Classical. 予求 should be interpreted as 'what I demand' (Qiu 1983). Elsewhere in texts, an equivalent alternative form 吾所求 is attested (Qiu 1983). Further, 予求 does not have an idiomized meaning. Therefore, there is no compelling reason to consider 予求 a grammatical word.

(24) 予 取　予 求
yú qǔ　yú qiú
1　take 1　seek
'I take [only] what I seek.' (左传・僖公七年)

Another apparent counterexample 天赐 in (26), which can be interpreted as 'what Heaven sends' in a patientive way. 赐 however has a regular nominal usage meaning 'gift', and an alternative form, 天之赐, is attested (26). 天赐 may indeed be a compound but it can be analyzed as a subordinative one.

(25) 一　年　之　中，　天賜　　四　至
yī　nián zhī　zhōng tiān-cì　　sì　zhì
one year GEN midst　heaven-gift four arrive
'Within one year, Heaven's gifts arrived four times.'

(26) 孤 賴 天 之 賜 也
gū lài tiān zhī cì yě
orphan rely.on heaven GEN gift SFP
'I rely on Heaven's gifts.' (吳越春秋・夫差內傳・十三年)

3.2.3.2 Gerund-like nominalization?

Sometimes a verb phrase is used to refer to the action it describes, and not who initiates this action (27). There however is insuffficient evidence to consider this a type of de-phrasal compounding: for one, these forms rarely have non-combinatorial meanings, and therefore do not appear to need to be analyzed as words.

(27) 相 人， 古 之 人 無 有 也
xiàng rén gǔ zhī rén wú yǒu yě
read.physiognomy person antiquity GEN person NEG have SFP
'As for judging people by their physiognomy, the ancients had no such practice.'

3.3 Derivation

Derivation is word formation by combining a content form and an abstract grammatical formative, which can be an affix or a process like reduplication (see also § 3.2).

3.3.1 Reduplication

Forms traditionally known as 双声叠韵叠字 are likely repudlicated forms.

螳螂 in (28) is analyzed as product of reduplication (Sun 2008, p. 93-95). As such, it was not truly simplex at a particular juncture of the history of Chinese.

(28) 黃雀 延 頸 欲 啄 螳螂
huáng-què yán jǐng yù zhuó tángláng
yellow-sparrow stretch neck want peck mantis
'The oriole stretched its neck, wanting to peck at the mantis.'

3.3.2 Deverbalization

3.3.2.1 Agent nouns

The verbs 出 ('go out'), 入 ('enter'), 亡 ('die, decay') are regularly derived to 出 ('what goes out'), 入 ('what comes in') and 亡 ('what dies').

3.4 Proper nouns

3.4.1 Personal names

A complete name of a person, during the Classical period, is quadripartite: there are two names, 姓 and 氏, that may be known as surnames, and there are two names, 名 and 字, that may be known as given names.

The two surnames rarely appear together and nor do the two given names. When both a surname and a given name are supplied, a surname-before-given name order can be seen.

3.4.1.1 Surnames

Before Han, the name of an aristocrat had two names and might be known as surnames: 姓 *xìng* 'ancestral clan name', and 氏 *shì* 'branch lineage name'.

The merger between the two was eventually finished during the Han dynasty, possibly because of the collapse of the old aristocracy and the old system of governance: people outside the Zhou dynasty nobility, now politically active, likely had neither ancestral clan names nor branch lineage names. It was common for them to adopt their birthplaces as their surnames, which did not fit into the double surname system, leading to the elimination of the distinction between *xìng* and *shì*.

Sima Qian himself does not clearly distinguish between the two.

(29) 及 生， 名 為 政， 姓 趙 氏
jí shēng míng wéi Zhèng xìng Zhào shì
when be.born name be NAME surname NAME lineage
'When he was born, his name was Zheng and his surname was Zhao.'

3.4.1.2 Given names

Besides the surnames, people also have 名 *míng* 'given name', the name given at birth. Classical given names are often monosyllabic (30) but disyllabic forms (31) exist.

(30) 賜 也， 始 可 與 言 詩 已 矣
Cì yě shǐ kě yǔ yán shī yǐ yǐ
NAME SFP only.now can with discuss Odes SFP SFP
'Ci, only now can I discuss the Odes with you.' (論語・學而)

(31) 小白 奔 莒
Xiǎobái bēn Jǔ
NAME flee NAME
'Xiaobai fled to Ju.'

Disyllabic given names are sometimes structurally analyzable

3.4.1.3 Courtesy names

Still it is considered impolite to use *this* given name when having conversations with strangers or not-so-close friends: in the latter cases, another name, the 字 *zì* 'courtesy name (lit. character)' is used. According to rites of Zhou, it was to be given at the adult ceremony of a man, after which addressing him by his "regular" given name (i.e. his *míng*) was disrespectful.

(32) 幼 名， 冠 字
yòu míng guàn zì
young give.name be.capped give.courtesy.name
'In childhood one is given a personal name; at the capping ceremony, a courtesy name.'

A courtesy name may be formed by prefixation: it was common before the Qin dynasty to first have a monosyllabic proper of the courtesy name and then prefix 子 (a respectful title) or a numeric index (伯 'first', 仲 'second', '叔' 'third', '季' 'last') expressing the man's birth order in his family.

For example, an important disciple of Confucius, usually known as 子路 in the modern literature, has 由 as his name given at birth (i.e. his *míng*): 子路 is his courtesy name, where 子 is a respectful title for a man. In *Analects*, Confucius usually calls him 由 because of their close relationship, while others, including the editors of *Analects*, call him 子路 (i.e. his *zì*). His *shì* is 仲, and therefore when both the surname and the given name are called, 子路 is known as 仲由.

(33) 子路 聞 之 喜。 子 曰：「由 也 好 勇 過 我，
Zǐlù wén zhī xǐ zǐ yuē Yóu yě hào yǒng guò wǒ
NAME hear 3 rejoice master say NAME SFP love courage surpass 1
無 所 取 材。」
wú suǒ qǔ cái
NEG.exist NMLZ obtain timber
'Zilu heard this and rejoiced. The Master said: "You surpasses me in his love of courage, but there is nowhere to get timber [for a raft]."'[3]

(34) 仲由 可 使 從 政 也 與
Zhòng-Yóu kě shǐ cóng zhèng yě yú
NAME can make engage.in government SFP SFP
'Can Zhong You be put to work in government?'

The practice to have courtesy names rapidly ceased in the twentieth century, when, quite similar to how the distinction between the ancestral clan name and the branch lineage name disappeared, due to rapid modernization, scholar-officials lost their positions in government, and rural land owners lost their properties both in mainland China and in Taiwan due to violent or non-violent land reforms. The "lay" citizens and peasants, now at the center stage of economy or even politics, had never had serious exposure to the tradition of courtesy names, and it is predictable that no one cared about courtesy names anymore.

3.4.1.4 Art names

Besides the courtesy name, certain people may have art names, or 號 *hào* 'lit. mark'. This is a post-Classical parctice: none of Classical prose authors had art names.

A person may have multiple art names. The great poet 李白, whose courtesy name is 太白 'lit. extreme-white', had 青蓮居士 as his art name or *hào*.

3.4.2 Place names

Some place names are composite forms formed by processes already discussed above. (35) is an example of attributive subordinative compounding.

(35) 曲阜
Qū-fù
bent-mound

[3] This follows the literal reading of 材 as timber; another interpretation reads it as 裁 'judgment'.

‘Qufu (lit. bent mound)’

3.4.2.1 Kra-Dai loanwords

Place names in modern day China, including the geographical locations of the former warring states, are often disyllabic and yet not analyzable. The city name 姑蘇 (36) for instane cannot be analyzed as a compound or an affixed form based on forms known in Trans-Himalayan languages.

(36) 吳子胥 不 磔 姑蘇 東 門 外 乎
Wú-Zǐxū bù zhé Gūsū dōng mén wài hū
NAME NEG dismember NAME east gate outside SFP
‘Was Wu Zixu not dismembered outside the eastern gate of Gusu?’ (荀子・宥坐)

These toponyms were likely borrowed from Kra-Dai languages spoken in China before Sinitic languages spread to the regions (Zhengzhang 1990; Zhengzhang 1996). The first syllable of 姑蘇 is likely a locative prefix (cf. 姑蔑; Zhengzhang 1996).

Zhengzhang largely bases his decipherment on modern Kra-Dai languages. These likely borrowed toponyms are perhaps better analyzed in comparison with the reconstructed common ancestor of modern-day Kra-Dai languages, and have recently gained attention for possibly revealing the early dialectology of the language family (Xia 2026).

Chapter 4

Pronouns

4.1 Personal pronouns

An unusual trait of Classical Chinese, which can be seen in Table 4.1, is the existence of at least two series of pronouns. There also lack subjective or objective prototypical third-person pronouns.

Table 4.1: Personal pronouns in Classical Chinese

Person	Forms	
First	吾, 我	余, 予
Second	汝, 女	尔

4.1.1 First-person pronouns

A corpus search in Chinese Text Project reveals that 我 and 吾 are far more frequent than 予 and 余 in Classical texts. This is consistent with the observation of Pulleyblank (1995, p. 76), who maintains that by classical times, the latter pronouns were "obsolescent". In post-Classical Literary Chinese, 予 and 余 do not fall out of use, presumably because of intentional archaism.

4.1.1.1 余 and 予

The grammatical distributions of 予 and 余 do not appear to have meaningful differences in prose. In the *Analects*, 余 does not appear, while in *Zuozhuan*, the overwhelming majority form is 余 (which is still considerably less frequent than 我 and 吾).

(1) 予所否者 (論語・雍也)
(2) 啟予足 (論語・泰伯)
(3) 天喪予 (論語・先進)
(4) 余死不汝見矣 (說苑・權謀)
(5) 事建如事余 (立節)

予 and 余 apparently have distributional difference in Chu poems, which are not covered in this grammar.

The contrast between 我 and 吾 is much more obvious in Classical texts, and yet a concise characterization of the exact grammatical environments they appear in is still challenging. It is sometimes claimed that 吾 is a subjective pronoun while 我 is an objective pronoun. This theory fails to explain the example (6).

Thus, below, we overview the distribution of 吾 and 我 in various grammatical functions.

4.1.1.2 吾 and 我 as subject

4.1.1.3 吾 and 我 as possessive

4.1.1.4 吾 and 我 as fronted object

(6) 不吾知也

(7) 子不我知而從我者乎 (說苑・雜言)

4.1.1.5 吾 and 我 as object

4.1.2 Second-person pronouns

Orthographically there are two sets of second-person pronouns.

4.1.3 Third-person pronouns

Classical Chinese has two third-person pro-forms (Table 4.2). Notably, no subjective third-person pronoun exists.

Table 4.2: Personal pronouns in Classical Chinese

Function	Form	Example
Object	之	(8)
Possessive	其	(9)

(8) 鳴 鼓 而 攻 之
míng gǔ ér gōng zhī
sound drum CONJ attack 3
'Beat the drum and attack him.' (論語・先進)

(9) 不 在 其 位
bù zài qí wèi
NEG be.at 3.GEN position
'[One] does not occupy that person's position.' (論語・泰伯)

之 seems to be the accusative pronoun in Old Chinese. 其 seems to be the genitive pronoun, and may be a phonological fusion of 之 and a possessive marker.

4.2 Deictic nouns

Some forms have developed deictic meanings in conversational contexts that resemble those of pronouns, but deviate from prototypical pronouns because the form being used strongly depends on the context and these deictic forms often at least retain some semantic transparency. The way some of these forms are used resembles the T-V distinction in that they imply that the person referred to is humble or respectable, but cannot be analyzed as such because of the abundance of these deictic forms and how the deictic forms are being used.

Classical Chinese deictic nouns are thus comparable to forms of address (*Your Honor*, *His Excellency*, etc.). They differ from prototypical forms of address in two ways: first, they can be used to refer to ordinary people without specific titles, and second, most Classical Chinese deictic nouns do not have regular alternation between second-person and third-person forms (*His Honor* vs. *Your Honor*).

4.2.1 Grammatical properties

4.2.1.1 Deictic nouns vs. ordinary nouns

In Classical Chinese, deictic nouns typically cannot be complemented or modified: thus (10a) is attested, but (10b) is not attested. We note that nothing prevents regular compound nouns from taking modifiers (e.g. 1 in Ch. 6, with a structure parallel to that of 10b). In this way one may argue that the deictic nouns in Classical Chinese are not simply simplex nouns or compounds with idiomatic deictic meanings, but have the pronoun-like grammatical property in that they themselves are full-fledged noun phrases. We note that this behavior is unlike the situation in Japanese, where "pronouns" do undergo typical noun phrase modifications.

(10) a. 惡 人
è rén
evil person
'an evil person'

b. * 惡 寡人
è guǎ-rén
evil humble-person
Intended: 'evil me (said by a ruler)'

Certain deictic forms with modification, like 愚臣, are attested (16). These forms however appear to have undergone at least idiomization. Changing the adjective modifying 臣 with a totally natural meaning, the frequency of the resulting form in the pre-Qin and Han sector of Chinese Text Project greatly drops (Table 4.3; the sequence 卑臣 is attested but always means 'make subjects humble').

Table 4.3: Frequencies of 微臣, 卑臣 and 愚臣 used as possible deictic nouns

Form	Occurrence
微臣	4
卑臣	0
愚臣	~ 40

(11)

4.2.1.2 Deictic nouns vs. pronouns

Apart from being highly context-dependent and being an open class, the deictic nouns however still differ from prototypical pronouns in that they are unable to undergo object fronting. The negator 不 attracts the object pronoun behind it (12), but a sequence like 不寡人 +V is never attested (13).

(12) 不我欺

(13) * 不寡人欺

4.2.2 General uses

Wang (2005, pp. 125-140) is a comprehensive survey on deictic nouns in Classical texts, which are known as 相互称呼 and 交际称呼 in the thesis. A thorough enumeration of deictic nouns is beyond the scope of this grammar, and below we only outline the basic patterns of the structure and use of deictic nouns, with several examples of deictic nouns used in specific conversational contexts.

An overall pattern in using the deictic nouns is for one to use a humble deictic noun to refer to himself or herself, and to use a respectful deictic noun to refer to other conversational participants. The former is true, even for those in power (15). When a person being referred to is *not* a conversational participant, the speaker tends to use a humble deictic noun to refer to him or her, if the person being referred to is in the same in-grouop with the speaker (19); otherwise a respectful deictic noun is used.

Deictic pronouns are not always the only appropriate option. In certain scenarios (e.g. a disciples referring to himself, talking to the master), person names are used (20). Plain personal pronouns are also proper in certain situations (21).

4.2.3 Royal and courtly register during the Classical period

In a conversational context in which the conversational participants are a ruler and his subjects, the deictic nouns used by the ruler and the subject to refer to themselves and each other are all different.

The noun 臣 'subject, minister, official' is used by a male, usually an official, to designate himself when talking to a ruler. Thus the occurence of 臣 in (14) means 'I (lit. [your] subject)', with a meaning of self-deprecation and humbleness, and not, say, 'some ministers'.

(14) 臣 聞 不 知 而 言 不 智
chén wén bù zhī ér yán bù zhì
minister hear NEG know CONJ speak NEG wise
'I have heard that speaking without knowing is unwise.' (韓非子・初見秦)

Rulers, interestingly, utilize *humble* deictic forms to refer to themselves. In (15), 寡人, literally meaning 'scant-person' i.e. a person of meager virtue, is the self-reference of the king.

(15) 寡人　　願　安　承　教
guǎ-rén　yuàn ān　chéng jiào
humble-person wish calmly receive instruction
'I wish to receive your instruction with a calm mind.' (孟子・梁惠王上)

The term a subject uses to refer to the ruler is typically 陛下 (16), 大王 (17) or simply 王.

(16) 願　陛下　幸　察　愚　臣　之　計，無　忽
yuàn bì-xià xìng chá yú chén zhī jì wú hū
wish steps-below graciously examine foolish minister GEN plan NEG neglect
'I hope Your Majesty will graciously examine your foolish servant's plan and not neglect it.' (韓非子・存韓)

(17) 愿大王图之 (戰國策・衛鞅亡魏入秦)

A ruler may choose to address his officials using seemingly equal deictic forms. (18) does not differ from how two members of the elite class have conversations.

(18) 吾不能蚤用子 (新序・善謀)

When the ruler is not a conversational participant but is being talked about in another conversatioanl context (in which how conversational participants address each other and refer to themselves may differ from what is described above), he is referred to by his subjects using a humble deictic noun (19) when talking with people from another country.

(19) 使者自稱曰寡君之老

4.2.4 Didactic dialogues

In didactic dialogues, in which the conversational participants are a master and his disciples, the disciples often uniformly refer to themselves (20) and are referred to by the master by their given names (TODO; § 3.4.1.2) or by using the second-person pronoun (21) and not their courtesy names.

(20) 赤　也　惑，　敢　問
Chì　yě huò　gǎn wèn
NAME SFP be.puzzled dare ask
'I, Chi, am puzzled and venture to ask.'

(21) 求！　爾 何　如？
Qiú　ěr hé　rú
NAME 2　what be.like
'Qiu! What about you?' (論語・先進)

The master, in such a conversational context, refers to himself either using ordinary first-person pronouns or his given name. (22) is an example of both.

(22) 吾 無 行 而 不 與 二 三 子 者，是 丘
wú wú xíng ér bù yǔ èr sān zǐ zhě shì Qiū
1 NEG.exist action CONJ NEG share.with two three master REL this NAME
也
yě
SFP
'There is nothing I do that I do not share with you, my disciples; that is how I, Qiu, am.' (論語・述而)

Deictic nouns primarily appear when the disciples need to refer to the master in a conversational context. The disciples refer to the master by terms like 子 or 夫子 (22).

(23) 子 如 不 言
zǐ rú bù yán
master if NEG speak
'If you, Master, do not speak, …' (論語・陽貨)

4.3 Demonstratives

Chapter 5

Prepositions

Prepositional complements are also placed after the verb (1).

(1) 君子 博 学 于 文
jūn-zǐ bó xué yú wén
lord-son broadly study at culture
'A gentleman studies cultural learning widely.'

The oblique clausal dependents can be post-verbal (2, 3, 4) or pre-verbal (5, 6, 7), with the pre-verbal order gaining popularity as time went by. The linear order of peripheral arguments is similar to that in Mandarin (He 2005, pp. 286-287).

(2) 侍 饮 于 长 者
shì yǐn yú zhǎng zhě
attend drink at elderly REL
'[One] attends an elder at a drinking party.'

(3) 孟孙 问 孝 于 我
Mèngsūn wèn xiào yú wǒ
NAME ask filial.piety at 1
'Mengsun asked me about filial piety.'

(4) 祷 尔 于 上 下 神 祇
dǎo ěr yú shàng xià shén qí
pray 2 to above below celestial.spirit earth.spirit
'[I] pray for you to the spirits above and below.'

(5) 韩生 南 向 坐
Hán-shēng nán xiàng zuò
NAME-scholar south face sit
'Scholar Han sat facing south.'

(6) 於 人 之 罪 无 所 忘
yú rén zhī zuì wú suǒ wàng
regarding person GEN offense NEG.exist NMLZ forget
'As for other people's offenses, [he] forgets none.'

(7) 为 人 谋 而 不 忠 乎
wèi rén móu ér bù zhōng hū
for person plan CONJ NEG loyal SFP
'Have I been disloyal when acting on behalf of others?'

Chapter 6

Adjectives

(1) 良　大夫
liáng dà-fū
good great-man
‘a good official’

Chapter 7

Noun phrase

7.1 Adjectives

(1) 斗 筲 之 人
dǒu shāo zhī rén
peck.measure bamboo.basket GEN person
'a person of limited capacity (lit. a person of peck-measure and basket capacity)'

Chapter 8

Verbs

8.1 Simplex verbs

8.1.1 Roots typically used as verbs

Most roots that typically are used as verbs are monosyllabic. Roots that in certain modern Sinitic languages cease to be able to be main verbs on their own can nonetheless be used independently in Classical Chinese (1). The situation is comparable to that of nouns (§ 3.1.1).

(1) 危 而 不 持
wēi ér bù chí
be.unsteady CONJ NEG support
'[When someone] is unsteady, [you] do not support him.' (論語・季氏)

8.1.2 Roots with multiple functions

Some roots have both nominal and verbal uses, and there is usually some semantic connection between the interpretations of the two uses, but this is not regularly inferable and thus cannot be analyzed as zero derivation.

Here we consider some examples in Yang, Kong, and Zhou (1991):

- 楚 may mean 'the Chu state' or 'do what Chu people do'.
- 床 may mean 'bed' or 'settle down your bed or sleep on a bed'.
- 城 may mean 'city, castle' or 'build a city'.

The interpretation of the verbal usage is usually *not* decided from the meaning of the root and that the root is used in a verbal environment; rather, it is instructed by the lexicon. Therefore, the verbal usage of 城市 only means 'build a city' although the 'do city-related things' reading in principle could make sense.

Therefore, roots like 城, 楚 and 床 have double functions: nominal and verbal, but the two functions are likely not related to each other by regular grammatical rules. This corresponds to the "multiple function" case in Dixon (2010, § 11.3). Moreover, what is stored in the lexicon is not the bare, non-categorized root 城, but one noun lexeme 城 'city' that specifies its nominal usage and one verb lexeme 'build a city' that specifies its verbal usage, and other seemingly possible ways to categorize the root, although attested elsewhere, are ruled out by their absence in the lexicon.

The boundary between roots with double functions and roots undergoing zero derivation (see below) is somehow blurry, as the nominal and verbal uses of 城 and 床 still seem to show a common pattern and may be understood as a rare derivation. This blurriness leads many grammatical works on Classical Chinese to simply refer to the two phenomena uniformly as “nouns used as verbs”.

8.1.3 Ad hoc categorization

There are sporadic verbal usages of nouns that are almost never attested elsewhere, like 军 in 沛公军霸上. This means that ad hoc re-categorization of roots is possible in Classical Chinese, and the meaning is to be decided from the context. This is also possible in informal English[1] but usually not accepted in formal texts. Alleged ad hoc categorized Classical Chinese roots are indeed a possibility, after all, although their frequency is not high enough and cannot be exaggerated to be the norm rather than the exception.

(2) 左右 欲 刃 相如
zuǒ-yòu yù rèn Xiàngrú
left-right want blade NAME
‘The attendants wanted to kill Xiangru with their blades.’

(3) 大 楚 兴，陈胜 王
dà Chǔ xīng Chén-Shèng wàng
great NAME rise NAME reign
‘Great Chu shall rise, and Chen Sheng shall be king.’

(4) 然 而 不 王 者，未 之 有 也
rán ér bù wàng zhě wèi zhī yǒu yě
so CONJ NEG reign REL NEG.yet 3 exist SFP
‘There has never been anyone who did this and yet failed to become king.’

8.2 Derivation

8.2.1 Reduplication

Certain forms, like 望洋, appear only in certain Warring States texts and can be analyzed as reduplication (Sun 2008, pp. 77-81)

8.2.2 Denominalization

The conventional term in Mandarin Chinese 名词作动词 ‘nouns used as verbs’ covers three phenomena, corresponding to (a) multiple functions and (b) zero derivation (Dixon 2010, § 11.3), and also (c) the less frequent case of ad hoc categorization of a root.

In other cases the meaning of the verbal use of a root usually appearing in a nominal context is regularly derived from the nominal meaning. This is because although

[1] As in *I might [guinea pig] it for you.* A longer example is a historic event being *“Sappho and her friend”’d.*

tropative or causative derivations in Classical Chinese are mainly verb-to-verb, they can also be applied to nouns. In this way, from 臣 'servant, official, minister' we have the causative verbal usage 'make sb. dependent on', and from 客 'guest' we have the tropative usage 'consider sb. as a guest'. These verbal usages are nothing different from noun-to-verb derivation observed in other languages, so we regard the relevant phenomena as zero derivation as in Dixon (2010, § 11.3).

In zero derivation, the meaning of the nominal usage has to be recorded in the lexicon, the meaning of the verbal usage can be automatically decided from the derivation rule. These derivations are however not completely regular and not for every word: the lexicon also controls whether a derivational rule applies.

8.2.3 Numeral-to-verb derivation

(5) 六 王　毕，四　海 一
liù wáng bì　sì　hǎi yī
six king　end　four sea unite
'The six kings were finished, and all within the four seas was united.'

8.2.4 Adjective-to-verb derivation

Box 8.1: Adjective-to-verb derivation

则其好游者不能穷也
　山多石，少土
　亲贤臣，远小人

8.3 Complex predication

Complex predication refers to the (often structually heterogeneous; Paul 2008) phenomenon that the verb word consists of multiple morphologically verbal stems, which sometimes is also known as serial verb construction(s).

Complex predication phenomena in Sinitic languages are typically collectively known as verb-complement constructions, or 述补结构, with the modifying verbal stem denoted as 补语 or *complement*.[2]

He (2005, pp. 212-240) is a comprehensive review of complex predication in Classical Chinese, based on a comparative study between *Zuozhuan* and *Shiji*. Her conclusion is that the resultative construction is largely absent in *Zuozhuan*, but can be more frquently found in *Shiji*.

He (2005) notes that verbs appearing in the resultative construction are mostly about wars or disasters (p. 225). This can be interpreted as a natural semantic consequence of the resultative construction, but may also be alternatively considered to be of a stylist origin

[2]Note that the term does not mean the verbal stem should be regarded as an argument. Some constituents, like those studied in (He 2005, pp. 126-177) sometimes also analyzed as 补语, are however indeed (core or peripheral) arguments. This section focuses on complex predication only.

Chapter 9

Verb frames

9.1 Intransitive constructions

9.1.1 Action verbs

9.1.2 Stative and change-of-state verbs

Known in Mei (2015) as 内动

A BE verb describes a state; a BECOME verb describes the change of a state. In both types of verb frames, the sole argument is a *theme*: the situation happening to it just happens, and usually it does not have much control over it nor any volition to trigger it (Li 2004, p. 345; Mei 2015, p. 275).

In Classical Chinese, just like in other languages, BECOME/BE verbs often have established causative usages, forming CAUSE-BECOME/BE argument structures with the *causer* argument being the subject and the *theme* argument being internal (§ 10.1.1). When a causer is absent, the structure of the clause is comparable to what sometimes is known as the middle voice in English (e.g. *the door opened*; cf. the transitive CAUSE-BECOME *I opened the door*). (1) is an instance: in its CAUSE-BE usage (2), the argument that is described as weak is an internal argument that appears after the verb, but in (1), the argument that is described as weak is the *subject*: the internal theme argument gets promoted to the subject position.

(1) 秦 强 而 赵 弱
Qín qiáng ér Zhào ruò
NAME strong CONJ NAME weak
'Qin is strong, whereas Zhao is weak.'

The "middle voice" construction exemplified in (1) (known as 内动 in Mei (2015)) has a subject, which corresponds to the argument that is the object in the CAUSE-BECOME/BE construction (i.e. the internal argument).

9.1.2.1 The alternation between BE and BECOME

Alternation between BE and BECOME verb frames is natural. Some BECOME verbs however do not have BE counterparts.

9.1.2.2 Impersonal stative verb

It is possible for the argument of a stative verb to be *not* promoted to the subject position. (2, 3).

(2) 蝼蛄　鸣
lóuguō míng
frog　call
'Frogs call.'[1]　(礼记・月令)

(3) [五 月]$_{\text{temporal}}$ [鸣]$_{\text{predicate}}$ [蜩]$_{\text{internal argument}}$
wǔ yuè　míng　tiáo
five month　chirp　cicada
'In the fifth (lunar) month, cicadas chirp.'

(3) can be analyzed as a situation if we assume that 鸣 is a BE verb, denoting a state where bugs continue to make noise, and that the sole argument 蜩 remains in-situ and is not promoted to the subject position.

No other analysis is available: for instance a DO verb can never have such a behavior (Mei 2015, p. 351).

9.2 Monotransitive constructions

9.2.1 Action verbs

The subject of a DO clause usually has to be animate, because it voluntarily initiates the event described by the clause (4). The subject is an *agent*, as opposed to a *causer* (§ 10.1) or a *theme* (§ 9.1.2).

Mei (2015, p. 272) lists some criteria to distinguish a transitive DO verb from a transitive CAUSE verb (§ 10.1.1).

(4) 桓公　杀　公子　纠
Huán-gōng shā gōng-zǐ Jiū
NAME-duke kill duke-son NAME
'Duke Huan killed Prince Jiu.'

9.2.1.1 S=A labile verbs

(5) 昭王　反　国
Zhāo-wáng fǎn guó
NAME-king return state
'King Zhao returned to his state.'

(6) 右师　反
yòu-shī fǎn
right-commander return
'The Commander of the Right returned.'

[1] The gloss of 蝼蛄 follows 鄭玄, who identifies it as a frog; its identification is disputed.

9.2.2 Causative verbs

Mei (2015, p. 272) summarizes the main difference between transitive, DO action verbs and causative verbs.

9.2.2.1 S=P labile verbs

(7) 大夫 溃 莒
dà-fū kuì Jǔ
great-man rout NAME
'The officials routed Ju.'

(8) 莒 溃
Jǔ kuì
NAME collapse
'Ju collapsed.'

9.2.2.2 Lexicalized synthetic causative construction

Some CAUSE verbs are fossilized, and do not have clear intransitive counterparts i.e. the un-causativized counterparts. For instance, 伤 'hurt' typically is a state transition verb meaning body, etc. being hurt, and it also has a causative (i.e. CAUSE-BECOME) meaning ('make ...hurt'). The CAUSE-BECOME verb frame of 伤 however has gained a separate lexicalized meaning that can't be transparently inferred from the meaning of the BECOME usage: it can mean 'let ...be damaged', in which the object is not necessarily body or a person. This usage of 伤 has no BECOME or other intransitive counterpart. The absence of a BECOME counterpart can be proven by the ability for this fossilized figurative usage of 伤 to undergo "passivization" (9), which is otherwise not possible (§ 10.2.1).

(9) 女红 伤 则 寒 之 原 也
nǚ-gōng shāng zé hán zhī yuán yě
woman-work damage then cold GEN source SFP
'When women's textile work is impaired, it is a source of exposure to cold.'

Verbs like 伤 in like (9) can easily be reanalyzed as DO verbs. This is likely a diachronic path of the creation of DO verbs. The verb 败 for example seems to be originally a BECOME verb ('to get corrupted') and later gained a specific meaning of 'to defeat' in its CAUSE-BECOME usage, which eventually evolved into a DO usage (Mei 2015, p. 285).

9.2.2.3 Causative verbs with non-agentive subjects

Some verbs license subjects that look like arguments of prototypical BECOME or BE verbs: the subject may be animate but it does not volitionally trigger the event. The situation "just happens to be the case", and the subject can be described as a *theme* and not an *agent*. What sets them apart from prototypical BECOME or BE verbs in § 9.1.2 is the fact that the subject seems quite unlike an internal argument. In (10), the subject 火 'fire' is definitely a theme and not an agent: the fire does not get to *decide* if it burns the flag (Mei 2015, p. 276). Still the clause is not a prototypical BECOME one

as there is an internal argument 其旗 in it, and the theme 火 is an *external* theme (Mei 2015, p. 353). These verbs therefore have difficulties participating in synthetic causativization (§ 10.1.1).

(10) 火 焚 其 旗
huǒ fén qí qí
fire burn 3.GEN flag
'Fire burned his flag.'

9.3 Ditransitive constructions

9.3.1 Verbs governing a locative object

Compare the prepositional (11) and the double-object example (12). The omission seems to be in line with the omission of the preposition in post-verbal locative arguments (cf. 13).

(11) 有 献 不 死 之 药 于 荆王 者
yǒu xiàn bù sǐ zhī yào yú Jīng-wáng zhě
exist offer NEG die GEN medicine to NAME-king REL
'There was someone who offered a medicine of immortality to the king of Jing.'

(12) 请 献 盆缶 秦王
qǐng xiàn pén-fǒu Qín-wáng
request offer basin-jar NAME-king
'Allow me to present a pottery vessel to the king of Qin.'

(13) [大-王]$_{\text{subject: NP}}$ 见 [臣]$_{\text{object: NP}}$ [列 观]$_{\text{locative: NP}}$
dà-wáng jiàn chén liè guàn
great-king receive minister ordinary hall
'Your Majesty received me, your servant, in an ordinary hall.'

9.3.2 Experience verbs

Some experience verbs, mostly verbs about emotions, behave like BECOME verbs (Mei 2015, p. 273): when used as transitive verbs, the subject does not look quite agentive and the clause is likely causative (15), and when used as intransitive verbs, there is a clear internal change-of-state meaning (14).

(14) 孔子 成 春秋， 而 乱 臣 贼 子
Kǒng-zǐ chéng chūn-qiū ér luàn chén zéi zǐ
NAME-master complete spring-autumn CONJ rebellious minister criminal son
惧
jù
fear
'Confucius completed the Spring and Autumn Annals, and rebellious ministers and wicked sons became afraid.' (孟子・滕文公章句下)

(15) 惧 之 以 怒
jù zhī yǐ nù
frighten 3 with anger
'[He] frightened them with his anger.' (左传・昭公十三年)

On the other hand, perception verbs (e.g. 见 'look') and cognition verbs (e.g. 知 'know') are often transitive, and therefore are not compatible with the synthetic causative construction (§ 10.1.1; Mei 2015, p. 274). Intuitively, these verbs are DO-like according to the criteria listed in § 9.2.1. For instance, they can appear in 所 construction (16).

(16) 異 乎 吾 所 聞
yì hū wú suǒ wén
differ from 1 NMLZ hear
'[This] differs from what I have heard.'

Certain perception verbs however have developed a figurative, fossilized meaning, and when intransitivized, can participate in synthetic causativization (§ 10.1.1, 3). This possibility indicates that these fossilized usages are BECOME- or BE-like: 见 'meet formally' therefore means 'in the state of regularly meeting an important figure'.

9.4 Experiential valency increasing

9.4.1 The affective constructions

Classical Chinese has two affective constructions, in which the subject is an experiencer suffering something bad from the situation described by the latter (Mei 2015, pp. 354-358).

The first affective construction simply attaches an experiencer to an argument structure (17). In (17a), 亡 appears as a BECOME verb (§ 9.1.2): it is intransitive and its subject, the *Odes*, did not have control over its being ignored. In (17b), a new argument – the experiencer subject – is introduced to the argument structure of 亡: the meaning of the sentence is 'the shepherds suffered from the sheeps getting lost.'

(17) a. 亡 as a BECOME verb
[诗]$_{\text{subject,theme: NP}}$ [亡]$_{\text{BECOME}}$ 然后 春秋 作
shī wáng rán-hòu chūn-qiū zuò
poem **get.lost** then Spring-Autumn compose
'The *Odes* got lost, and then the *Spring and Autumn Annals* was composed.'

b. 亡 as a AFFECTIVE-BECOME verb
[二 人]$_{\text{subject: NP}}$ [相 与 牧 羊，而 俱
èr rén xiāng yǔ mù yáng ér jù
two person mutually go.together herd sheep CONJ all
[亡]$_{\text{AFFECTIVE-BECOME}}$ 其 羊]$_{\text{coordinated VP}}$
wáng qí yáng
get.lost **3.GEN sheep**
'Two people herded their sheep together, and they both lost their sheep (lit. suffer from their sheep's missing).'

The second affective construction *obligatorily* has an object, which is in possession of the subject. The meaning of (18), for example, is that Confucius was frustrated by the fact that his tree was cut in Song.

The fact that the object (树 and 胁 here) should not contain any possessive markers and has to be interpreted as something being possessed by the subject suggests that the second affective construction is an external possession construction.

(18) 吾 再 逐 於 鲁， 伐 树 於 宋
wú zài zhú yú Lǔ fá shù yú Sòng
1 twice expel at NAME cut.down tree at NAME
'I was expelled twice from Lu and had my tree cut down in Song.'

(19) 范睢 折 胁 於 魏
Fàn-Jū zhé xié yú Wèi
NAME break rib at NAME
'Fan Ju had his ribs broken in Wei.'

> **Box 9.1: External possession**
>
> External possession as subject

9.5 Applicative constructions

> **Box 9.2: Applicative constructions**
>
> benefactive; the claim that double object constructions are similar to benefactive constructions; what object gets passivized. See Mei (2015, p. 421).

(20) 子 使 漆雕開 仕
zǐ shǐ Qīdiāo-Kāi shì
master make NAME take.office
'The Master urged Qidiao Kai to take office.'

(21) 雍 也 可 使 南 面
Yōng yě kě shǐ nán miàn
NAME SFP can make south face
'Yong could be made to face south [as a ruler].'

9.6 Direct quotation

《詩》云：『如切如磋，如琢如磨。

(22) [子禽]$_{\text{subject: NP}}$ [[问]$_{\text{verb}}$ [于 子贡]$_{\text{target}}$]$_{\text{VP branch 1}}$
Zǐqín wèn yú Zǐgòng
NAME ask at NAME
[[曰：“…”]$_{\text{direct quotation}}$]$_{\text{VP branch 2}}$
yuē
say

‘Ziqin asked Zigong: …’

(23) 季康子 患 盜， 問 於 孔子。
Jì-Kāngzǐ huàn dào wèn yú Kǒng-zǐ
NAME worry.about theft ask at NAME-master
‘Ji Kangzi was concerned about theft and asked Confucius about it.’

9.7 Control

In (24), the constituent introduced by 曰 is a proper name, and clauses with 曰 as the main verb where the subject is a person and the object is a proper name are rare if not impossible. If 曰 in (24) is understood as a marker of a direct quotation, however, the sentence makes sense: the second argument has a *metalinguistic* usage, whose semantic interpretation is the quoted 灵台 itself, without referring to anything in the real world.

(24) [谓]$_{\text{verb}}$ [其 台]$_{\text{object: NP}}$ [曰 [灵 台]$_{\text{proper name}}$]$_{??}$
wèi qí tái yuē líng tái
call 3.GEN terrace say spirit terrace
‘[They] called his terrace “Spirit Terrace.”’

Therefore, in Classical Chinese, 曰 is both used as a lexical verb (17) and a grammaticalized marker of direct quotations. What is quoted can be a sentence (18) or a noun phrase (24). Under this analysis, examples like (25) are probably clauses with nominal predicates, with the predicate being a direct quotation.

(25) 其 名 曰 觙
qí míng yuē Jí
3.GEN name say NAME
‘His name was Ji.’

Box 9.3: In search of auxiliaries?

See e.g. here.

- 克”“能”“堪”“可”“可以”“可得”“得”“足”“足以”等
- 欲”“肯”“將”“宁”“敢”“忍”“愿”“屑”“慇（yìn）”等，其中“敢”“忍”“屑”通常用于否定句，“肯”“（不）敢”“（不）忍”“（不）屑”沿用至今。“欲”“將”“愿”“慇”表示主观的希望或打算，可以翻译成“希望”或“打算”。例如：
- 当”“如”“宜”“任”“合”“应”，其中“合”“应”
-

能 does not seem to be a prototypical auxiliary. The main argument against its status as an authentic auxiliary is the prevalence of sequences like 不能不 (as in 人主不能不有遊觀安燕之時). Here the two negations mean ‘not doing …is impossible’, and the structure is routinely used to emphasize the core verb phrase (VP). The structure of these examples seems to be better analyzed as a complement clause construction, as nowhere else in Classical Chinese can we find two identical negators in a single clause. (非不 is possible, but not 不不)

cf. Negation in Cartography

TODO:
坠车、队于车

(26) 公　惧，　队　[于　车]$_{\textbf{source}}$
gōng jù　zhuì yú　chē
duke afraid fall　from carriage
'The duke was afraid and fell from the carriage.'

Chapter 10

Valency alternation

10.1 Causative constructions

A *causer* makes a situation to be the case, but does not always do so intentionally. It can therefore be inanimate, as opposed to how an *agent* behaves (§ 9.2.1).

We can divide causative constructions in Classical Chinese into synthetic and analytic ones. In the synthetic causative construction, there is only one verb in the surface form: the causative valency alternation is supposedly marked by a prefix **s-*, which is invisible in the written texts but is reflected by tonal changes of the verb. If a root develops a lexicalized usage in the synthetic causative construction, then CAUSE verb is formed.

10.1.1 Synthetic causative

The synthetic causative construction applies to existing argument structures, or sometimes bare roots. The synthetic causative construction cannot be applied to a DO construction: the reason is probably because a DO construction is too "big", already having a full-fledged wannabe subject (Mei 2015, p. 363-364).

On the other hand, the syntactic causative construction can be applied to a pseudo-passive structure (1) and BECOME or BE (2) argument structures. Certain intransitivized experience verbs, possibly having an argument structure comparable to a BECOME/DO verb (§ 9.3.2) with a wannabe subject also have causative usages (3), but their transitive counterparts are never compatible with the synthetic causative construction (Mei 2015, p. 274).

Finally, the synthetic causative construction can be directly applied to a root (4): the word 妻 'wife' is sometimes used as a verb, meaning 'to marry daughter to …', inconsistent with the meaning of (4). Therefore, in (4), 妻 is ad hoc categorized into a CAUSE verb, its usual verbal usage being irrelevant here.

(1) 是 夭 子蛮， 杀 御叔……
shì yāo Zǐmán shā Yùshū
this die.young NAME kill NAME
'This woman made Ziman die at a young age, and got Yushu killed…'

(2) 以 弱 天下 之 民
yǐ ruò tiān-xià zhī mín
PURPOSE weaken heaven-below GEN people
'…to weaken the people.'

(3) 子尾 见 疆
Zǐwěi xiàn Jiāng
NAME cause.to.visit NAME
'Ziwei had Jiang pay a formal visit [to Xuanzi].' (左传・昭公二年)

(4) 妻 帝 之 二 女
qì dì zhī èr nǚ
make.wife emperor GEN two daughter
'[He] took the emperor's two daughters as wives.'

The labile S/O alternation between the BE/BECOME usage and the CAUSE-BE/BECOME usage is quite regular in Classical Chinese; verbs allowing this alternation are sometimes known as *ergative verbs* (Mei 2015, p. 378), although the phenomenon is about the core argument structure and has nothing to do with ergativity in alignment. It should be noted that not all BECOME/BE verbs are compatible with the synthetic causative construction. For instance, 鸣 'chirp' in (3) does not have a transitive CAUSE-BE usage. More examples are given in Mei (2015, p. 276). On the other hand, some clauses that look like CAUSE-BE/BECOME clauses actually do not have BE or BECOME counterparts (§ 9.2.2.2).

10.1.2 Analytic causative

Classical Chinese has an analytic construction to express the causative meaning. In (5), the word 使 is applied to the stative structure 渚者居中原 'people living on small lands reside inland', meaning 'let people living on small lands in water live inland'.

(5) 不 使 [渚 者]$_{\text{shared object: NP}}$ [居]$_{\text{BE}}$ [中-原]$_{\text{locative object: NP}}$
bù shǐ zhǔ zhě jū zhōng-yuán
NEG let small.land.in.water NMLZ reside middle-land
'...do not let people living on small lands in water live inland'

The word 使 can be replaced by 令 (6) or 俾 (7) (Mei 2015, p. 376). Note that in (6), 令 seems to be applied to a DO argument structure.

(6) 令 军 勿 敢 犯
lìng jūn wù gǎn fàn
order army NEG dare violate
'[He] ordered the army not to dare violate [the prohibition].'

(7) 俾 民 不 迷
bǐ mín bù mí
make people NEG go.astray
'[This] keeps the people from going astray.'

A question is whether 使 is a lexical verb, or just a non-incorporated causative marker. The fact that (6) involves a DO verb deviates from the behavior of the synthetic causative construction, which is not compatible with DO verbs (§ 10.1.1). Further, the fact that we can choose among 使, 令 and 俾 is rather unusual for a grammatical marker: we also note that 使 retains the meaning of 'send sb. to do sth.' and 令 retains the meaning of 'command sb. to do sth.' Therefore, we consider 使, 令 and 俾 to be *lexical verbs*, and not grammaticalized causative markers. What is grammaticalized is the causative clause with the form of V_1 NP VP, with V_1 being one of these verbs.

10.2 “Passive” constructions

10.2.1 The agent-less pseudo-passive construction

What is often known as the passive in Classical Chinese is not really a passive construction comparable to the English or Latin passive. The main problem is the lack of a grammaticalized way to say the agent: in a “true” passive construction, the original subject is somehow demoted (represented by the appearance of *by* in English or the ablative case in Latin) and sometimes omitted, and an internal argument is promoted to the subject position, while the so-called “passive” constructions in Classical Chinese – which we name the *pseudo-passive* here – involve an agent that is *obligatorily removed* (Mei 2015, p. 287-289).

10.2.1.1 Passivization vs. information packaging

In a language that allows multiple topicalization and focalization, whether omission of the agent is to be analyzed as argument structure alternation instead of information packaging needs discussion. One may want to analyze the promotion of the patient argument to the subject position as pro-drop of the agent plus topicalization of the patient. This analysis essentially eliminates the necessity of the concept of subject in this construction. However, we contend that this analysis is not viable.

First, we note that Classical Chinese has explicitly marked passive constructions like the 见-construction. Their existence cannot be easily captured within an information structure-only framework.

Second, we have both passivization after causativization and causativization after passivization (§ 10.2.1.2), and the semantics and the word order in the two constructions are regularly determined. This strongly indicates that we are encountering valency alternation.

Third, information structure-driven omission of arguments, i.e. pro-drop in Classical Chinese mainly applies to arguments representing *known* entities. Yet the agent omitted in a pseudo-passive construction is frequently *unknown*.

10.2.1.2 Relation with causative constructions

It is not frequent but not impossible to apply pseudo-passivization to a causative construction (Mei 2015, p. 283). Suppose we have a bivalence causative construction. If pseudo-passivization is applied to it, we are to suppress the external argument and let the internal argument to be the subject. But such a bivalence causative construction usually has a CAUSE-BECOME structure, and removing the causer leaves us a clause that looks just like a BECOME clause. So there are two competing analyses, and since usually if a verb root is lexically licensed to head a CAUSE-BECOME clause, then its usage in a BECOME clause is also in the lexicon, the simpler BECOME analysis is preferred. However, where this preference is eliminated, pseudo-passivization of a causative construction is possible (Mei 2015, pp. 284,370-372; § 9.2.2.2).

Given the strict conditions of pseudo-passivization *after* causativization, alternate application of pseudo-passivization and causativization that results in a PASS-CAUSE-PASS-… argument structure is not attested, although such a combination is not semantically unfeasible (e.g. consider ‘someone is made to be made to do sth.’), and is occasionally attested, probably in a playful way, in living languages like modern

Japanese (8). This indicates that the argument structure in Classical Chinese is subject to considerable lexical specification and valency alternation constructions are not completely productive. In this particular case of the impossibility of a CAUS-PASS-CAUS chain, what is revealed probably also includes the decay of productive voice affixes in Old Chinese.

(8) 見つかる と 削除させられさせられる
mitsukar-u to sakujo-sase-rare-sase-rare-ru
be.found-PLAIN.NONPAST COND delete-CAUS-PASS-CAUS-PASS-PLAIN.NONPAST
'If it is discovered, someone will be made to be made to delete it.' (Modern Japanese, from Internet.)

10.2.1.3 The case with multiple internal arguments

When the pseudo-passive construction is applied to double object clauses, it is the *recipient* that is promoted to the subject position (9) (Mei 2015, p. 421). On the other hand, when the pseudo-passive construction is applied to the corresponding prepositional argument construction, it is the *theme* that is promoted to the subject position (10). These phenomena establish a hierarchy of *pivotality* of arguments (§ 2.2.5.2).

(9) 诸侯， 赐 弓 矢 然后 征
zhū-hóu cì gōng shǐ rán-hòu zhēng
various-lord grant bow arrow so-after campaign
'The feudal lords are granted bows and arrows and only then undertake military campaigns.'

(10) 药 言 先 献 于 贵， 然后 闻 于 卑
yào yán xiān xiàn yú guì rán-hòu wén yú bēi
medicinal word first offer to high.ranking so-after hear among low.ranking
'Remedial advice is first offered to those of high rank, and then heard among those of low rank.'

TODO: 杀御叔

10.2.2 Tropative

Tropative is a construction which attaches an experiencer to a BE argument structure, with the meaning being '*A* consider *B* to be ...' (Mei 2015, pp. 413-414) The Classical Chinese tropative is actually not limited to stative verbs: it also applies to nouns. It is however not likely that this construction comes from transformation of the nominal predicate construction (§ 2.2.1). The main difference is that in the nominal predicate construction, the predicate is a noun *phrase*, but the tropative construction never takes a nominal predicate as input.

Chapter 11

Nominal predication

Box 11.1: The possessive nominal predicate construction

It seems the predicate in the possessive nominal predicate construction can never be a bare noun without any modification. The modification can be a numeral or an attributive.

(1) 王　六 军，大　国　三　军
wáng liù jūn　dà　guó　sān　jūn
king six army large state three army
'The king has six armies; a large state has three armies.'

(2) 秦王　[为 人]Box 20.1，蜂　准，长　目，挚鸟　膺，豺
Qín-wáng wéi rén　fēng zhǔn cháng mù　zhì-niǎo　yīng　chái
NAME-king be person　high nose long eye　fierce-bird breast jackal
声，少　恩　而　虎狼　心
shēng shǎo ēn　ér　hǔ-láng　xīn
voice little kindness CONJ tiger-wolf heart
'The king of Qin, as a person, has a high nose, elongated eyes, a hawk's breast, and a jackal's voice; he shows little kindness and has the heart of a tiger or wolf.'

Another problem is that the 者-也 construction seems to be incompatible with the possessive nominal predicate.

Chapter 12

Tense, aspect, modality

12.1 Time adverbs

It is probably more appropriate to recognize these TAM markers as (semi-)lexical adverbs instead of auxiliaries or particles, because the class of TAM markers seems to be open during the Classical period. We believe that they have been incorporated into a grammaticalized TAM system, because depending on their meanings, they have a fixed linear order, consistently with the order of TAM markers observed cross-linguistically (§ 2.2.3). For instance, from (1), we observe that the epistemic adverb 或 appears before the future adverb 將. Still, multiple time adverbs are rare in truly Classical texts, and whether an adverb is a part of the TAM system or a temporal peripheral argument like *at that time* sometimes has to be determined by its semantics.

Box 12.1: List of time adverbs

list

- Are some of them peripheral arguments?

12.1.1 Subject-oriented adverbs

The pre-verbal particle 或 seems to be an epistemic modality marker. In (1), we see the coexistence of the epistemic 或 and a future adverb 將.

(1) 或 將 豐 之，不 亦 難 乎
huò jiāng fēng zhī bù yì nán hū
perhaps FUT enrich 3 NEG also difficult SFP
'Perhaps [Heaven] will enrich him; would that not make things difficult?'

必 indicates the speaker's strong confidence towards the statement being made; similar adverbs include 固, 實, and 誠 (Li 2004, pp. 376-377). These adverbs also appear before the future 將, further confirming their status as epistemic modality or evidentiality markers.

(2) 知 惠 之 必 將 至 也
zhī huì zhī bì jiāng zhì yě
know favor GEN certainly FUT arrive SFP
'[He] knows that the favor will certainly come.'

(3) 諾，吾固將圖之
nuò wú gù jiāng tú zhī
yes 1 certainly FUT plan 3
'Yes, I shall certainly consider it.'

We can further observe both the linear order 固 > 必 and the reverse.

(4) 固必通乎性命之情者
gù bì tōng hū xìng mìng zhī qíng zhě
indeed necessarily comprehend concerning nature destiny GEN reality REL
'one who indeed necessarily understands the realities of nature and destiny'

(5) 將欲弱之，必固強之
jiāng yù ruò zhī bì gù qiáng zhī
FUT want weaken 3 necessarily first strengthen 3
'If one wishes to weaken it, one must first strengthen it.'

12.1.2 The tense adverbs

嘗 seems to be a marker of the simple past.

The concept of future events can be expressed by 將, but it is not compatible with 嘗. On the other hand, in English, we have *would do sth.* or even *would have done sth.* The incompatibility of 嘗 and 將 is evidence for the syntactic distinction between how future is expressed in English and in Classical Chinese: in English, we have both a past/non-past distinction and a future/non-future distinction, beyond the standard Reichenbach system (Vikner 1985), while in Classical Chinese, the past future tense is non-existent and we have a tripartite past/present (related to 矣; § 12.3.1)/future distinction.

12.1.3 The modality adverbs

(6) 又將能忍吾子乎
yòu jiāng néng rěn wú zǐ hū
then FUT can bear.to.harm 1.GEN son SFP
'Then could he bear to harm my son?'

12.2 (Semi-)auxiliaries

In English, it is possible to have both *want to be able to do sth.* and *be able to want to do sth.*[1] A corpus search reveals that 可 'should, ought to' always appears before clearly lexical verbs, while the reverse order is not attested.

[1]This may sound awkward, but it makes sense in certain contexts, like *you have to be a special kind of nerd to be able to want to drive 28 hours for a video game.*

(7) 見 其 可 欲 也，則 必 前 後 慮 其 可
jiàn qí kě yù yě zé bì qián hòu lǜ qí kě
see 3.GEN can desire SFP then necessarily before after consider 3.GEN can
惡 也 者
wù yě zhě
hate SFP SFP

'When one sees what is desirable about something, one must consider, beforehand and afterward, what may be objectionable about it.'

Further, consider 罪當可赦: 當 > 可.

12.3 The role of sentence final particles

12.3.1 矣 and the tense system

The appearance of the sentence final particle 矣 seems to have a tense effect. When 矣 appears and there are no other TAM markers, the relation between the event time and the speech time is still largely arbitrary, but the event always seems to have some relevance to the speech time: the meaning of English *somebody did something* or *somebody had done something* is not admissible. It seems that these observations can be summarized as follows: the grammar of Classical Chinese does (although implicitly) recognize Reichenbach's speech time, reference time and action time, and when 矣 appears, the latter two are identical, hence prohibiting the simple past meaning (Mei 2015, pp. 437-446). However, after considering its combinations with other markers, we do not consider 矣 itself to be a tense marker (§ 14.1.3).

Chapter 13

Negation

(1) 我 未 之 能 易 也
wǒ wèi zhī néng yì yě
1 not.yet 3 can change SFP
'I have not yet been able to change it.' (禮記・檀弓上)

Chapter 14

Sentence final particles

Grammatical categories marked by sentence final particles in Classical Chinese include (a) the illocutionary force (interrogative in 13), (b) TAM, especially the so-called sentential aspect, and (c) what is traditionally known as *judgement*, which more or less involves the speaker's commitment towards the faithfulness of the clause (4, § 14.1).

The analysis above is largely based on Li (2004). We take the term *judgement particles* from Li (2004, p. 220), who distinguishes 决断词 'judgement particles' from 语气词 'mood particles'. He observes that

> [the two] have points of resemblance: they are all frequently attached to the end of the whole clause [...] however, judgement particles and mood particles still have considerable differences [...] judgement particles [...] usually are able to turn situation, action, changes into a subjective reception.

He further points out that the judgement particles frequently correlate with the internal structure of clauses, which he denotes as 句型 'clause (structural) patterns' as opposed to 句类 'clause types', i.e. the illocutionary force, marked by the real 语气词 'mood particles'.[1] He thus concludes that the judgement particles and what he calls mood particles (i.e. the illocutionary force particles) are different grammatical systems.

We agree with his analysis. We want to further add that certain particles, like 矣, appear to have TAM functions as well, though what they precisely represent is debatable. A clear line cannot always be drawn between TAM particles and judgement particles. Thus, we retain the bipartite division of particles of Li (2004).

(1) 温 故 而 知 新，可 以 为 师 矣
wēn gù ér zhī xīn kě yǐ wéi shī yǐ
review old CONJ know new can thereby be teacher SFP
'By reviewing what is old and learning what is new, one can become a teacher.'

14.0.1 Sentence final particles and coordination

It seems a sentence final particle can be shared by two conjuncts.

[1] However, Li (2004, p. 58) appears to consider all nominal sentences somehow subjective (which he names 论断句).

(2) 虎 者， 戾 虫； 人 者， 甘 饵 也
hǔ zhě lì chóng rén zhě gān ěr yě
tiger TOPIC fierce creature person TOPIC sweet bait SFP
'The tiger is a fierce creature; a human is tasty bait.'

In early texts, a sentence final particle can be inserted after the main verb.

(3) 巧 言 令 色， 鲜 矣 仁
qiǎo yán lìng sè xiǎn yǐ rén
clever speech pleasing expression scarce SFP benevolence
'Clever speech and a pleasing expression seldom go with benevolence.'

14.1 Judgement and TAM particles

14.1.1 The particle 也

The sentence final particle 也 indicates that the speaker is making a judgement. The clause may be verbal (4) or nominal. This typically means that the clause is more or less "static", and

(4) 人 而 无 信， 不 知 其 可 也
rén ér wú xìn bù zhī qí kě yě
person CONJ NEG.exist trustworthiness NEG know 3.GEN acceptable SFP
'If a person is untrustworthy, I do not know how that can be acceptable.'

14.1.1.1 TAM values

14.1.2 The particle 矣

14.1.3 TAM values

The particle 矣 is contrasted with 也 in the example (5). From the example it can be immediately see that the sentence final particle 矣 is used in a clause describing a concrete event, while 也 is for clauses describing something happening regularly, not a single concrete event (Mei 2015, pp. 443-445).

Its function is close to the sentence final 了 in Modern Standard Mandarin (Paul 2014b; Pan 2021): it is a wide-scope category indicating that the reported situation of affair is new (Pulleyblank 1995, p. 116), and has been compared with inchoativity/evidentiality (Tsai 2008) and is comparable to the iamitive or the new situation marker in Southeastern Asian languages (Mazzitelli 2025).

(5) 余 助 苗 长 矣
yú zhù miáo zhǎng yǐ
1 help seedling grow SFP
'I have helped the seedlings grow.'

(6) 將 发 命 也
jiāng fā mìng yě
FUT issue order SFP
'[He] is about to issue an order.'

The particles however are different from prototypical aspect markers in several aspects.

First, that 矣 appears predominantly in direct quotations and not ordinary complement clauses strongly indicates that the scope of the particle is wider than most TAM categories and likely involves conversational and speaker-oriented categories.[2]

Second, it seems that 矣 and 也 can be shared by two coordinated conjuncts with different subjects, which is rather unusual for an aspect marker (2). A prototypical TAM category is in relation with a nucleus clause, while 矣 and 也 are in relation with a sentencial clause, i.e. an arbitrarily complex clause that is one utterance in a conversational context.

We therefore disagree with Mei (2015)'s analysis.

(7) 亦 各 言 其 志 也 已 矣
yì gè yán qí zhì yě yǐ yǐ
merely each speak 3.GEN aspiration SFP SFP SFP
'Each is simply stating his aspirations, that is all.'

Box 14.1: Alternative analysis

First, 矣 is compatible with both 將 and 嘗 (8, 9; § 12.1.2), which probably means it is not directly related to the tense system.

(8) 難 將 至 矣
nàn jiāng zhì yǐ
disaster FUT arrive SFP
'Disaster is about to arrive.'

[2] This is consistent with the usual analysis of sentence final particles in modern Standard Mandarin (Paul 2014b; Pan 2021).

On the other hand, Zhu (2009, p. 233) acknowledges the widespread analysis that sentence final particles are in relation with the whole sentence, not the nucleus clause, but insists that certain sentence final particles are a part of the predicate. Yet no convincing argument is provided. Among the three distributional classes he recognizes in Mandarin, two (marking the interrogative/imperative force, and attitude of the speaker) are uncontroversially attached to *sentences* and not nucleus clauses. The remaining class, which is called the "tense" class by Zhu (2009) and is structurally the innermost, resembles the class of 矣 and 也 discussed here, seems to be forbidden in most embedded clauses (Deng 2010), just like the other two classes do. Therefore all the three classes of Mandarin sentence final particles described in Zhu (2009) are indeed in relation with the sentence and not the nucleus clause, which is consistent with the structural status of sentence final particles in Classical Chinese.

We also note that it is not completely impossible for two independent nucleus clauses to share one TAM marker. In Japhug, for example, a series of nucleus clauses with different subjects can be coordinated with the TAM categories being marked at the end of the compound clause (Jacques 2021, pp. 1090-1091). However, Japhug lacks clear clause-level subject (Jacques 2021, § 2.5.3), and therefore coordinated nucleus clauses with a shared TAM marker but different "subjects" is less strange in Japhug than it is in Classical Chinese: in the latter, we have a well-defined clausal pivot grammatical relation (§ 2.2.5.3) whose scope is over all TAM categories, making clauses sharing the TAM marker but not subjects rather unusual, but in the former this is probably not the case.

Mei (2015, pp. 443-445) relies solely on semantic criteria. Although we do not believe his analysis of 也 and 矣 as tense markers is completely correct, the two clearly have non-trivial interaction with TAM categories, a phenomenon also observed in Mandarin, where the lowest sentence final particle has access to TAM categories of the nucleus clause (Paul 2014a, p. 258).

(9) 牛山 之 木 嘗 美 矣
Niú-shān zhī mù cháng měi yǐ
NAME-mountain GEN tree once beautiful SFP
'The trees on Mount Niu were once beautiful.'

Second, 矣 seems to have a weak exclamative function. For example, in (10), the speaker has something to say but has waited for quite a while for a change to say it. In (11), Confucius is impressed by how well cultivated 韶乐 '*Shao* music' is.

(10) 吾 欲 言 之 久 矣
wú yù yán zhī jiǔ yǐ
1 want say 3 long SFP
'I have wanted to say this for a long time.'

(11) 尽 美 矣
jìn měi yǐ
entirely beautiful SFP
'[It] is perfectly beautiful!'

We therefore conclude that 矣 is not a tense marker, and instead analyze it as a "sentential aspect" marker, which informs the listener that a new piece of information arrives. This is consistent with the function of the sentence final 了 in modern Standard Chinese.

Box 14.2: Alternative analysis

In Ch. 11 of Mei (2015), it is claimed that 矣 is a prototypical tense marker. We agree with Mei (2015) in that Classical Chinese has a (rather weak) tense system, and we basically reiterate his arguments above. His arguments for the claim that 矣 itself is a tense marker however we do not consider valid. On p. 447, he argues that 矣 appears in imperative sentences and therefore cannot be related to aspectual categories. But the category of tense also rarely appears in imperative sentences. On the other hand, the sentential aspect category is essentially a discourse-level category, and has no inherent conflict with the imperative speech act. On p. 448, he notes that in coordination, when 矣 appears at the end of the first clause, it sets up a reference time for the following clauses. Yet he states clearly that the reference time is actually the event time of the first clause, whose relation with the speech time is *not* specified by 矣.

On the other hand, if we accept the analysis that 矣 hints at new information that the speaker wants to express to the listener at the speech time, then we can easily understand why 矣, because a generic fact like *the Earth goes around the Sun* rarely is new information, and a piece of new information usually involves a specific situation at a specific time, which is inevitably compared with the speech time, and hence the tense system. The fact that 矣 is often related to the present tense or the present perfect tense may simply be because of the absence of other TAM markers: cf. (8, 9).

Other sentence final particles, like 也, do not have this effect: hence an alternation between narrations about events and descriptions on what these events mean on the battlefield (Mei 2015, pp. 444-445).

Box 14.3: The relation between topic and sentential aspect

A question is whether we can claim that the category of the sentential aspect is somehow higher than the category of topic. If so, then the sentential aspect belongs to CP. See

Mandarin peripheral construals at the syntax-discourse interface, page 30. 是北京昨天下雪了：the focus can be on 北京 and 昨天下雪了 modifies 北京, or 了 modifies

Mei (2015, p. 445) and Pulleyblank (1995, p. 19) both claim that form 也矣 is impossible. It turns out that it is rare, but not non-existent. It however has to be admitted that 也已矣 is much more frequent than 也矣. Pulleyblank (1995, p. 20) analyzes 已 (when used as a sentence final particle) as a fusion of 也 and 矣.

14.1.4 Combinatorial possibilities

(12) 是 善 惡 之 分 也 矣
shì shàn è zhī fēn yě yǐ
this good evil GEN distinction SFP SFP
'This is the distinction between good and evil.'

Box 14.4: Other combinations

寡人之於國也，盡心焉耳矣

14.2 Discourse particles

14.2.1 Question particles

(13) 大 车 无 輗， 小 车 无 軏， 其 何
dà chē wú ní xiǎo chē wú yuè qí hé
large cart NEG.exist yoke.pin small carriage NEG.exist yoke.pin MOD what
以 行 之 哉
yǐ xíng zhī zāi
with move 3 SFP
'If a large cart has no yoke-pin and a small carriage has no yoke-pin, how could one drive them?'

Chapter 15

Clause types

Box 15.1: Clause types

In Mei (2015, p. 131), he classifies clauses into 说明句, 描写句, and 叙事句. The classification is comparable to that given in http://area.hcjh.tn.edu.tw/noise/hcjh-ca/4-b.htm#0303. Mei doesn't mention on which basis he makes this distinction. In the latter source, it seems the distinction is made based on the type of the predicate. Thus a 描写句 is a stative (adjectival?) clause, and a 判断句 is a nominal predicate construction, and a 叙事句 is a verbal predicate construction that is not a 描写句. So what does Mei mean by 说明句? The term appears in Li (2004) as well.

We can go to places where he mentions the term. p. 445: 矣 is for 叙事, and 也 is for 说明. pp. 264-265: 事件句 (叙事句) 和非事件句（描写句和说明句）The distinction is also mentioned in http://paper.wenweipo.com/2018/02/14/ED1802140024.htm

So it's related to the event structure. We need to know where the event structure resides in the vP-TP-CP hierarchy. Particularly, we need to identify *where* the category of this distinction lies. I think probably that's related to the aspect: consider the distinction between a habitual clause and a prototypical "event" clause.

The distinction has syntactic significance. We note that certain topicalization constructions seem to be only compatible with 说明句 (Box 20.1).

Box 15.2: Adjectival predication of a non-finite clause?

Consider 其為人也，發憤忘食，樂以忘憂，不知老之將至云爾: the subject of 發憤忘食，樂以忘憂，不知老之將至云爾 is clearly 其 i.e. Confucius. But then consider the fact that the subject *cannot* appear after 也. This seems to indicate that we are dealing with some sort of subject sharing construction.

Also, consider 为文斐然可观矣: here 斐然客观 semantically modifies 文?

See also Box 20.1.

15.1 也

We also note that the 者…也…framework is not limited to topicalization of nominal predicate clauses (i.e. "judgmental clauses"). For instance, we have (1), in which the sole argument in a existential clause is topicalized, and the comment receives 也 as its sentence final particle.

(1) 然 而 不 王 者，未 之 有 也
rán ér bù wàng zhě wèi zhī yǒu yě
so CONJ NEG reign REL NEG.yet 3 exist SFP
'There has never been anyone who did this and yet failed to become king.'

Chapter 16

Information structure

Verbal judgemental clause

(1) 单父 人 吕公…… 吕公 者， 好 相 人，
Shànfǔ rén Lǚ-gōng Lǚ-gōng zhě hào xiàng rén
NAME person NAME-lord NAME-lord TOPIC like read.physiognomy person
……

‘Lord Lü, a man from Shanfu, …As for Lord Lü, he liked reading people’s physiognomies, …’

Box 16.1: Fronted SFP

甚矣汝之不惠 can be analyzed in multiple ways. We may assume that (38) is essentially some sort of cleft construction, in which the subject is first separated from the rest of the clause and then the rest of the clause is focalized. Or we can analyze 矣 as a TAM marker, and not a marker from the CP layer. Or maybe we can argue that markers from CP layers are morphologically verbal and have to be attached to either the main verb or the verb phrase. Which analysis works best depends on whether they are consistent with other phenomena.

- Can we prove that the SFPs are very “high-level” and are above the topic layer? For example, can two clauses with different subjects share a SFP?
- Semantically do 矣 carry TAM meanings?

(2) 以 五十 步 笑 百 步，则 [何 如]$_{\text{SOV interrogative clause}}$
yǐ wǔ-shí bù xiào bǎi bù zé hé rú
with five-ten pace laugh.at hundred pace then what be.like
‘What would it be like to use [a retreat of] fifty paces to laugh at [a retreat of] a hundred paces?’ (孟子・梁惠王上)

We also note that topicalization can happen multiple times (3).

(3) [萬 乘 之 國]$_{\text{topic:NP}_i}$， [弑 [其]$_i$ 君 者]，
wàn shèng zhī guó shì qí jūn zhě
ten.thousand chariot GEN state murder.ruler 3.GEN lord REL
必 千 乘 之 家
bì qiān shèng zhī jiā
necessarily thousand chariot GEN household

‘In a state of ten thousand chariots, the one who murders its ruler will surely be a household of a thousand chariots.’ (孟子・梁惠王上)

Chapter 17

Relative clauses

者 The particle 者 most frequently appears as a relativizer, a complementizer, or in the *zhe-ye* construction. The three functions can be uniformly analyzed as the function of a low-level determiner (Aldridge 2009).

(1) 马 之 死 者
mǎ zhī sǐ zhě
horse GEN die REL
'the horses that died'

Box 17.1: *zhi-zhe* construction

The structure of the 之-者 construction may cause some debates. It can be analyzed as a possessive construction on top of a fused relative clause construction and translated word-to-word into English as '[those who go one thousand miles] of horses'. An interesting question then is whether we have any other instances of the N 之 V 者 construction where the relation between N and [V 者] is prototypically possessive. It seems this is indeed possible: 城北徐公，齐国之美丽者也.

Under this analysis, 楚人有吹箫于市者 is composed by applying the external possessive construction to 楚人之吹箫于市者

One fact (or is it really a fact?) supporting the determinative analysis of 之-者 is that the construction seems to be unable to receive a further determiner: * 此马之千里者. The sequence 此马之千里者 does appear but it is almost always a nominal predication construction.

Chapter 18

Clause linking

The marker 乃 is also available as a marker of the consequence clause (Mei 2015, p. 87).

Other markers for the condition clause are also available (Mei 2015, Ch. 3). We note that the marker 若 is able to appear *after* the subject of the condition clause (Mei 2015, p. 94).

Sometimes the marker 则 is dropped (1) but putting it back should never render a sentence ungrammatical (Mei 2015, p. 86).

(1) [杀 女]$_{\text{condition}}$，[我 伐 之]$_{\text{consequence}}$
shā rǔ wǒ fá zhī
kill 2 1 attack 3
'If [he] kills you, I shall attack him.'

condition (2) or temporal clause (3) can be a gerundive one (§ 2.2.7.2).

(2) 我 之 不 德， 民 將 弃 我
wǒ zhī bù dé mín jiāng qì wǒ
1 GEN NEG virtuous people FUT abandon 1
'If I am not virtuous, the people will abandon me.'

(3) 臣 之 壯 也， 犹 不 如 人
chén zhī zhuàng yě yóu bù rú rén
minister GEN vigorous SFP still NEG match person
'Even when I was in my prime, I was not as capable as others.'

In some condition clauses, the marker 而, instead of the otherwise genitive marker 之, appears between the subject and the predicate, forming a clause type that is not gerundive and only appears as an irrealis condition clause (Mei 2015, pp. 100-102).

Chapter 19

Genres and formulae

19.1 Idioms

(1) 非 请 勿 入
fēi qǐng wù rù
NEG invite NEG enter
'Do not enter unless invited.'

19.2 Classical-like formal documents

19.2.1 Background

Box 19.1: Background

- 各代公文
- 晚清民国

19.2.2 Pseudo-Classical documents

The text shown in (2) is an article from the Chinese version of the Geneva Conventions, an example of the pseudo-Classical Chinese used in official documents. The text is Classical-like when we inspect the grammatical markers appearing in it: 之 in place of 的 as a relative clause marker and also a possessive marker, 是 in place of 这 as a demonstrative, and 彼等 in place of 他们 as a third person pronoun. Yet several features can be immediately noticed that deviate from the Classical standard.

(2) 战俘 不 得 放弃 本 公约 或 上 条 所
zhàn-fú bù dé fàng-qì běn gōngyuē huò shàng tiáo suǒ
war-captive NEG may relinquish this convention or preceding article NMLZ
述 之 特别 协定—— 如 其 订 有 是 项 协定—— 所
shù zhī tèbié xiédìng rú qí dìng yǒu shì xiàng xiédìng suǒ
state GEN special agreement if 3 conclude EXIST this CL agreement NMLZ
赋予 彼等 权利 之 一 部 或 全部
fùyǔ bǐ-děng quánlì zhī yī bù huò quán-bù
confer that-PL right GEN one part or all-part

‘Prisoners of war may not renounce any or all of the rights conferred on them by this Convention or by the special agreements referred to in the preceding article, if such agreements have been concluded.’

First, polysyllabic content words are prevalent. The verb 放弃 ‘to give up’ is not unheard of in Classical texts (cf. 3), but it is usually analyzable as a verb-level coordination of 放 ‘to put down’ and 弃 ‘to abandon’. 放弃 as a whole with the meaning of giving up something abstract is a feature of Modern Standard Mandarin, where 弃 is no longer acceptable when used alone as a verb, which is indeed the case in the Chinese version of the Geneva Conventions.

Nouns like 战俘 or 协定 are never attested in common Classical texts. Their structures can be explained with the Classical grammar. Still, the fact that roots within them do not seem to appear freely suggests that the translator likely had the lexicon of Mandarin in their mind.

(3) 放 棄 詩 書， 極 意 聲 色
fàng qì shī shū jí yì shēng sè
put.aside abandon Odes Documents exhaust desire music beauty
‘[They] put aside and abandoned the Odes and Documents, indulging their desires in music and sensual pleasure.’

(TODO: add the second pseudo-Classical feature.)

It is not impossible for Classical grammar to appear in a pseudo-Classical document, though.

(4) 有 组织 之 抵抗 运动 人员 之 在 其
yǒu zǔzhī zhī dǐkàng yùndòng rényuán zhī zài qí
have organization GEN resistance movement personnel GEN at 3.GEN
本国 领土 内外 活动 者
běn-guó lǐngtǔ nèi-wài huódòng zhě
own-country territory inside-outside operate REL
‘members of organized resistance movements operating inside or outside their own country’s territory’

19.2.3 Formulae

Box 19.2: Formulae

- 此令

19.3 Traditional letters

Box 19.3: Formulae

- 钧鉴
- 谨呈
- 母亲大人膝下，敬禀者，日前寄上海婴照片一张，想已收到

The last example can be parsed with Classical grammar: ‘ At the knees of my mother, what is respectively reported below is: …’ Yet the formula X 鉴 is kind of hard to parse. Further, consider 道席 in 某公道席: it is similar to an epithet, or something like *your Majesty*.

So we have three types of 提称语: one is related to the request that the recipient reads the letter (the X 鉴 formula), one is related to the profession or status of the recipient (like 道席), and the other is essentially a locative expression (膝下). The origin of these formulae remains unknown.

Chapter 20

Discussions on quirky examples

Box 20.1: A 之于 B 也

(1) 寡人 之 于 国 也，尽 心 焉 耳 矣
guǎ-rén zhī yú guó yě jìn xīn yān ěr yǐ
humble-person GEN toward state SFP exhaust heart therein only SFP
'As for my dealings with the state, I simply devote my whole heart to them.'

The original structure is probably 寡人于国尽心. The fact that 寡人 and 于国 are subtracted from the nucleus clause is rather unusual. And similar examples exist. Consider 史记夫乐之与音，相近而不同. The original structure is 乐与音，相近而不同, and the insertion of 之 is rather strange.

(2) 其 为 人 也，发 愤 忘 食，乐 以 忘 忧，不
qí wéi rén yě fā fèn wàng shí lè yǐ wàng yōu bù
3.GEN be person SFP arouse zeal forget eat rejoice CONJ forget worry NEG
知 老 之 將 至 云 尔
zhī lǎo zhī jiāng zhì yún ěr
know old.age GEN FUT arrive say thus
'As a person, he is so zealous that he forgets to eat, and so joyful that he forgets his worries, unaware that old age is approaching—that is all.'

A 之于 B 也 (or A 之为 B 也)，predicate, or A 为 B 也, predicate. The structure seems to be parallel to the English *I, as a concerned citizen, want to emphasize that …*, where *as a concerned citizen* obligatorily modifies the subject *I*. 其为人也 here seems to be a *frame*, somehow comparable to the "global" temporal or locational phrase. Another issue is that the sentence seems to be unable to represent a specific event: * 昨日，孔子为人也，发愤忘食, while we have *yesterday, as a concerned citizen, I …*

(3) 水 之 积 也 不 厚，则 其 负 大 舟 也 无
shuǐ zhī jī yě bù hòu zé qí fù dà zhōu yě wú
water GEN accumulate SFP NEG deep then 3.GEN carry large boat SFP NEG.exist
力。
lì
strength
'If the accumulated water is not deep, it has no strength to carry a large boat.'

(4) [良人 者]$_{\text{subject}}$ [所 仰望 而 终 身]$_{\text{predicate}}$ 也
liáng-rén zhě suǒ yǎng-wàng ér zhōng shēn yě
good-person TOPIC NMLZ look.up-gaze CONJ finish life SFP
'A husband is someone one looks up to and depends on for the rest of one's life.'

It seems subjects of ordinary verbal clauses cannot be topicalized (5), but if the verb phrase is emphasized, topicalization is possible.

(5) a. 三 王 既 以 定 法度
sān wáng jì yǐ dìng fǎ-dù
three king already with establish law-standard
'The three great kings had already established social institutions by [it].'

b. * 三 王， 既 以 定 法度
sān wáng jì yǐ dìng fǎ-dù
three king already with establish law-standard
'As for the three great kings, [they] had already established social institutions by [it].'

(6) 秦， 虎狼 之 国， 不 可 信， 不 如 毋 行
Qín hǔ-láng zhī guó bù kě xìn bù rú wú xíng
NAME tiger-wolf GEN state NEG can trust NEG be.as.good.as NEG go
'Qin, a state of tigers and wolves, cannot be trusted; it would be better not to go.'

What's the structure of the sentence below?

(7) 將 者， 欲 伐 而 未 成， 见 其 臣 尚
jiāng zhě yù fá ér wèi chéng xiàn qí chén shàng
FUT TOPIC want attack CONJ NEG.yet accomplish show 3.GEN minister still
可 以 谏， 而 季氏 尚 可 以 止 也
kě yǐ jiàn ér Jì-shì shàng kě yǐ zhǐ yě
can thereby remonstrate CONJ NAME-lineage still can thereby stop SFP
"'About to" means that they wished to attack but had not yet done so; it shows that his ministers could still remonstrate and that the Ji family could still be stopped.'

In the example below, the subject of the second clause is 'other people', and yet it seems the three clauses share a topic (i.e. 人).

(8) 人 而 不 仁， 疾 之 已 甚， 亂 也
rén ér bù rén jí zhī yǐ shèn luàn yě
person CONJ NEG benevolent hate 3 too extreme disorder SFP
'If a person is not benevolent and others hate him too intensely, disorder results.'

What's the structure of the sentence below? 'But as for Qiu, is it about state affairs?'

(9) 唯 求 则 非 邦 也 與
wéi Qiú zé fēi bāng yě yú
FOC NAME then NEG state SFP SFP
'But as for Qiu, was he not speaking about a state?'

The sentence below seems to be due to omission: 宗庙会同，非诸侯而何为之？

(10) [宗庙 会 同]$_i$， 非 诸侯 而 何 —$_{\text{main verb}}$ —$_i$
zōng-miào huì tóng fēi zhū-hóu ér hé
ancestor-temple meeting assembly NEG various-lord CONJ what
'As for ancestral-temple ceremonies and diplomatic assemblies, who other than the feudal lords would take part in them?'

(11) 赤 也 为 之 小， 孰 能 为 之 大
Chì yě wéi zhī xiǎo shú néng wéi zhī dà
NAME SFP act.as 3 small who can act.as 3 great
'If Chi were to serve as a junior official there, who could serve as the senior one?'

(12) 其 为 人 也 孝 弟， 而 好 犯 上 者
qí wéi rén yě xiào tì ér hào fàn shàng zhě
3.GEN be person SFP filial respectful.to.elders CONJ like offend superior REL
'those who are filial and respectful to their elders, yet like to offend their superiors'

(13) 且 夫 水 之 积 也 不 厚，则 其 负 大
qiě fú shuǐ zhī jī yě bù hòu zé qí fù dà
moreover PTCL water GEN accumulate SFP NEG deep then 3.GEN carry large
舟 也 无 力
zhōu yě wú lì
boat SFP NEG.exist strength
'Moreover, if the accumulated water is not deep, it has no strength to carry a large boat.'

(14) (人) 如 礼 何
rén rú lǐ hé
person deal.with ritual how
'What can [a person] do with ritual?'

(15) 如 之 奈何
rú zhī nài-hé
deal.with 3 deal.with-how
'What can be done about it?'

(16) 父母 唯 其 疾 之 忧
fù-mǔ wéi qí jí zhī yōu
father-mother only 3.GEN illness OBJ worry.about
'Let your parents worry only about your illness.'

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Appendix A

Theoretical framework

The theoretical framework of this work is strongly inspired by Distributed Morphology (for an introduction see Siddiqi (2009)) plus Cartographic Syntax (e.g. Cinque (1999)), although I intentionally choose to reuse the terminology in descriptive grammars. We will see that the structuralist tradition of Chinese linguistics agrees incredibly well with the generative framework.

A.1 Lexicon, grammar, wordhood

The architecture of a language is assumed to be in line with the basic assumptions of Distributed Morphology. The so-called "lexicon" consists of a list of roots (List A), the phonological forms of the roots and formatives, and whether and how some of the roots are only realizable in combination with grammatical formatives (List B), and grammatical objects of arbitrary sizes with custom meanings (List C).

We note that List B has functionalities beyond specifying the concrete forms of the morphemes. By dictating that some root-formative complexes spellout while the individual morphemes do not, List B is able to dictate the grammatical properties of these forms: thus a transitive verb stem is a root-transitivity head complex in List B, which does *not* allow the bare root to spell out without the transitivity (see Siddiqi 2009 for details). Therefore, we should say that List B dictates grammatical lexical specification i.e. morphosyntactic listedness of grammatical properties of forms, including both how a form looks like in the surface structure, and lexically specified grammatical properties, such as subcategorization of valency (cf. § 9.2.2.2). Because morphological words (i.e. grammatical words) are built by applying vocabulary insertion to complex heads, morphosyntactic listedness is related to wordhood (§ A.5). We note that this does not entail the necessity of a lexicalist grammatical framework where structure building mechanisms for words and phrases are categorically different (cf. Bruening 2018). Lexical specification may be called (grammatical) lexicalization, but the latter term is much more frequently used to refer to idiomization, a List C phenomenon.

List C dictates idiomization, which may also be known as semantic lexicalization (we avoid using the term *lexicalization* to denote idiomization in this work: it is commonly used to refer to idiomization but idiomization has no direct relation with wordhood). The distinction between (syntactic) lexical specification and (semantic) idiomization is discussed in e.g. McInnerney (2023).

Lexical specification and/or idiomization of a complex structure may eventually lead to the collapse of its internal makeup, causing (*syntactic*) fossilization. An example is the collapse of verbs taking complement clauses into compound verbs. Fossilization may end up producing a single *root*, of which only diachronic analysis is possible. Syntactic fossilization is important in Classical Chinese due to its long history.

Parts of speech labels hold two different meanings in different contexts, as is said in the main text. The noun may just be defined as the head word of a noun phrase, i.e. a grammatical word with an n head, roughly corresponding to a *distributional class* in descriptive linguistics; hence notions of "verbs used as nouns". Note that what is a head noun or a head verb also has cross-linguistic variances: we may define the head noun as the grammatical word that contains the innermost functional projections of a DP, and obviously what a head noun contains has cross-linguistic variances. In the second definition, people refer to *lexical* form classes in List B that share similar lexicalized properties as parts of speech;[1] note that these List B entries strictly speaking are *not* grammatical words, as they may further undergo regular affixation (as abstract functional heads have their own Vocabulary Insertion forms.)

This second definition of parts of speech sees more cross-linguistic variances. To say that the Latin word *canis* is a noun means that *canis* belongs to a form class in which every member usually appear as the head of an NP, carries an inherent gender feature and a number feature, and that its inflection pattern follows one of Latin nominal declensions. (It's easy to imagine how such a form looks like in Distributed Morphology.) To say that *dog* is an English noun means something else: it means the root *dog* regularly appears in the nominal modification construction (as in *a* $[dog]_{nominal\ (not\ NP)}$ *tag*), which is not attested in Latin.

A.2 High-level constituents; grammatical relations

The analyses in § 2.2 and § 2.3 are inspired by the extended CP and DP structures proposed in Cartography. To avoid confusion caused by technical terms in generative syntax, I intentionally use terms like *sentence*, *nucleus clause* and *argument structure* in place of CP, TP and vP. Similarly we talk about *noun phrases* instead of DPs.

Concepts like SpecTP and SpecvP, in a descriptive context, correspond to concepts like *clausal subject* and *pivot in the argument structure* (§ 2.2.5). Structural positions definable at the vP level we call argument slots below (the much more frequently used term in generative syntax is thematic roles; the correspondence between traditionally semantically motivated thematic role classification and the light verb shell has been extensively discussed, as in the literature on UTAH), and structural positions definable at the TP level we call *grammatical functions*. The "core" of CPs and DPs (i.e. the roots at their centers plus the categorizers) – instead of the grammatical markers – should be known as heads.

In CPs and DPs recurring patterns or *constructions* and *grammatical relations* in them can be observed: subject-predicate and verb-object relations, subordination and coordination structures.[2] The fine details of the light verb shell I replace by concepts like *DO clause* or *CAUSE clause* (§ 2.2.2.3).

Grammatical markers can be distinguished from content roots and words in that grammatical markers are from a closed class and mark grammatical categories and functions in certain constructions, and have no meaning outside these constructions. This often means that forms containing grammatical markers that seemingly semantically make sense are not acceptable (e.g. § 2.2.3).

[1] This obviously begs the question: where does the rigidity of these form classes come from? Why can't, say, a Latin noun have no gender feature? This may be explained by cross-linguistic variances in List A: we may assume that the nominal categorizer is gendered (Kramer 2015). Alternatively this may be explained by a language-specific well-formedness condition. The competition between the two analyses is a part of the wider macroparameter vs. microparameter debate, which we do not touch in this work.

[2] Here the term is used *without* the implication that a construction is somehow understood as a whole and its internal structures should not be further analyzed, contrary to the fundamental hypothesis of e.g. various Construction Grammars.

The correspondence between descriptive notations and theoretical linguistic concepts has been demonstrated in Deng (2010), which shows that Minimalist generative syntax and the constituency-based analysis seen in the Chinese linguistics tradition outlined in § 1.3.3, which appears strongly inspired by American structuralism exemplified in Huddleston and Pullum (2002), are compatible to each other, with the former being a more concise form of the latter and the latter being a logical consequence of the former under certain assumptions.[3] Further, as shown in § A.3, the constituency-based analyses are consistent with the Basic Linguistic Theory (the de facto unified framework in modern linguistic description of underdocumented languages).

The three approaches can be seen as three "representations" of the same grammatical complexity class, and their differences are mostly notational.[4] Which framework to use is determined by the properties of the language. For instance, although the definition of the verb phrase does not alter the grammar system substantially, Dixon's definition works more smoothly for a language with a lot of auxiliaries but rather infrequent subject-sharing coordinations. It turns out that for most constructions, the method in Huddleston and Pullum (2002) is a good choice for Classical Chinese.

We note that we do not consider adjunction an independent structure-building process that is different from Merge (which is often invoked to explain the argument/adjunct distinction). We in fact do not utilize the concept *modifier* as a primitive of our analysis framework at all, because as Cinque has shown, adjectival positions are not interchangeable. The discussions above mean that we should identify far more grammatical relations than those identified in Huddleston and Pullum (2002). For practical reasons, though, this is not always possible as whether two constituents occupy different structural positions is not always possible, and inevitably, we have to cluster grammatical relations and categories with similar structural properties into one. This is what we do in Fig. A.1, in which we use the term *TAM adverb*.

Regarding phenomena typically attributed to the argument/adjunct distinction, discussed in Ch. 2, we distinguish between (a) structural prominence, i.e. the core/peripheral distinction,

[3]Note that Deng (2010), Huddleston and Pullum (2002) and works in § 1.3.3 are all lexicalist, which we have argued is not necessary to account for phenomena purportedly supporting the lexicalist hypothesis. These works are also based on the more traditional X-bar theory: the syntactic differences between the subject and the object, for example, are explained by the difference between specifier and complement. In Cartography, what is recognized as a complement may be a lower specifier: thus [V O]$_{V'}$ is replaced by a "transitivity phrase" [O [TRANS $\sqrt{V}$]$_{TRANS'}$]$_{TRANSP}$ (Siddiqi 2009), where the category-free root is first combined with a transitivity functional head and then combined with the object. The resulting tree diagram is quite like the X-bar scheme before, say, the DP hypothesis: the only difference being that there are many X'-like projections, all of which may appear in other constructions. For instance, the English *nominal* mentioned in Huddleston and Pullum (2002), which is often recognized as N', can actually be a modifier in the DP, as in *a [Fuji apple] tree*. Deng (2010) shows that the subject-predicate structure or the verb-object structure can all be derived from the traditional X-bar scheme: the Cartographic approach pushes this decomposition even further and demonstrates that the traditional X-bar scheme itself can be further decomposed into a series of functional heads and roots.

[4]A more important controversy is the mental status of grammar. Generativism treats grammar as a semi-autonomous component of human cognitive abilities. But it has been argued that grammatical constructions originate from domain-general cognitive abilities, and there is actually no such thing as an autonomous mental grammar. Detailed discussions on this topic are far beyond the scope of this work, and can only be finally settled down with the assistance of neurological studies. Here we just note that currently no comprehensive description of a language has been successfully attempted under this line of thinking. On the contrary, in physicists' terms, the grammatical framework adopted here is at least a good *effective theory*. So-called functional-typological grammars do not differ significantly in the way they carry out analyses from structuralist works.

Another topic is whether natural language-like "grammars" appear in systems that are demonstrably independent to the human neurological linguistic capacity, which may also challenge the domain-specific status of the latter. This is partly discussed in § 1.4.5. Note however that such discussions are largely irrelevant to linguistic description, for reasons mentioned above.

if the terms are taken literally, (b) whether a constituent's presence is lexically specified (by List B; see above), which is considered the most important criterion in Huddleston and Pullum (2002, p. 219), and (c) case marking and agreement (note that a constituent bearing an inherent case is unlikely to participate in verbal agreement, presumably because constituents with inherent cases are not probing targets).

The first two parameters are somehow strongly related to each other, as the way post-syntactic operations work means that List B-specification of the necessity of presence of arguments structually close to the verb root involves less post-syntactic operations than that of arguments structurally far from the verb root (which reside in high specifier positions). The third parameter should be better known as obliqueness, especially considering that some oblique constituents are likely structurally close to the verb root (Mei 2015, p. 404). There is a perception that structurally core arguments are typically not oblique, but it is an arfifact due to terminology: a non-oblique argument in a structurally high position is simply *called* a core argument, even when it is not lexically specified and is added into the clause quite freely. We should expect that constituents bearing structural cases in a clause are limited in number, but this should not be explained by assuming that these constituents are always structually closer to the verb root, as it can simply be explained by the number of available probes that assign structural cases.[5]

A.3 Dependency

Grammars can be written in terms of *dependency relations*, instead of *constituency relations*. The two however are largely equivalent (Boston, Hale, and Kuhlmann 2009). In his Basic Linguistic Theory, Dixon (2009) fervently argues against constituency analysis (and also other aspects of generative syntax) and advocates for a "flat" constituency structure, possibly with the levels of clause and noun phrase only, where the rest of the grammatical information is represented by dependency relations (e.g. Dixon 2009, p. 49). Yet the primacy of noun phrases and clauses in syntax is indeed acknowledged by generative syntax (for instance, they are *phases*; it is also not uncommon to see terms like *CP syntax* or *DP syntax* in generative works).

The binary constituency relations in generative syntax and in American structuralism are not arbitrary and have consequences. For instance, when applied to the argument structure, they are related to properties of arguments in valency alternation (§ 2.2.5.2), and the subject-predicate binary division is directly related to the clausal pivotal status of the subject. These phenomena of course have to be taken into account by Basic Linguistic Theory, and labels like "clausal pivot" and "surface S, deep O" have to be attached to dependency arcs between the verb and the arguments, essentially labeling the "distance" between the two. As a parallel, the ancient Indian grammarian Pāṇini initially proposes a grammatical framework in which arguments are all equal, but later commentators still effectively set up a pivot position in the argument structure.[6]

[5] Or similar mechanisms. Whether abstract Case is needed in accounting for licensing of subject and object is always disputed. See e.g. Sheehan and Wal (2016).

[6] The main difference between Pāṇini's treatment of Sanskrit and modern linguistic theories is that Pāṇini apparently treats all dependency relations equally and there is, for example, no concept of the pivot or the "external argument" of a clause. This is however modified in the commentaries on his *Aṣṭādhyāyī*, which explicitly allow an argument to be promoted to the agent position because of the intentions of the speaker (Keidan 2017). The agent position thus becomes a subject position at least in the argument structure, consistent with modern practices (§ 2.2.5.2).

The Paninian tradition therefore is extremely close to modern linguistic description practice; the most important difference probably is that modern linguistic description, practically, may even be less rigorous than *Aṣṭādhyāyī*, because of possible competing "mind grammars" among speakers with mutual intelligibility or even within the mind of one speaker, and also the fact that a description as detailed as

Other phenomena that can be attributed to binary constituent structures include ordering of tense and aspect auxiliaries and adverbs à la Cinque, attested in languages completely unrelated to the Germanic or Romance languages that have been thoroughly investigated in Cartography (e.g. Grimm 2021, pp. 166-167), and also available clause sizes: it is rare that in a language with tense, aspect and modality marking, there exists a clause type with discourse devices but no tense, aspect and modality marking, which can be explained simply by the fact that CP is built on TP.

A.4 On the term *phrase*

Besides the theoretical problems outlined above, there are also some minor, largely notational inconsistencies between the grammatical theories mentioned here. One such inconsistency is the definition of the *phrase*. Dixon (2009) calls the main verb plus auxiliaries – without any argument – the *verb phrase*. This notion of verb phrase has a generative syntax counterpart: the CP phase. In a CP, arguments are phases themselves, and the tense, aspect and modality categories marked by the auxiliaries are thus in some senses closer to the main verb. On the other hand, the term *verb phrase* generally means the verb plus internal arguments in constituency-based analyses (as in e.g. Danckaert 2017).

A.5 Wordhood

Another problem is the definition of *word*. We need to distinguish phonological wordhood from morphological wordhood (where the boundary of the morphological template is the boundary of the morphological word). Further, since we reject the lexicalist hypothesis, we eseentually deny that the level *word* plays any primitive role in morphosyntax. Any wordhood definition therefore has to be derivative and the possibility of domain phenomena with conflicting boundaries cannot be ruled out.

We first note that it is not possible to define wordhood purely on (abstract) syntactic structures. We can define a word to be a very small constituent: a compound is a rather small constituent in the hierarchy of the structure of the DP, and hence a word. But in this way *sinned* is *not* a syntactic word as it involves a rather high-scope clausal category (i.e. the past tense). For *sinned* to be a syntactic word, syntactic wordhood can be based on Dixon's verb phrase, and inevitably *have been eating* is a syntactic word.

Therefore, (abstract) properties of structure building processes cannot be the only criterion used in defining wordhood.[7]

The closest notion of morphological wordhood, i.e. grammatical wordhood in Distributed Morphology is post-syntactic aggregation of roots and formatives into complex heads, or, more precisely, *maximal* complex heads.[8] In descriptive linguistics, this is related to existence of morphological templates that can't be motivated by purely combinatorial syntax, boundaries of forms in phonology that lack pure prosodic motivation, etc.

Aṣṭādhyāyī requires corpus data whose quality and quantity exceed the capacity of most field linguists.

[7] This is to be contrasted with the criteria distinguishing compounds and nominal complementation in Huddleston and Pullum (2002), which does rely heavily on abstract properties of perceived word-like and phrase-like structure building processes. In theories with syntactic (and not just post-syntactic) complex X^0 heads, the X^0 heads may be regarded as "syntactically defined words", but this essentially allows a word to contain a word as its constituent, and thus this definition still suffers from the boundary problem discussed right above. See also the discussion on X^0 and wordhood in Bruening (2018), which does not distinguish between syntactic and post-syntactic complex X^0.

[8] Whether this process is guided by (the three parts of) the lexicon or by language-specific rules is a part of the macroparameter/microparameter controversy, which is not covered here.

A complex head may have several Vocabulary Insertion exponents, corresponding to multiple morphemes in a grammatical word, which may undergo Local Dislocation processes. Regardless of how many exponents it has, because Vocabulary Insertion is guided by List B, which is the origin of morphosyntactic listedness (§ A.1), in descriptive terms, the presence of lexical specification of grammatical behaviors is strong indication that we're dealing with words.

In many words, only one Vocabulary Insertion exponent contains content roots, and in this case we know that the content (and not functional) entries in List B with unpredictable morphosyntactic behaviors in the List B may thus be known as *lexemes*, from the obvious relation between them and the functional List B entries. This term may be misleading because it often assumes a clear inflection/derivation distinction (see below). Further, List B also contains both irregular word forms like *feet* and half-words (see next paragraph) which probably should not be known as lexemes, because the term *lexeme* implies a regular morphological pipeline, while List B entries can be of different sizes.

The criterion of wordhood based on lexical specification of grammatical properties faces a major problem, because it does not tell us the upper bound of where the word boundaries may be. Note that a form with lexically specified grammatical properties may only be a half-word, like English neo-classical combining form *geo-* (which prefers an adjective-like position but is not a word). We know *geo-* is not a word, while *geography* is, because *geo-* does not take the plural suffix. Thus the determination of wordhood in this case ultimately relies on morphology-based word boundary test. Without this piece of evidence it is surprisingly hard to prove that *geography* is a single grammatical word if we disgard all phonological evidence. It is entirely possible for other languages to have grammatical constructions with similar complexity constraints with those of English neoclassical compounds, in which neither constituents can be used independently as the head of a noun phrase and yet there is evidence suggesting that the two constituents are all morphologically independent.

The distinction between derivation and phrasal grammar follows from the definition of word. Ideally, the direction of inference should not be reversed: a structure building process being structurally limited in complexity (e.g. impossibility of coordination) should not automatically qualify it as a word-internal process and hence a derivation process (though in the existing literature, the derivation/phase grammar distinction is often dealt with in this way). These structural criteria are also rarely decisive: the form *pre- and post-revolutionary* obviously mixes phrasal coordination with affixation, but the latter is prototypically derivational.[9]

The derivation/inflection distinction involves multiple parameters that do not always coincide. The distinction may be defined based on abstract structural criteria (thus tense marking is inflectional while argument structure alternation is perhaps not); this inevitably has to rely on a pre-defined concept of wordhood (or otherwise auxiliary verb constructions have to be considered inflectional; cf. the same problem faced in syntactic definition of wordhood). The distinction sometimes just means the synchronic/diachronic distinction: derivation processes are supposed to be less productive, a criterion largely consistent with the structural criterion (as a smaller grammatical unit also tends to fossilize faster), but productive derivation processes exist. The most uncontroversial definition of the derivation/inflection distinction relies on the morphological template of words, when the inflectional ending can be relatively clearly set apart from the stem (this, though, means stem alternation marking of aspect in Latin is derivational).

Wordhood is sometimes also considered related to productivity, as word formation processes are considered less productive than syntax. We consider this an epiphenomenon: a

[9]In this concrete example, we consider *pre-revolutionary* a grammatical word because phonologically the boundary between *pre-* and *revolutionary* doesn't matter: the form is read as a single phonological word and there is a single stress. This is hard to explain if there are two grammatical words in it.

word formation process having lower productivity is equivalent to the functional heads involved in it having difficulties being unconditionally spelt out, and such a diachronic change is much less disruptive if its outputs are already largely recorded in List B. The productivity criterion therefore is just the criterion of lexical specification of grammatical properties.

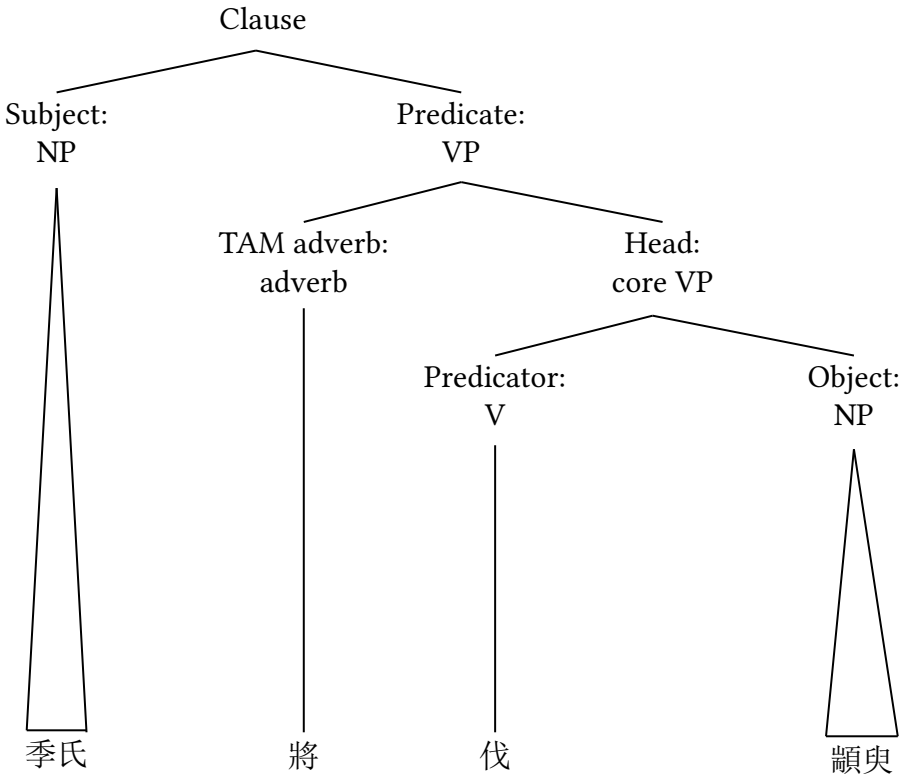
Distributed Morphology also has Local Dislocation operations, which correspond to some types of cliticization in descriptive linguistics (Embick et al. 2007). Not all clitics are due to Local Dislocation, though: the guilding intuition is that a clitic is added to a word *in a later stage*, which can be before or after Vocabulary Insertion, and be morphological or phonological.

Ultimately, we find that a single notion of grammatical wordhood is not completely necessary in a generative description of languages: the only things that matter are roots, functional projections, idioms, language-particular linearization, dislocation rules, and finally prosody and phonology. Multiple domain effects may emerge in the complex interplay between these processes. The "classic" perception of grammatical word as something defined by morphological templates can be formalized in a Distributed Morphology-like theory as complex head formation, and the boundary of the morphological template is determined by how large the complex heads can be in post-syntactic operations. Thus, a maximal complex head may be *defined* as a grammatical word (Embick et al. 2007). Multiple analyses may still be possible, and sometimes we can only proceed by adopting the simplest analysis.

A.6 Notations

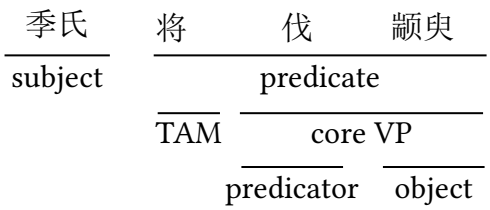
We have mentioned that the framework of Huddleston and Pullum (2002) seems to be a good starting point of the description of Classical Chinese. An example of the tree diagram representation of the constituency analysis (e.g. Huddleston and Pullum 2002, p. 954 [9]) is shown in Fig. A.1. This representation however does not clearly distinguish function words from content words.

Figure A.1: Tree diagrammatic analysis of 季氏將伐顓臾 'Jishi (a noble family which controls the politics of the Kingdom of Lu) is going to invade Zhuanyu (a vassal state of Lu).' (*Analects*)



Researchers in modern China conventionally draw lines below constituents with labels to represent constituency relations. Fig. A.2 contains exactly the same information in Fig. A.1 and is more compact. It also highlights the lexical/functional distinction. For instance, if 將 were a future tense particle and not an adverb, we could refrain from giving a label to it (and eliminating the "TAM" layer in Fig. A.1), implying that it is a grammatical marker and not a (content) constituent of the predicate.

Figure A.2: More compact version of Fig. A.1



The problem of Fig. A.2 is also its compactness: the label of a constituent cannot be too long, or otherwise the whole diagram cannot be fit within one line. In this work, we represent examples in a way that is more aligned to modern description grammars: we place constituents in brackets, and then label these brackets by subscripts. (1) is an example. When we have grammatical markers, we do not put them (e.g. the sentence final particle 矣 in 2) in brackets to highlight the lexical/functional distinction, consistent with the suggestion of Dixon (2009, p. 49).

(1) [[季氏]$_{\text{subject:NP}}$ [[將]$_{\text{TAM adverb}}$ [伐 顓臾]$_{\text{core VP}}$]$_{\text{predicate:VP}}$]$_{\text{verbal clause}}$
Jì-shì jiāng fá Zhuānyú
NAME-lineage FUT attack NAME
'The Ji family is going to invade Zhuanyu.' (*Analects*)

(2) [[[民$_{\text{subject,A:NP}}$ [鮮 $-_{\text{P}}$]$_{\text{predicate:VP}}$]$_{\text{subject: complement clause}}$
mín xiǎn
people lack
[久]$_{\text{predicate:AdjP}}$]$_{\text{verbal nucleus clause}}$ 矣$_{\text{sentential aspect}}$]$_{\text{sentence}}$
jiǔ yǐ
long SFP
'The people have lacked this for a long time!' (*Analects*)

One further issue is phonological transcription of the texts. The standard practice is to apply the Modern Mandarin phonology (romanized in *pinyin*), following the reading tradition (Pulleyblank 1995); this is comparable to how ancient Greek texts are read in modern Greece. Historically, it was also possible to read Classical Chinese with different phonological systems.

This standard practice does not give us much insights into the phonological reality of these texts. Some choose to transcribe texts in reconstructed Old Chinese (Jacques 2022). Details in reconstruction of the phonology of Old Chinese remains a highly debated topic, and no mutually accepted lossless romanization that reflects the historical phonology exists currently.

A.7 Procedure of analysis

We adopt a top-down order in writing this book. We start with high-level constituents, i.e. clauses and noun phrases. Grammatical systems are identified when these forms are analyzed, and are introduced in the rest of the sections of § 2.2 and § 2.3. Parts of speech divisions are made based on grammatical evidence and not semantics.

This organization is in contrast with some descriptive grammars, which start with e.g. the nominal and verbal morphology, and then discuss grammatical relations like possession, modification, and argument structure, and then outline the structure of the simple clause and complex clauses. This is not the approach taken here, partly because Classical Chinese is a rather analytic language and there is simply no inflectional morphology to describe, and partly because a well organized study of its grammar has to have the whole syntactic environment in mind: otherwise it will likely collapse into a dictionary of the various usages of function words.

This work also pays attention to the combinatorial possibilities of dependents in clauses and noun phrases and their relative orders and scope, which is often skipped in *ethno-philological* works mainly aiming to prepare the reader to comprehend *natural texts* in the language in question. We aim to study the grammar of Classical Chinese as a machine that takes lexical items and produces *arbitrarily* complex utterances, and to understand the structure of this machine. The approach is admittedly not unproblematic for a dead language, which has no native speaker with acceptability intuitions and may also have strong internal variances historically.